

TokTickIT

Lab 3 Final Submission Report

Users, roles, authentication, IT Staff ticketing, administrator screens, and Zen Green responsive evidence

Student: Jitvoottikrai | Repository: book6349/toktickit | Reviewer: Punge089

Committed main: 0353a4052c4a6d7ead9ddfb9b7c682d3e7958d02 | Local audit: 2026-10-01 | Base HEAD: 1fb8df040eb12ac443bb4246a314babfde1564e9 | Local fixes are uncommitted

The report records the reviewed Lab 3 workflow and the repaired local implementation. Lab 2 remains the completed baseline. REVIEWED INTEGRATION AND FINAL-MAIN VERIFICATION ARE STILL REQUIRED BEFORE SUBMISSION.

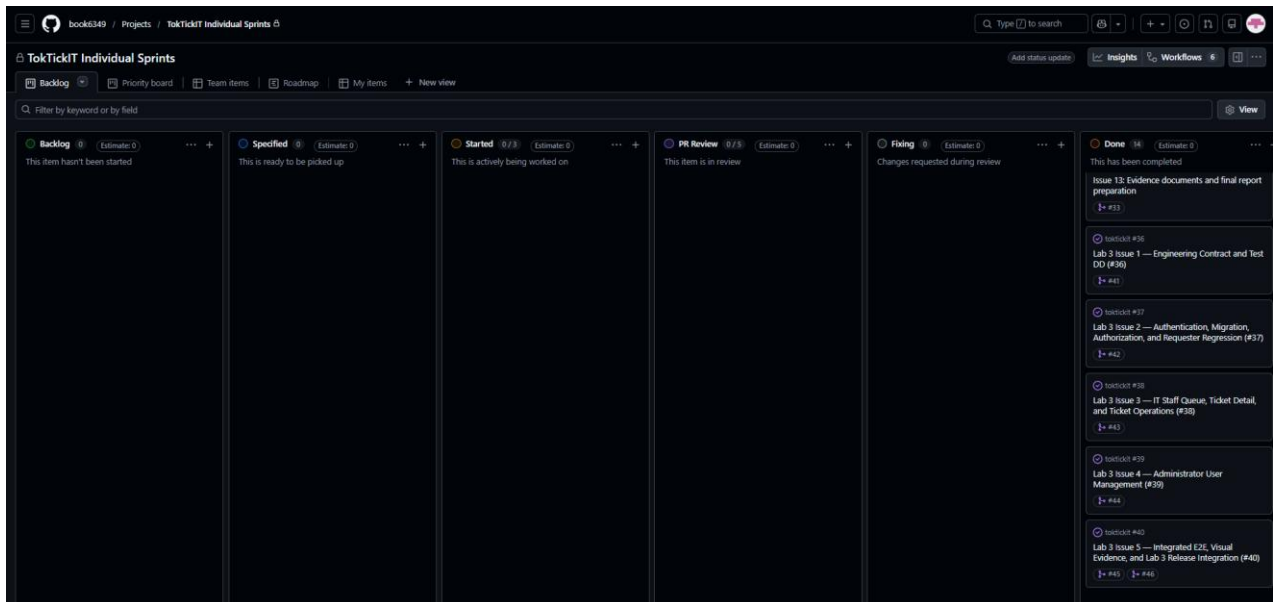
Answer Part 1 - Git Use with Engineering Workflow

The five approved units used Issues #36-#40 and feature PRs #41-#45 into lab3-staging. Reviewer Punge089 approved and merged them. Release PR #46 initially promoted staging to main; reviewed correction PR #47 and release PR #48 produced committed main 0353a40. No additional GitHub action was made during this local audit.

The workflow appendix gives actual review, reply, approval and merge links through PR #48. The retained board screenshot shows all five Issues in Done; the recorded 2026-09-30 live check confirmed Done/closed. Current reviewer access has not been independently reverified in this local report update.

Remote main was checked read-only on 2026-10-01 and remains 0353a40. The local repaired code, tests, documentation and screenshots are not yet reviewed or integrated. The report distinguishes these local results from committed-main history.

Unit	Issue	Branch	PR to target	Merge
Contract and Test DD	#36	feature/lab3-specification	#41 -> lab3-staging	95cbdfc
Auth / migration / Requester	#37	feature/lab3-auth-requester	#42 -> lab3-staging	dbc2511
Staff Queue / Detail	#38	feature/lab3-staff-workflow	#43 -> lab3-staging	3b55635
Admin Users	#39	feature/lab3-admin-users	#44 -> lab3-staging	160343f
Integration / evidence	#40	feature/lab3-integration-evidence	#45 -> lab3-staging	71139cc
Release	#40	lab3-staging	#46 -> main	493758b
Post-audit correction	#40	feature/lab3-submission-corrections	#47 -> lab3-staging	7d8ee7a
Reviewed release	#40	lab3-staging	#48 -> main	0353a40



GitHub Project board evidence showing Lab 3 Issues #36-#40 in Done status.

Answer Part 2 - Spec DD

The approved specification.md, api-spec.md and ui-spec.md define numbered requirements, business rules, role authorization, migration decisions, acceptance criteria and Product Definition of Done. Their rendered appendices are preserved below.

The contract and Test DD were merged in PR #41 before the implementation PRs. This preserves the required specification-before-implementation evidence. The final report includes rendered specification, API, and UI contract appendices rather than relying only on a link.

Answer Part 3 - Test DD and Traceability

Current Test DD traceability points to actual assertions and files. The 2026-10-01 dirty local tree passed 190 server tests, 36 client tests and 16 Chromium E2E scenarios; both builds and Prisma validation passed. Actual PostgreSQL migration snapshots, seed-twice assertions and attachment-byte checks passed on new disposable databases. Complete passing output is rendered at the end of this report.

These results do not certify committed main 0353a40. The clean-main checks recorded on 2026-09-30 had legacy regression failures; the current audit repaired those tests for session-based authentication. Reviewed staging/release integration and a clean-main rerun are still required. Earlier failed tests and the Staff indicator red test are retained with subsequent passing runs.

Command / check	Observed result	Boundary
server: npm test -- --reporter=verbose	190 tests / 19 files passed; exit 0	2026-10-01 dirty local tree; not committed-main evidence
client: npm test -- --reporter=verbose	36 tests / 14 files passed; exit 0	2026-10-01 dirty local tree; not committed-main evidence
server/client: npm run build	Both passed; exit 0	2026-10-01 dirty local tree; not committed-main evidence
server: npx --no-install prisma validate	Schema valid; exit 0	2026-10-01 dirty local tree; not committed-main evidence
root: npm run test:e2e:prepare	Migration baseline preserved; seed twice stable; exit 0	New disposable run2 database only
root: npx --no-install playwright test e2e/lab-03	16 passed; exit 0; responsive and keyboard journeys	2026-10-01 dirty local tree; not committed-main evidence
Live authorization and attachment download	Self-removal/role denial; attachment bytes match	Step 3 real API output; separate mocked last-admin branch
Final-main verification	NOT RUN after these new local changes	Requires reviewed integration

Answer Part 4 - AI Use with Reflection

Codex was the development assistant. Seven recorded representative prompts and My Reflection are rendered in the AI-use appendix. Specification-agent work clarified identity, authorization, migration and workflow before coding; coding-agent work supported implementation and verification. The audit found obsolete regression assumptions, an owner-filter defect and a missing Staff resolution display, showing why generated summaries must be checked against assertions and actual UI. Human approval and Punge089 peer review remain mandatory. The exact earlier model identifier was not recorded and is not guessed.

Answer Part 5 - Working Login and Password Change UI

Real database-backed browser checks demonstrate valid/invalid/inactive login, mandatory password replacement, authenticated identity/role, logout and direct API blocking. Busy feedback is captured while the real login request is deliberately delayed. The responsive Login and Change Password figures now use actual API sessions, not mocked authentication responses.

Status: Passed locally in tests and actual captures at desktop/tablet/mobile widths. Current code/evidence is not yet committed-main evidence.

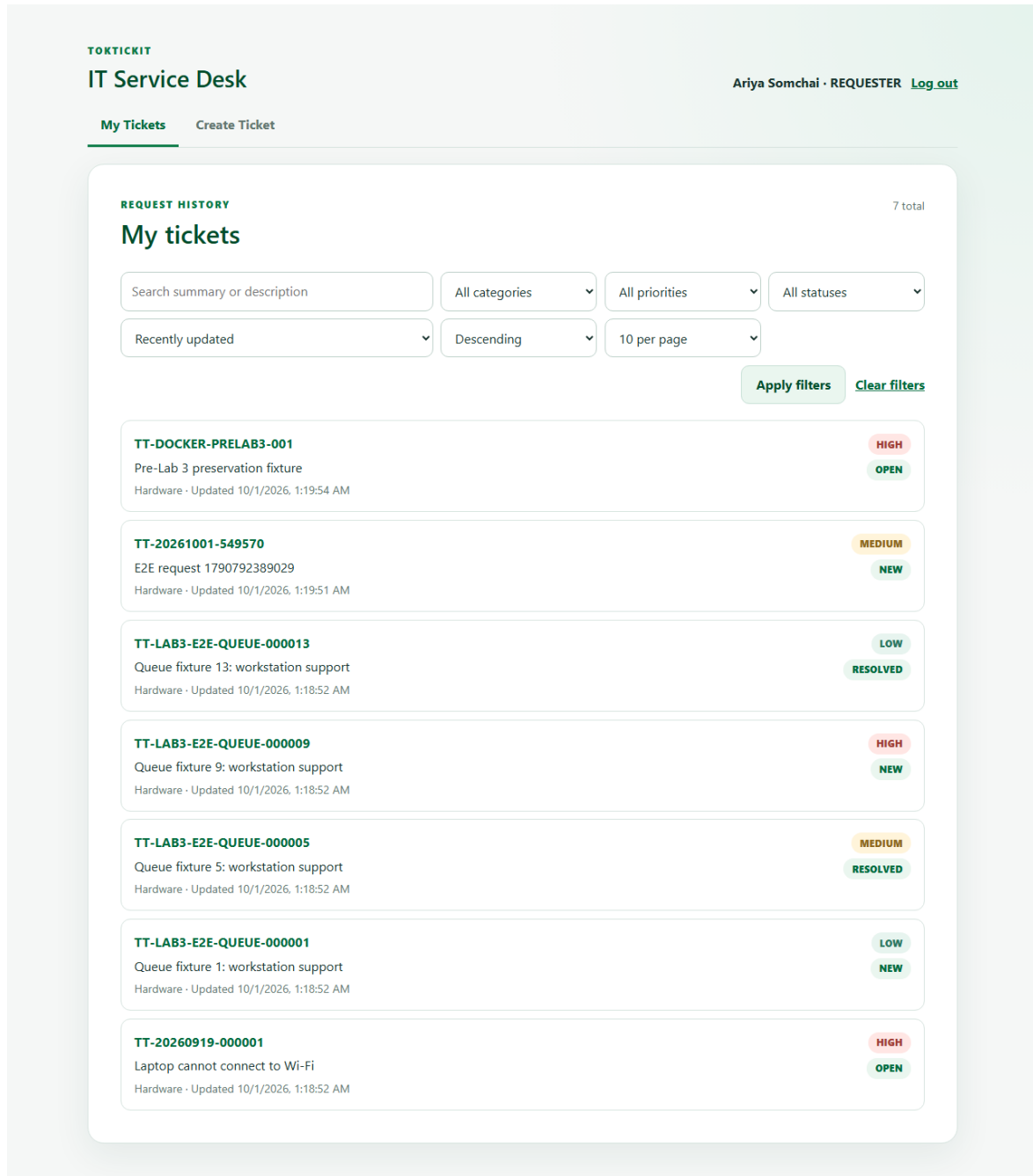


Figure 1. Requester My Tickets shell after authentication at desktop width Captured 2026-10-01, dirty local tree.

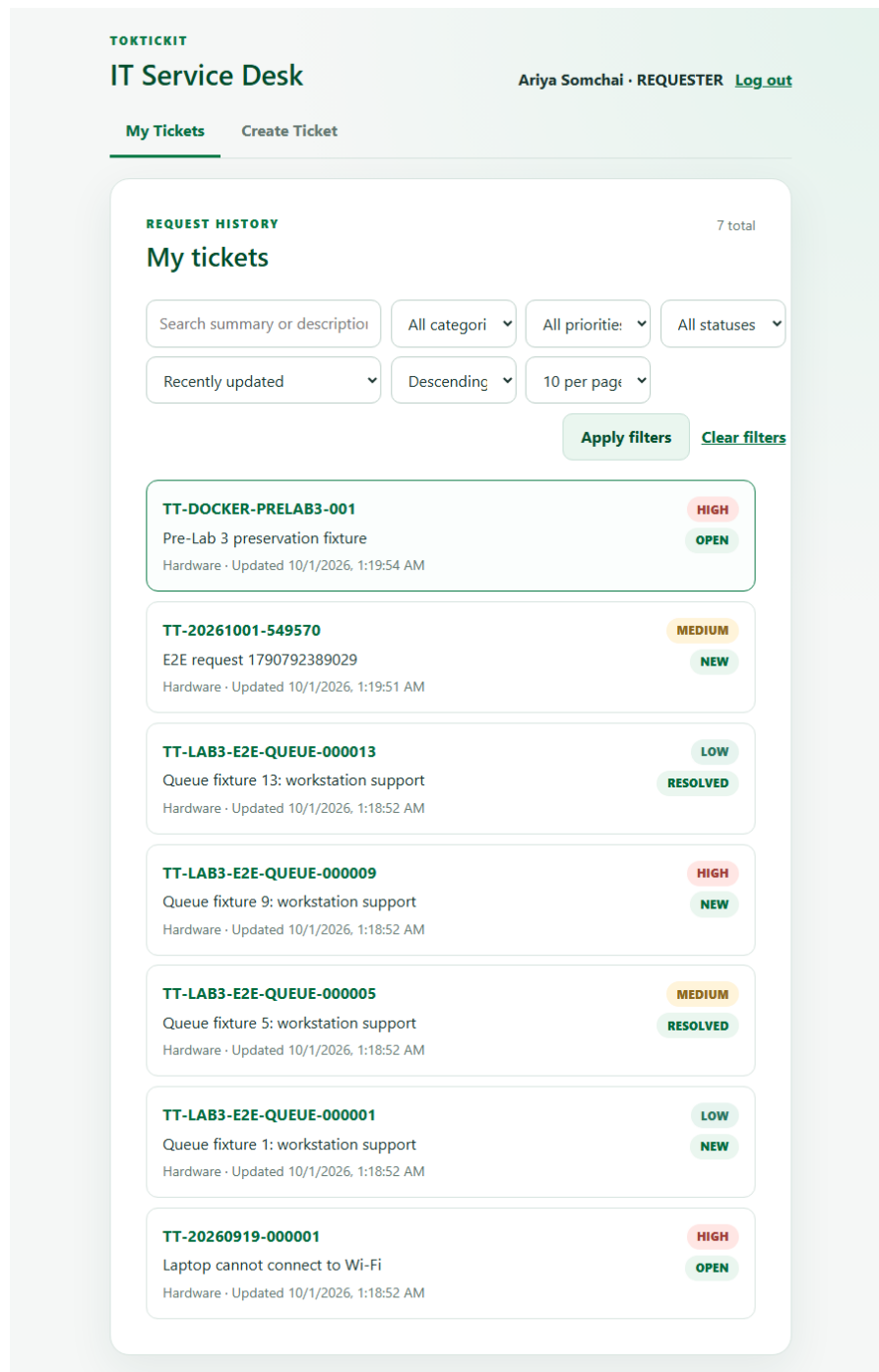


Figure 2. Requester My Tickets shell at tablet width Captured 2026-10-01, dirty local tree.

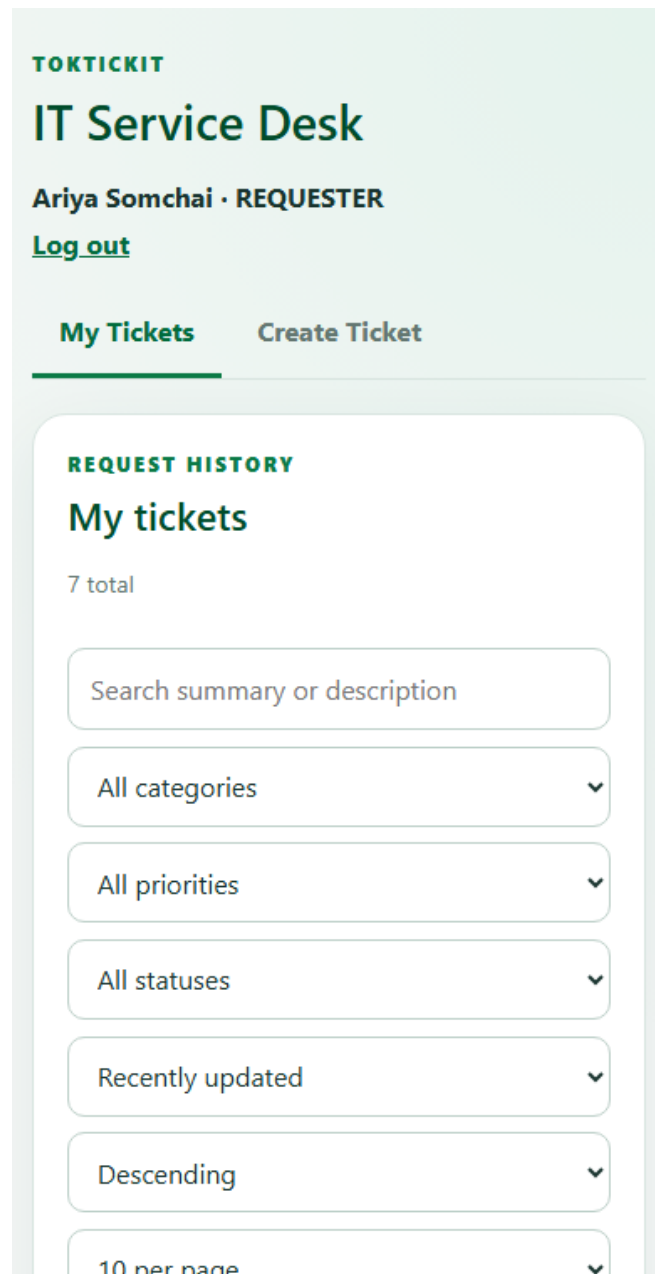


Figure 3. Requester My Tickets shell at mobile width; native readable section, full originals/continuations retained.
Captured 2026-10-01, dirty local tree.

IT SERVICE DESK

TokTickIT Sign in

Sign in with your TokTickIT account to continue.

Email or password is incorrect.

Email

Password

Sign in

Additional state capture — Safe invalid-login failure Captured 2026-10-01, dirty local tree.

IT SERVICE DESK

TokTickIT Sign in

Sign in with your TokTickIT account to continue.

Email or password is incorrect.

Email

Password

Sign in

Additional state capture — Safe inactive-account failure Captured 2026-10-01, dirty local tree.

ACCOUNT SECURITY

Change your password

Your initial password must be replaced before you can open the service desk.

Signed in as narin.kittisak@example.com. Use 12–128 characters with at least one letter and one number.

Current password

New password

12–128 characters; include a letter and a number.

Confirm new password

[Log out](#) Save password

Additional state capture - Real mandatory password gate; 2026-10-01 local capture, not keyboard-traversal proof.

IT SERVICE DESK

TokTickIT Sign in

Sign in with your TokTickIT account to continue.

Email

Password

Sign in

Additional state capture — Login gate after logout Captured 2026-10-01, dirty local tree.

VERIFIED LOCALLY - LOGIN BUSY FEEDBACK: client test and delayed-login browser capture show the disabled busy button; API response is real in E2E.

Answer Part 6 - Working IT Staff Ticket Queue UI

The Queue shows synthetic realistic priorities, statuses and assigned/unassigned owners. The disposable preparer produces 20 tickets before E2E; Requester creation adds another. Search, filters, all seven sort fields in both directions, owner combinations and populated second-page navigation are tested. Real no-results and labelled controlled empty/failure/loading captures are included.

Status: Passed locally for tested API/query combinations, populated pagination and responsive overflow. Controlled response screenshots prove UI presentation, not real database outage behavior.

TOKTICKIT
IT Service Desk
Somchai Staff · IT_STAFF [Log out](#)

Ticket Queue

OPERATIONS
Ticket Queue
Find shared work, inspect the detail, and keep ownership visible.

Search tickets
Number, summary, description, requester

Status
All statuses

Requested priority
All requested priorities

IT priority
All IT priorities

Owner
All ownership

Sort by
Last updated

Sort direction
Descending

Tickets per page
10

Apply filters **Clear filters**

TT-DOCKER-PRELAB3-001
Pre-Lab 3 preservation fixture
Hardware · Ariya Somchai · Updated 10/1/2026, 1:45:15 AM
Requested HIGH
IT LOW
OPEN
Owner: Kanya Staff
Open Detail

TT-20261001-158960
E2E request 1790793910044
Requested MEDIUM
IT MEDIUM

Figure 4. Queue native viewport at 1280x900, captured 2026-10-01 locally. Full-page originals and readable mobile continuations retained separately; not final-main.

Ticket Queue

OPERATIONS

Ticket Queue

Find shared work, inspect the detail, and keep ownership visible.

Search tickets	Status	Requested priority	IT priority
Number, summary, description	All statuses	All requeste	All IT priorit
Owner	Sort by	Sort direction	Tickets per page
All ownership	Last update	Descending	10
Apply filters			Clear filters

TT-DOCKER-PRELAB3-001	Requested HIGH
Pre-Lab 3 preservation fixture	IT LOW
Hardware · Ariya Somchai · Updated 10/1/2026, 1:45:15 AM	OPEN
Owner: Kanya Staff	Open Detail

TT-20261001-158960	Requested MEDIUM
E2E request 1790793910044	IT MEDIUM

Figure 5. Queue native viewport at 900x900, captured 2026-10-01 locally. Full-page originals and readable mobile continuations retained separately; not final-main.

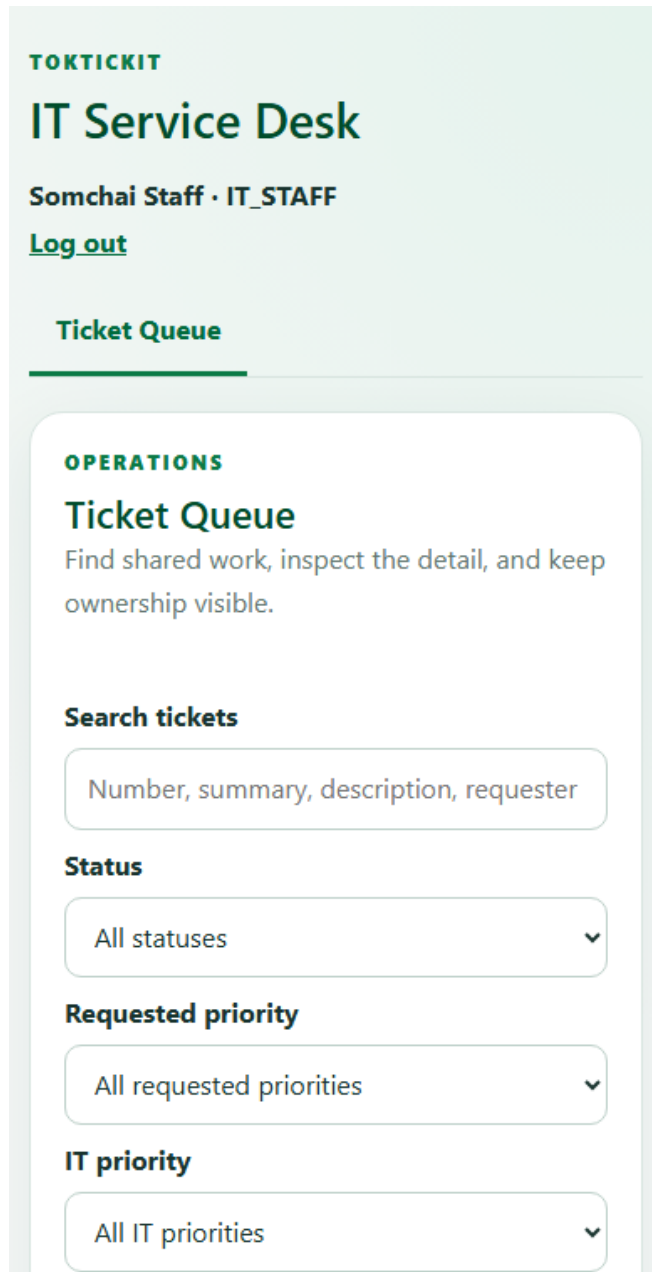


Figure 6. IT Staff Queue at mobile width (390 x 844 viewport); native readable section, full originals/continuations retained. Captured 2026-10-01, dirty local tree.

Ticket Queue

OPERATIONS

Ticket Queue

Find shared work, inspect the detail, and keep ownership visible.

Search tickets

Status

Requested priority

IT priority

Owner

Sort by

Sort direction

Tickets per page

Apply filters

Clear filters

No matching tickets

Clear or adjust the filters to broaden the Queue search. [Clear filters](#)

Additional state capture — Filtered queue with no matches Captured 2026-10-01, dirty local tree.

Ticket Queue

OPERATIONS

Ticket Queue

Find shared work, inspect the detail, and keep ownership visible.

Search tickets

Status

Requested priority

IT priority

Owner

Sort by

Sort direction

Tickets per page

Apply filters

Clear filters

No tickets yet

New tickets will appear here when they are submitted.

Additional state capture — Empty queue state Captured 2026-10-01, dirty local tree. CONTROLLED MOCKED API STATE; UI presentation only.

OPERATIONS

Ticket Queue

Find shared work, inspect the detail, and keep ownership visible.

Search tickets

Status

Requested priority

IT priority

Owner

Sort by

Sort direction

Tickets per page

[Apply filters](#)[Clear filters](#)

Controlled evidence: Queue service unavailable. [Retry](#)

Additional state capture — Queue API failure with retry Captured 2026-10-01, dirty local tree. CONTROLLED MOCKED API STATE; UI presentation only.

PAGINATION EVIDENCE: Figure 40 shows the actual populated second Queue page from the passing 2026-10-01 E2E run. The native test output records the executed Next-page assertion.

Answer Part 7 - Working IT Staff Ticket Detail UI

Staff Detail groups ownership, IT Priority/status, attachment metadata, Public Comments and Internal Notes. The 2026-10-01 audit added the omitted Requester-resolution indicator, which now displays the reported timestamp and explicitly preserves formal status. Four client assertions and the new real Staff E2E scenario verify it; Figures 43-45 show the fix.

Status: Passed locally. API tests exercise all 64 status pairs with mocked delegates, terminal confirmation, role restrictions and concurrency; real browser flows exercise representative transitions, claim/reassignment and content submission. A real Requester attachment download matched the original 17 bytes and SHA256. This is not exhaustive browser coverage of every transition.

TOKTICKIT

IT Service Desk

Somchai Staff · IT_STAFF [Log out](#)

Ticket Queue

← Back to Ticket Queue

TT-20261001-158960

E2E request 1790793910044

Requester

Ariya Somchai ·
ariya.somchai@example.com

Category

Hardware

Related system

Corporate Laptop

Created

10/1/2026, 1:45:10 AM

Updated

10/1/2026, 1:45:12 AM

Requested priority

MEDIUM

IT priority

MEDIUM

Owner

Somchai Staff ·
somchai.staff@example.com

Description

Integrated E2E requester regression flow.

Requester resolution indication

Requester indicates that the problem appears resolved.

Reported 10/1/2026, 1:45:10 AM. Formal ticket status is still NEW. Only IT Staff can formally resolve or close the Ticket.

Figure 7. Staff Detail native viewport at 1280x900, captured 2026-10-01 locally. Full-page originals and readable mobile continuations retained separately; not final-main.

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Ticket Queue

[← Back to Ticket Queue](#)

TT-20261001-158960

NEW

E2E request 1790793910044

Requester	Category	Related system	Created
Ariya Somchai · ariya.somchai@example.com	Hardware	Corporate Laptop	10/1/2026, 1:45:10 AM
Updated	Requested priority	IT priority	Owner
10/1/2026, 1:45:12 AM	MEDIUM	MEDIUM	Somchai Staff · somchai.staff@example.com

Description

Integrated E2E requester regression flow.

Requester resolution indication

Requester indicates that the problem appears resolved.

Reported 10/1/2026, 1:45:10 AM. Formal ticket status is still NEW. Only IT Staff can formally resolve or close the Ticket.

Figure 8. Staff Detail native viewport at 900x900, captured 2026-10-01 locally. Full-page originals and readable mobile continuations retained separately; not final-main.

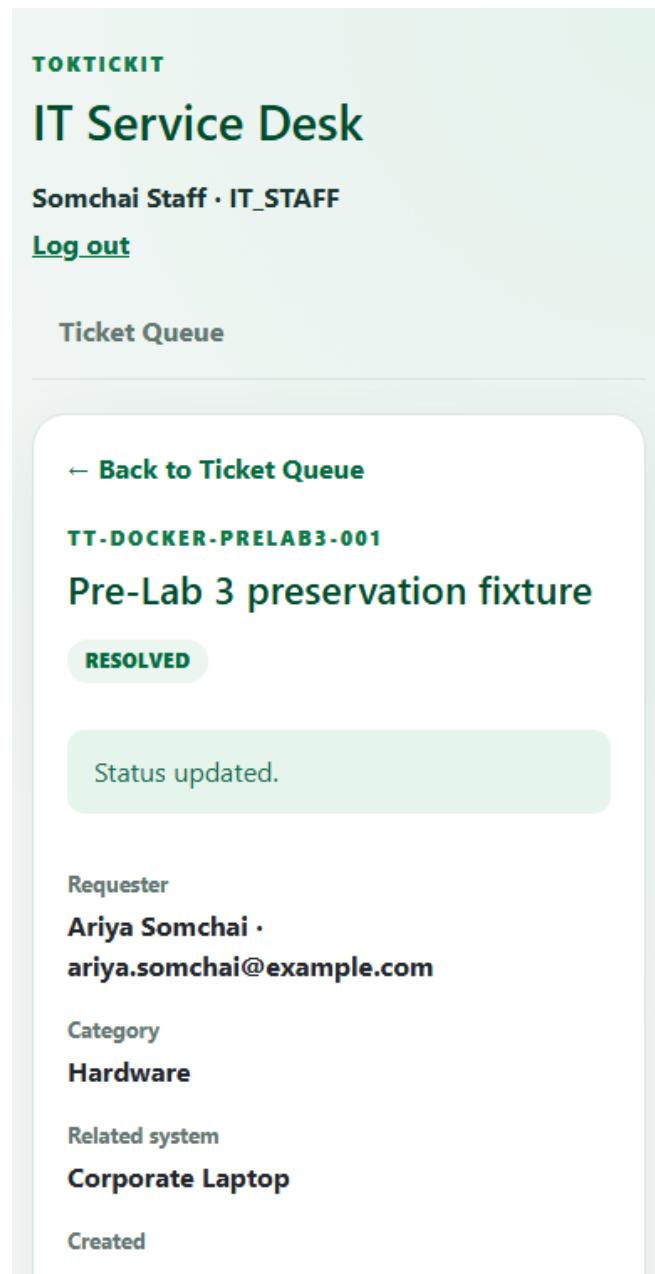


Figure 9. IT Staff Ticket Detail at mobile width (390 x 844 viewport); native readable section, full originals/continuations retained. Captured 2026-10-01, dirty local tree.

TOKTICKIT

IT Service Desk

Somchai Staff · IT_STAFF [Log out](#)

Ticket Queue

[← Back to Ticket Queue](#)

TT-DOCKER-PRELAB3-001

OPEN

Pre-Lab 3 preservation fixture

Ticket claimed.

Requester

Ariya Somchai · [ariya.somchai@example.com](#)

Category

Hardware

Related system

Corporate Laptop

Created

10/1/2026, 1:18:48 AM

Updated

10/1/2026, 1:31:47 AM

Requested priority

HIGH

IT priority

LOW

Owner

Somchai Staff · [somchai.staff@example.com](#)

Description

Synthetic migration fixture for local verification.

Ownership

Only active IT Staff or Administrators may own a Ticket.

Claim as me

Unassign

Assign owner ID

7

Assign owner

IT Priority and status

IT Priority

LOW

Save IT Priority

Next status

Choose a permitted transition

Update status

Attachments

Existing Ticket files.

baseline.txt

Uploaded 10/1/2026, 1:18:48 AM

Public Comments

Visible to Requesters, IT Staff, and Administrators.

No public comments.

Add a public comment

Add public comment

Internal Notes

Private to IT Staff and Administrators: never shown in the Requester workspace.

No Internal Notes.

Add an Internal Note

Add Internal Note

Additional state capture — Claim, IT Priority, public comment, and Internal Note operations Captured 2026-10-01, dirty local tree.

TOKTICKIT

IT Service Desk

Somchai Staff · IT_STAFF [Log out](#)

Ticket Queue

← Back to Ticket Queue

TT-DOCKER-PRELAB3-001

OPEN

Pre-Lab 3 preservation fixture

Owner not found.

Requester

Ariya Somchai ·
ariya.somchai@example.com

Category

Hardware

Related system

Corporate Laptop

Created

10/1/2026, 1:18:48 AM

Updated

10/1/2026, 1:19:54 AM

Requested priority

HIGH

IT priority

LOW

Owner

Kanya Staff ·
kanya.staff@example.com

Description

Synthetic migration fixture for local verification.

Ownership

Only active IT Staff or Administrators may own a Ticket.

Claim as me

Unassign

Assign owner ID

999999

Assign owner

IT Priority and status

IT Priority

LOW

Save IT Priority

Next status

Choose a permitted transition

Update status

Attachments

Existing Ticket files.

baseline.txt

Uploaded 10/1/2026, 1:18:48 AM

Public Comments

Visible to Requesters, IT Staff, and Administrators.

No public comments.

Add a public comment

Add public comment

Internal Notes

Private to IT Staff and Administrators: never shown in the Requester workspace.

No Internal Notes.

Add an Internal Note

Add Internal Note

Additional state capture — Safe owner-validation failure Captured 2026-10-01, dirty local tree.

Answer Part 8 - Working Administrator User Management UI

User Management displays Name, Email, Role, Active/Inactive and Edit. Real captures show search/role filtering, create, invalid password, duplicate email, edits of name/email/role/activation, persistent reset feedback and required next-login change. Self-deactivation is disabled in the UI and rejected by the real API; Requester/Staff management denial is recorded.

Status: Passed locally for executed scenarios. The separate last-active-Administrator error branch is tested with mocked delegates; live self-removal reaches the earlier self-protection guard and leaves one active Administrator. Do not confuse these two proof forms.

The screenshot shows the 'IT Service Desk' interface for an 'Administrator'. The top navigation bar includes the 'TokTickIT' logo, the title 'IT Service Desk', and the user's role 'Admin User · Administrator' with a 'Log out' link. Below this, the 'User Management' section is highlighted. The main content area is divided into two panels. The top panel, titled 'Administrator workspace', contains a 'Ticket ID' input field and an 'Open Ticket Detail' button. The bottom panel, titled 'User Management', features a 'Create user' button, a search bar for 'Search name or email', a dropdown for 'All roles', and 'Apply filters' and 'Clear filters' buttons. Below these filters, a table lists users, with the first entry being 'Admin User' (admin@example.com, Updated 10/1/2026, 1:45:16 AM), which has 'Administrator', 'Active', and 'Edit' buttons next to it.

TOKTICKIT
IT Service Desk
Admin User · Administrator [Log out](#)

User Management

ADMINISTRATOR
Administrator workspace
The shared IT Staff Queue is not available to Administrators. Open an explicitly permitted Ticket Detail directly when you have a Ticket ID.

Ticket ID

Open Ticket Detail

ADMINISTRATOR
User Management
Manage the minimum account fields required by Lab 3. Password reset is a separate action and requires a change at the user's next login.

Apply filters [Clear filters](#)

Admin User
admin@example.com
Updated 10/1/2026, 1:45:16 AM

Administrator **Active** **Edit**

Figure 10. Administrator User Management native viewport at 1280x900, captured 2026-10-01 locally. Full-page originals and readable mobile continuations retained separately; not final-main.

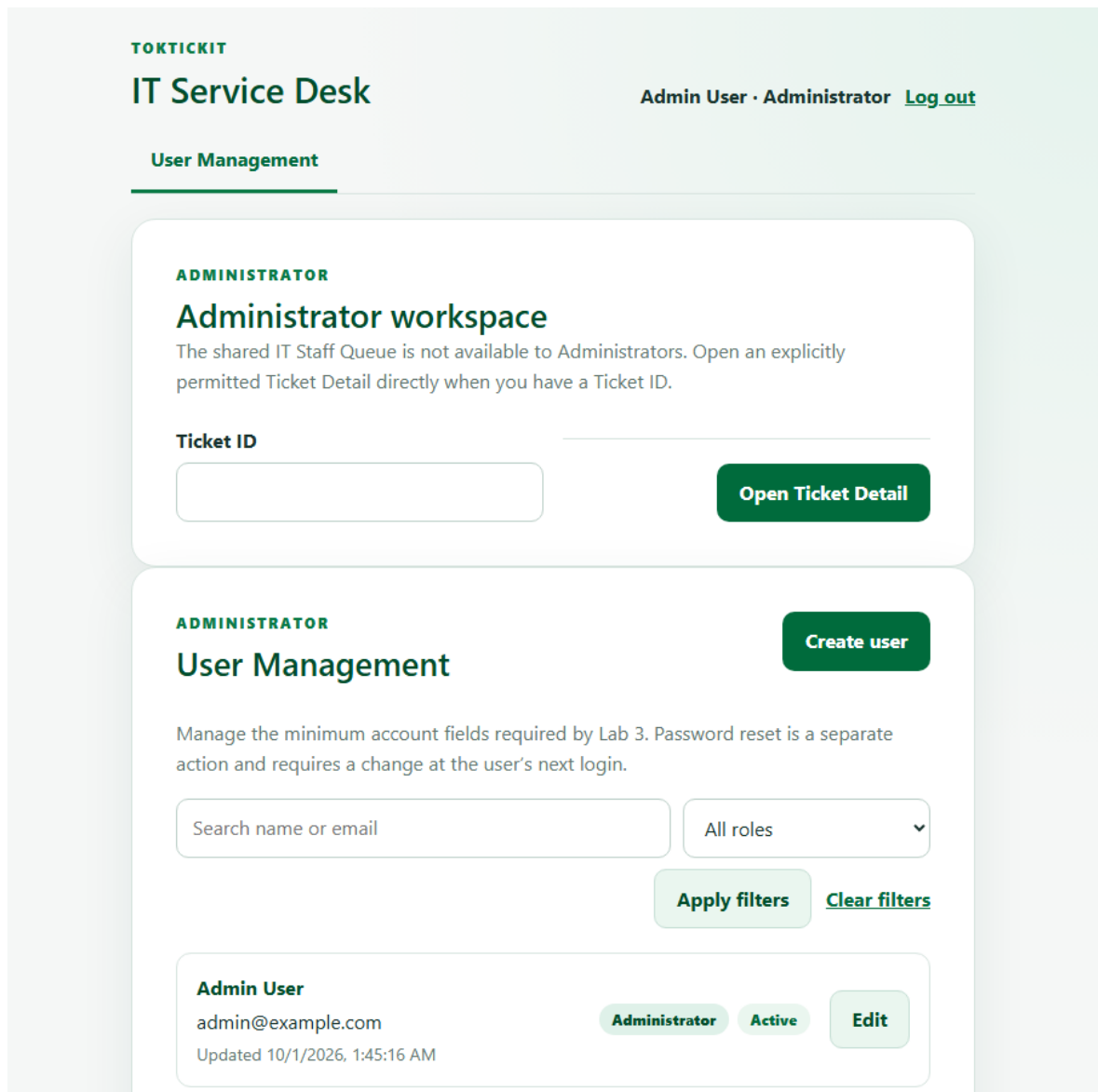


Figure 11. Administrator User Management native viewport at 900x900, captured 2026-10-01 locally. Full-page originals and readable mobile continuations retained separately; not final-main.

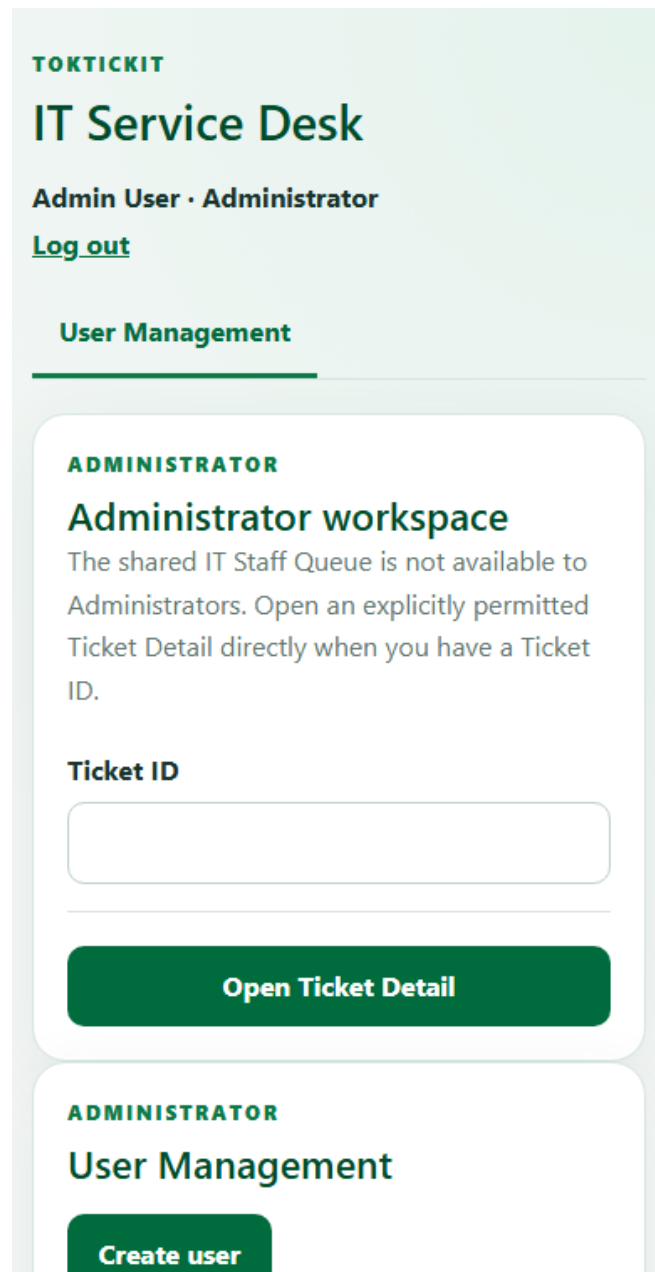


Figure 12. Administrator User Management at mobile width (390 x 844 viewport); native readable section, full originals/continuations retained. Captured 2026-10-01, dirty local tree.

User Management[← Back to User Management](#)

NEW ACCOUNT

Create user

Name

Email

Role

☒ Active account

Initial password

Passwords use 12–128 characters with at least one letter and one number. New and reset passwords require a change at next login.

Create user

[Cancel](#)

Additional state capture — Administrator create-user form Captured 2026-10-01, dirty local tree.

User Management[← Back to User Management](#)

NEW ACCOUNT

Create user

That email address is already in use.

Name

Email

Choose a different email address.

Role

☒ Active account

Initial password

Passwords use 12–128 characters with at least one letter and one number. New and reset passwords require a change at next login.

Create user

[Cancel](#)

Additional state capture — Duplicate-email validation Captured 2026-10-01, dirty local tree.

TOKTICKIT

IT Service Desk

Admin User · Administrator [Log out](#)

User Management

ADMINISTRATOR

Administrator workspace

The shared IT Staff Queue is not available to Administrators. Open an explicitly permitted Ticket Detail directly when you have a Ticket ID.

Ticket ID

Open Ticket Detail

ADMINISTRATOR

User Management

Create user

Manage the minimum account fields required by Lab 3. Password reset is a separate action and requires a change at the user's next login.

zz-no-user-20261001

All roles

Apply filters

[Clear filters](#)

No matching users

Try a different search or role filter.

Additional state capture — Administrator search with no matches Captured 2026-10-01, dirty local tree.

TOKTICKIT

IT Service Desk

Admin User · Administrator [Log out](#)

User Management

← Back to User Management

EDIT ACCOUNT

User #8

Edit user

Name

Kanya Staff

Email

kanya.staff@example.com

Role

IT Staff

☒ Active account

Passwords use 12–128 characters with at least one letter and one number. New and reset passwords require a change at next login.

Save changes

Cancel

Reset initial password

This is separate from account editing and forces a password change at next login.

New initial password

Reset password

Additional state capture — Administrator edit form Captured 2026-10-01, dirty local tree.

TOKTICKIT

IT Service Desk

Admin User · Administrator [Log out](#)

User Management

← Back to User Management

EDIT ACCOUNT

User #22

Edit user

Name

Evidence Edited Account

Email

evidence-edited-20261001@example.com

Role

Requester

Active account

☒

Passwords use 12–128 characters with at least one letter and one number. New and reset passwords require a change at next login.

Save changes

Cancel

Reset initial password

This is separate from account editing and forces a password change at next login.

Initial password reset. The user must change it at next login.

New initial password

Reset password

Additional state capture - Persistent reset confirmation, real API; 2026-10-01 local capture.

ACCOUNT SECURITY

Change your password

Your initial password must be replaced before you can open the service desk.

Signed in as evidence-edited-20261001@example.com. Use 12–128 characters with at least one letter and one number.

Current password

New password

12–128 characters; include a letter and a number.

Confirm new password

Save password

[Log out](#)

Additional state capture — Reset user reaches the mandatory password-change screen at next login Captured 2026-10-01, dirty local tree.

Answer Part 9 - Zen Green UI and Responsive Evidence

The UI specification and checklist are rendered below. Real major-screen captures cover desktop 1280x900, tablet 900x900 and mobile 390x844. Tall mobile views use native readable browser sections with overlap; full-page originals are retained. Automated document-width assertions and three keyboard journeys passed in E2E. Inspection covers the captured states, not every possible device or assistive technology.

Checklist item	Evidence	Status
Desktop/tablet/mobile	Actual major-screen captures and native mobile sections	Inspected locally; not final-main
Authentication gates	Real API-backed Login/Change Password at three widths	Captured and overflow checked
Role navigation/badges/read-only fields	Requester/Staff/Admin views, readonly Admin Detail	Inspected; operation authorization also tested
Validation and feedback	Real invalid login/owner/password/duplicate; controlled failure states labelled	Observed in identified captures
Public/private and resolution display	Separate content panels; Figures 43-45 after Staff fix	Locally verified; formal status unchanged
Keyboard/focus	Three real E2E journeys plus busy flow	Passed locally; not complete accessibility certification
Clipping/overflow/readability	Width assertions and native sections; QA inspection	Passed for captured states; no numeric contrast audit

VERIFIED LOCALLY - KEYBOARD AUDIT: three keyboard-only E2E journeys passed; the separate busy-state browser test also passed.

Current local E2E and responsive evidence (dirty working tree)

The following figures are actual browser captures from the 2026-10-01 local audit. Step-4 screenshots show unchanged screens; replacement Staff indicator captures come from the post-fix 16-test run. All remain dirty local evidence, not final-main.

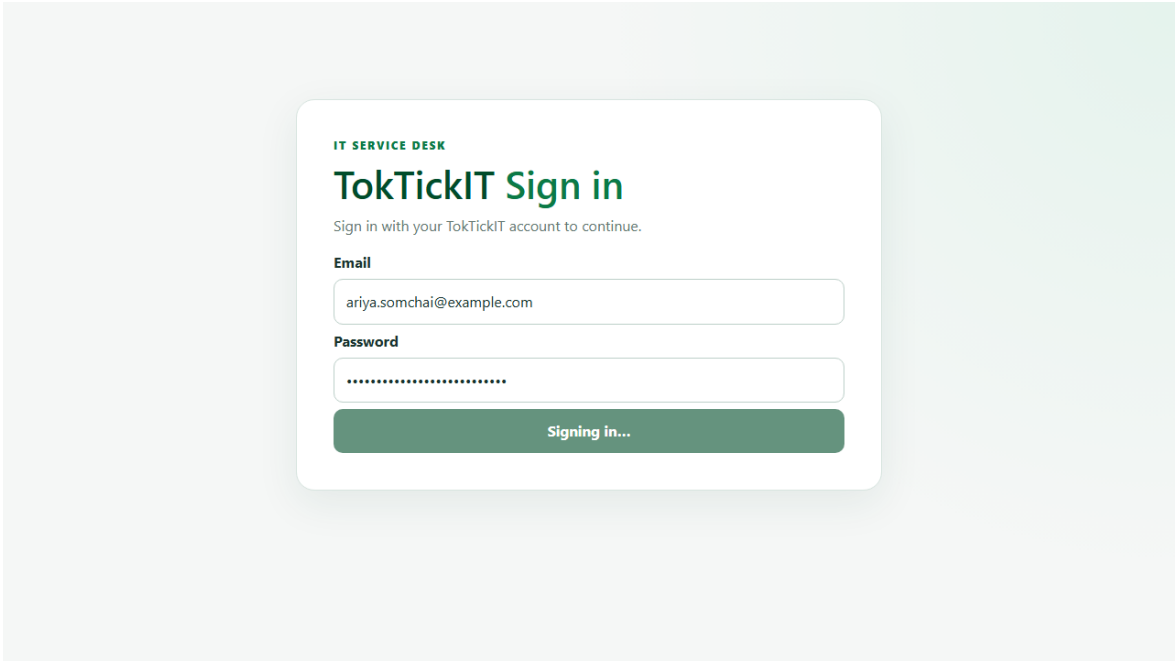


Figure 13. Disabled Signing in feedback during a delayed authentication request. Captured 2026-10-01, dirty local tree.

REQUEST HISTORY

7 total

My tickets

All categories ▾

All priorities ▾

All statuses ▾

Recently updated ▾

Descending ▾

10 per page ▾

Apply filters

[Clear filters](#)

TT-DOCKER-PRELAB3-001

Pre-Lab 3 preservation fixture

Hardware · Updated 10/1/2026, 1:45:15 AM

HIGH

OPEN

TT-20261001-158960

E2E request 1790793910044

Hardware · Updated 10/1/2026, 1:45:12 AM

MEDIUM

NEW

TT-LAB3-E2E-QUEUE-000013

Queue fixture 13: workstation support

Hardware · Updated 10/1/2026, 1:43:57 AM

LOW

RESOLVED

TT-LAB3-E2E-QUEUE-000009

Queue fixture 9: workstation support

Hardware · Updated 10/1/2026, 1:43:57 AM

HIGH

NEW

TT-LAB3-E2E-QUEUE-000005

Queue fixture 5: workstation support

Hardware · Updated 10/1/2026, 1:43:57 AM

MEDIUM

RESOLVED

TT-LAB3-E2E-QUEUE-000001

Queue fixture 1: workstation support

Hardware · Updated 10/1/2026, 1:43:57 AM

LOW

NEW

TT-20260919-000001

Laptop cannot connect to Wi-Fi

Hardware · Updated 10/1/2026, 1:43:57 AM

HIGH

OPEN

Figure 14. Successful Requester destination after authentication. Captured 2026-10-01, dirty local tree.

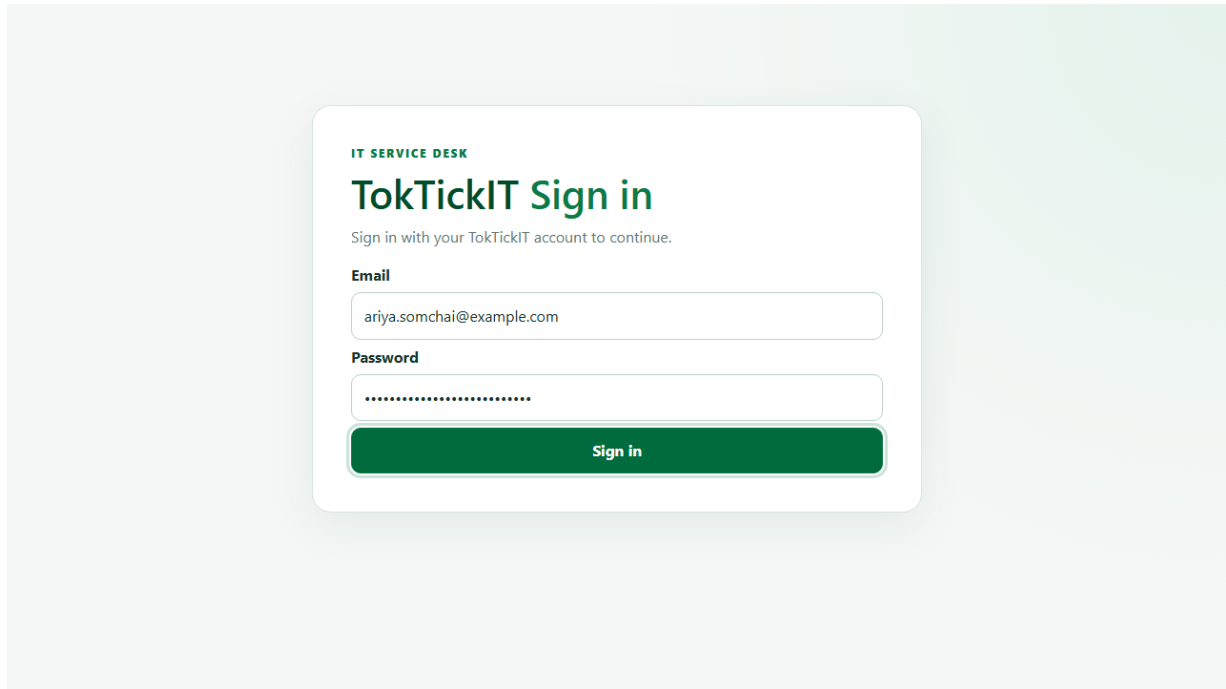


Figure 15. Sign-in submit control reached through keyboard navigation. Historical 2026-09-29 capture; current keyboard journeys are recorded separately.

[My Tickets](#)[Create Ticket](#)

REQUEST HISTORY

7 total

My tickets

All categories

All priorities

All statuses

Recently updated

Descending

10 per page

Apply filters

[Clear filters](#)

TT-DOCKER-PRELAB3-001

Pre-Lab 3 preservation fixture

Hardware · Updated 10/1/2026, 1:45:15 AM

HIGH

OPEN

TT-20261001-158960

E2E request 1790793910044

Hardware · Updated 10/1/2026, 1:45:12 AM

MEDIUM

NEW

TT-LAB3-E2E-QUEUE-000013

Queue fixture 13: workstation support

Hardware · Updated 10/1/2026, 1:43:57 AM

LOW

RESOLVED

TT-LAB3-E2E-QUEUE-000009

Queue fixture 9: workstation support

Hardware · Updated 10/1/2026, 1:43:57 AM

HIGH

NEW

TT-LAB3-E2E-QUEUE-000005

Queue fixture 5: workstation support

Hardware · Updated 10/1/2026, 1:43:57 AM

MEDIUM

RESOLVED

TT-LAB3-E2E-QUEUE-000001

Queue fixture 1: workstation support

Hardware · Updated 10/1/2026, 1:43:57 AM

LOW

NEW

TT-20260919-000001

Laptop cannot connect to Wi-Fi

Hardware · Updated 10/1/2026, 1:43:57 AM

HIGH

OPEN

Figure 16. Requester Create Ticket navigation reached by keyboard. Captured 2026-10-01, dirty local tree.

TOKTICKIT

IT Service Desk

Somchai Staff - IT_STAFF [Log out](#)

Ticket Queue

← Back to Ticket Queue

TT-DOCKER-PRELAB3-001

Pre-Lab 3 preservation fixture

OPEN

Requester

Ariya Somchai -
ariya.somchai@example.com

Category

Hardware

Related system

Corporate Laptop

Created

10/1/2026, 1:43:53 AM

Updated

10/1/2026, 1:45:15 AM

Requested priority

HIGH

IT priority

LOW

Owner

Kanya Staff -
kanya.staff@example.com

Description

Synthetic migration fixture for local verification.

Ownership

Only active IT Staff or Administrators may own a Ticket.

Claim as me

Unassign

Assign owner ID

8

Assign owner

IT Priority and status

IT Priority

LOW

Save IT Priority

Next status

Choose a permitted transition

Update status

Attachments

Existing Ticket files.

baseline.txt

Uploaded 10/1/2026, 1:43:53 AM

Public Comments

Visible to Requesters, IT Staff, and Administrators.

No public comments.

Add a public comment

Add public comment

Internal Notes

Private to IT Staff and Administrators; never shown in the Requester workspace.

No Internal Notes.

Add an Internal Note

Add Internal Note

Figure 17. IT Staff opened Ticket Detail and reached the Back control by keyboard. Captured 2026-10-01, dirty local tree.

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TOKTICKIT
IT Service Desk
Admin User · Administrator [Log out](#)

User Management

[← Back to User Management](#)

NEW ACCOUNT

Create user

Name

Email

Role
Requester

☒ **Active account**

Initial password

Passwords use 12–128 characters with at least one letter and one number. New and reset passwords require a change at next login.

[Create user](#) [Cancel](#)

Figure 18. Administrator opened Create User and focused the Name field by keyboard. Captured 2026-10-01, dirty local tree.

TOKTICKIT
IT Service Desk
Admin User · Administrator [Log out](#)

User Management

[← Back to User Management](#)

EDIT ACCOUNT

User #22

Edit user

Name
Evidence Edited Account

Email
evidence-edited-20261001@example.com

Role
Requester

☒ **Active account**

Passwords use 12–128 characters with at least one letter and one number. New and reset passwords require a change at next login.

[Save changes](#) [Cancel](#)

Reset initial password

This is separate from account editing and forces a password change at next login.

Initial password reset. The user must change it at next login.

New initial password

[Reset password](#)

Figure 19. Persistent administrator reset-success confirmation before returning to User Management. Captured 2026-10-01, dirty local tree.

Required Submission Evidence Audit - Pages 16-18

The following editable audit is the direct checklist against the official Required Submission Evidence table. It distinguishes repository/test evidence from screenshot evidence and makes every unavailable proof visible.

Lab 3 Required Submission Evidence Audit

This audit maps labsheet pages 16-18 to current local evidence. Functional gaps were repaired, but reviewed integration, final-main passing output, current reviewer-access confirmation and the final submission acceptance audit remain outstanding.

Evidence Status

The board screenshot is present. Workflow records contain actual Issues #36-#40 and PRs #41-#48, with Punge089 reviews, replies, approvals and merges. Docker-backed local migration, seed and E2E checks now passed. Remote main is still 0353a40, so the new repairs are not yet published and must not be described as final-main results.

Page 16 - Submission format and Part 1

Required evidence	Evidence location	Verification	Status
One concise PDF to submit	output/docs/Lab3_Final_Report.pdf	Regenerated from the editable DOCX; final page count, hash, links, and every rendered page are checked after export	Updated editable report; final acceptance audit pending
Headings Answer Part 1 through Answer Part 9 in exact order	Final report body pages 1-32	Extracted heading check; Part 1-9 headings occur in order	Verified locally / recorded workflow; final-main rerun pending
Working links	Final report annotations and GitHub links	Issues #36-#40 and PRs #41-#48 and review/comment/reply/approval anchors were fetched and confirmed; permission endpoint returned 403	Destinations retained; final access/link audit pending
Readable screenshots	Final report Figures 1-51 and artifacts/lab-03/screenshots/	Current evidence: 2026-10-01 local logs/captures; see current Parts 3-9 and full-output appendix. Historical Git events through PR #48 remain distinct.	Real captures and native sections inspected; local only
Final repository and main as source of truth	Part 1, local reviewer.md, committed main 0353a40	Report records local corrections; remote main remains 0353a40. These local repairs have not been published.	Updated editable report; final acceptance audit pending
Feature branches merged into lab3-staging, then main	Part 1 table and local reviewer.md section 2	Issues #36-#40 and PRs #41-#48 confirmed through read-only GitHub fetches; all merged/closed	Verified locally / recorded workflow; final-main rerun pending
Final Kanban board with all Issues in Done	reviewer.md, Part 1, screenshot embedded on report page 2	Image shows #36-#40 Done; current live board access not rechecked	Verified locally / recorded workflow; final-main rerun pending

Required evidence	Evidence location	Verification	Status
Rendered reviewer.md with reviewer identity, PR links, comments, replies, approvals	Final report workflow appendix and reviewer.md	Eight review/reply/approval/merge rows are rendered	Verified locally / recorded workflow; final-main rerun pending
README and .gitignore evidence	Final report workflow appendix and repository links	Historical rendered files retained; current local changes require reviewed integration and final synchronization.	Verified locally / recorded workflow; final-main rerun pending
Repository directory structure	Final report repository appendix	git ls-tree -r --name-only 0353a4052c4a6d7ead9ddf b9b7c682d3e7958d02: 218 committed paths; untracked and ignored local files excluded.	Verified locally / recorded workflow; final-main rerun pending

Page 17 - Parts 2 through 8

Part	Required evidence	Evidence location and check	Status
2	Rendered docs/lab-03/specification.md	Full source rendered in the specification appendix; numbered requirements, business rules, authorization rules, ACs, migrations, and DoD are visible	Verified locally / recorded workflow; final-main rerun pending
2	Specification existed before implementation PRs	Part 2 chronology: contract/Test DD in PR #41 before PRs #42-#45	Recorded chronology: PR #41 preceded implementation PRs #42-#45.
3	Rendered docs/lab-03/tests.md	Full Test DD rendered in the test appendix; traceability and file paths are visible	190 server / 36 client / 16 E2E passed locally; not main
3	Planned tests, AC traceability, actual paths, final status	Test DD appendix plus docs/lab-03/tests.md	190 server / 36 client / 16 E2E passed locally; not main
3	Complete passing unit, API/integration, UI, authorization, regression, and E2E output from committed main	Current full server/client logs and 16-test E2E run are from the dirty tree; historical main log is summarized only	190 server / 36 client / 16 E2E passed locally; not main
4	Rendered docs/lab-03/ai-use.md, LLM name, 6-10 prompts, reflection	Current evidence: 2026-10-01 local logs/captures; see current Parts 3-9 and full-output appendix. Historical Git events through PR #48 remain distinct.	Codex recorded; exact earlier model identifier unavailable
5	Valid/invalid login and inactive-account handling	Authentication tests, fresh 16-test E2E run, and browser captures	190 server / 36 client / 16 E2E passed locally; not main

Part	Required evidence	Evidence location and check	Status
5	Busy and safe failure feedback	Client tests and fresh delayed-login browser capture	190 server / 36 client / 16 E2E passed locally; not main
5	Login and mandatory Change Password at desktop/tablet/mobile	Figures 34-39; actual React client with actual API sessions	Verified locally / recorded workflow; final-main rerun pending
5	First-password change, user/role display, logout, direct access blocked	Authentication client/API tests and E2E	190 server / 36 client / 16 E2E passed locally; not main
6	Realistic queue data, search, filters, sorting, pagination, ownership, badges, open detail	Queue tests, 21 real E2E fixture records, Page 2 of 3 assertion, and screenshot	190 server / 36 client / 16 E2E passed locally; not main
6	Empty, no-results, and failure feedback	Real filtered no-results capture; controlled-empty-1280x900.png and controlled-failure-1280x900.png separately labelled UI-state captures.	Verified locally / recorded workflow; final-main rerun pending
6	Responsive behavior	Required desktop/tablet/mobile queue figures and Figures 20-24	Real captures and native sections inspected; local only
7	Claim/reassign, IT Priority, permitted status changes	Staff API/client tests and migration/operations E2E capture	190 server / 36 client / 16 E2E passed locally; not main
7	Public Comments, Internal Notes, attachment continuity, requester resolution	Detail tests, migration/operations capture, and Requester resolution screenshot	190 server / 36 client / 16 E2E passed locally; not main
7	Role restrictions, validation, safe failure, direct API authorization	Current API/client tests and E2E	190 server / 36 client / 16 E2E passed locally; not main
8	List with Name, Email, Role, Status, Edit	Responsive user-management figures	Real captures and native sections inspected; local only
8	Search and optional role filter	Admin E2E and no-results capture	Verified locally / recorded workflow; final-main rerun pending
8	Create user with permitted role and initial password	Create-form capture and admin E2E	Verified locally / recorded workflow; final-main rerun pending
8	Duplicate-email and invalid-input validation	Duplicate-validation capture and API tests	190 server / 36 client / 16 E2E passed locally; not main

Part	Required evidence	Evidence location and check	Status
8	Edit name, email, role, activation	Edit-form capture and API/client tests	190 server / 36 client / 16 E2E passed locally; not main
8	Reset initial password and required next-login change	Persistent reset-success-1280x900.png plus reset-first-login-1280x900.png	Verified locally / recorded workflow; final-main rerun pending
8	Self-deactivation and last-active-Administrator protection	Named test cases in users-admin.api.test.ts and complete server test output	190 server / 36 client / 16 E2E passed locally; not main
8	Forbidden non-Administrator access	Named denial cases in users-admin.api.test.ts and staff-queue.api.test.ts	190 server / 36 client / 16 E2E passed locally; not main
8	Responsive Zen Green and safe failure feedback	Responsive figures, API/client tests, and report checklist	190 server / 36 client / 16 E2E passed locally; not main

Page 18 - Part 9 and PDF format

Required evidence	Evidence location	Verification	Status
Rendered ui-spec.md	UI-spec appendix and repository link	Full UI specification rendered	Verified locally / recorded workflow; final-main rerun pending
Desktop, tablet, and mobile screenshots for all major Lab 3 screens	Figures 1-12, 20-33, responsive auth Figures 34-39, and interaction Figures 40-51	Current evidence: 2026-10-01 local logs/captures; see current Parts 3-9 and full-output appendix. Historical Git events through PR #48 remain distinct.	Updated editable report; final acceptance audit pending
Design consistency and Zen Green tokens	UI specification and visual checklist	Token and responsive rules rendered	Real captures and native sections inspected; local only
Role navigation	Authentication/staff/admin tests and captures	Functional role navigation passed	190 server / 36 client / 16 E2E passed locally; not main
Badges	Queue, detail, and user-management captures	Status, priority, role, and active-state badges visible	Verified locally / recorded workflow; final-main rerun pending
Editable/read-only fields	Staff detail and Administrator direct-detail evidence	API/client tests and detail captures	190 server / 36 client / 16 E2E passed locally; not main
Validation placement	First-login, queue/detail, and administrator validation captures	Field and alert placement visually inspected	Verified locally / recorded workflow; final-main rerun pending
Keyboard focus	z-accessibility-evidence.spec.ts and fresh browser captures	Three keyboard-only flows and one delayed-login busy-state flow passed in the local E2E run	Verified locally / recorded workflow; final-main rerun pending

Required evidence	Evidence location	Verification	Status
Clipping and overlap	Fresh viewport screenshots, responsive assertions, and mobile crops	E2E viewport bounds passed for four main screens; final PDF pages are visually inspected after export	Updated editable report; final acceptance audit pending
Horizontal overflow	E2E scrollWidth <= clientWidth assertions	Passed for Requester, Queue, Ticket Detail, and Admin at required viewports	Verified locally / recorded workflow; final-main rerun pending
One PDF with exact Answer Part format	output/docs/Lab3_Final_Report.pdf	Re-exported from editable DOCX; exact section order, figure numbering, links, and all page renders checked	Updated editable report; final acceptance audit pending

Evidence classification

- **Verified:** the command, source, screenshot, or real GitHub event was observed.
- **Partially evidenced:** useful evidence exists, but one requested proof form is incomplete.
- **Not verified:** the evidence could not be obtained from the current environment and is not claimed as complete.
- **Unavailable:** the required external view or permission was not accessible.

Verification Commands and Observed Results

Command / check	Observed result	Boundary
server: npm test -- --reporter=verbose	190 tests / 19 files passed; exit 0	2026-10-01 dirty local tree; not committed-main evidence
client: npm test -- --reporter=verbose	36 tests / 14 files passed; exit 0	2026-10-01 dirty local tree; not committed-main evidence
server/client: npm run build	Both passed; exit 0	2026-10-01 dirty local tree; not committed-main evidence
server: npx --no-install prisma validate	Schema valid; exit 0	2026-10-01 dirty local tree; not committed-main evidence
root: npm run test:e2e:prepare	Migration baseline preserved; seed twice stable; exit 0	New disposable run2 database only
root: npx --no-install playwright test e2e/lab-03	16 passed; exit 0; responsive and keyboard journeys	2026-10-01 dirty local tree; not committed-main evidence
Live authorization and attachment download	Self-removal/role denial; attachment bytes match	Step 3 real API output; separate mocked last-admin branch
Final-main verification	NOT RUN after these new local changes	Requires reviewed integration

Appendix - Rendered Specification DD

Rendered source: docs/lab-03/specification.md

Lab 3 Sprint Engineering Specification

Status: Approved final Lab 3 engineering contract. Implementation, test, and evidence status is traced in docs/lab-03/final-report.md and docs/lab-03/submission-evidence.md; remaining limitations are explicitly labelled there.

1. Sprint Goal

Deliver the next TokTickIT increment with secure, real user authentication and backend-enforced role authorization. Requesters continue their Lab 2 ticket work through their authenticated identity, IT Staff receive an operational queue and ticket workflow, and Administrators receive a deliberately small user-management screen.

2. Stakeholder Request

Replace the temporary Development Requester selector with login using an email address and password. Enforce a mandatory password change for accounts using an initial password. Preserve the Lab 2 requester ticket and attachment capabilities, add public communication and requester resolution indication, give IT Staff a queue and ticket operations, and give Administrators only the user-account operations needed for the lab.

3. Scope

Included

- Email/password login, logout, current-user retrieval, and first-login password change.
- Active/inactive accounts and safe authentication errors.
- Exactly one role per user: REQUESTER, IT_STAFF, or ADMINISTRATOR.
- Server-side authentication, role checks, and ownership checks.
- Migration of existing Lab 2 RequesterUser records into the canonical User model without losing Tickets or Attachments.
- Requester regression using authenticated identity rather than a client supplied requesterId.
- Requester Public Comments and a Problem Appears Resolved indication.
- IT Staff Ticket Queue with search, filters, sorting, pagination, and meaningful processing states.
- IT Staff Ticket Detail with ownership, IT Priority, permitted status transitions, Public Comments, Internal Notes, and existing Attachments.
- Minimal Administrator User Management: list, search, optional role filter, create, basic edit, activation/deactivation, and new initial password.
- Zen Green responsive and accessible UI extensions.
- Unit, API/integration, UI component/style, authorization, migration, regression, responsive, and end-to-end tests.

Explicitly excluded

- Email invitations, password-reset email, MFA, social login, and SSO.
- Self-registration or Requester-created accounts.
- Actions Taken, formal SLA calculation, escalation, notifications, and KPI dashboards.
- Multi-tenant organizations, departments, profile photos, and account-history screens.
- Multiple roles per user, user deletion, bulk operations, import/export, and advanced identity-management workflows.
- Production deployment or cloud-infrastructure changes.

4. Engineering Decisions

4.1 Authentication and sessions

- Passwords are hashed with Node's built-in asynchronous scrypt implementation using a per-password random salt. Plaintext passwords never enter the database.
- Authentication uses a server-side opaque session. The browser receives only a random session token in an HttpOnly, SameSite=Lax cookie named toktickit_session. The database stores only a hash of that token.
- Sessions expire after eight hours and are revoked on logout. A revoked or expired session cannot authenticate any protected request.
- Unsafe cross-origin requests require the approved local origin and a double-submit CSRF token. The CSRF token is not a credential and is never used as authorization.
- Client code uses fetch credentials and never stores session tokens in localStorage, sessionStorage, URLs, or rendered page data.
- Login returns safe user identity and role data only. It never returns a password, password hash, session token, or secret.

4.2 Password policy

- A new password must contain 12–128 characters, at least one letter and one number, and must not be whitespace-only.
- The new password must differ from the current initial password.
- Initial passwords are local-development test credentials only and are documented as such. No real personal password or secret may be committed.

4.3 User model migration

- The Prisma model becomes User, with the existing SQL table retained through an explicit mapping to RequesterUser during this increment. This preserves existing IDs, Ticket requesterId values, and Attachment relationships.
- Existing RequesterUser rows become role REQUESTER, receive a seeded or migration-generated initial password hash, and are marked mustChangePassword=true.
- The Development Requester selector, Change Requester action, and X-Requester-Id ownership mechanism are removed from the normal application flow. The backend derives requester ownership from the authenticated User.
- Existing Tickets, Attachments, Categories, and Related Systems remain valid. Existing Tickets retain status NEW and receive IT Priority equal to their Requested Priority.

4.4 Role separation

Requester owns and manages only their own requester-facing Ticket resources. IT Staff operate the shared Ticket workflow. Administrators manage accounts and have only the explicit read/priority/note permissions listed in the authorization matrix below; Administrator does not automatically receive the IT Staff queue.

5. Functional Requirements

Authentication and shell

- FR-01: The system shall authenticate an active user by email and password.
- FR-02: The system shall reject inactive users without revealing unnecessary account state.
- FR-03: The system shall establish a server-side session after valid login.
- FR-04: The system shall provide current authenticated user identity and role.
- FR-05: The system shall require a valid password change before a user with mustChangePassword=true can enter normal application screens.
- FR-06: The system shall revoke the session and clear authenticated access on logout.
- FR-07: The shell shall display the authenticated user's name and role and expose only permitted navigation.

- FR-08: Direct access after logout or without a valid session shall be blocked.

Requester regression

- FR-09: A Requester shall create Tickets using the authenticated identity.
- FR-10: A Requester shall list and open only owned Tickets.
- FR-11: A Requester shall use only permitted owned Attachment operations.
- FR-12: A Requester shall post append-only Public Comments on owned Tickets.
- FR-13: A Requester shall set or clear a Problem Appears Resolved indication

on an owned Ticket.

- FR-14: A Requester shall not set formal Resolved or Closed status.

IT Staff operations

- FR-15: IT Staff shall retrieve a shared Ticket Queue.
- FR-16: The Queue shall support documented search, filters, sorting, and pagination.

• FR-17: IT Staff shall open Ticket Detail and see ownership, requested priority, IT Priority, status, comments, notes, and Attachments.

- FR-18: IT Staff shall claim, assign, or reassign Ticket ownership.
- FR-19: IT Staff shall update IT Priority.
- FR-20: IT Staff shall perform only the documented status transitions.
- FR-21: IT Staff shall post Public Comments and Internal Notes.

Administrator user management

- FR-22: Administrators shall view a user list containing name, email, role, status, and an Edit action.
- FR-23: Administrators shall search users by name or email and may filter by one role.
- FR-24: Administrators shall create a user with one permitted role, active state, and an initial password.
- FR-25: Administrators shall edit name, email, role, and activation state.
- FR-26: Administrators shall set a new initial password that is required at the user's next login.
- FR-27: The system shall reject duplicate email addresses and invalid roles.
- FR-28: The system shall prevent self-deactivation and deactivation of the last active Administrator.

Cross-cutting behavior

- FR-29: Every protected API operation shall enforce authentication and authorization on the server.
- FR-30: Public Comments and Internal Notes shall be separate in the API and visually distinct in the UI.
- FR-31: The UI shall show loading, saving, success, validation, empty, no-results, forbidden, not-found, conflict, and safe failure feedback where meaningful.
- FR-32: Lab 3 screens shall preserve the Lab 2 Zen Green tokens, component conventions, keyboard access, and responsive behavior.
- FR-33: Seed behavior shall be idempotent and shall provide the required active and inactive role fixtures without real secrets.

6. Business Rules

Authentication and identity

- BR-01: Only an active user with valid credentials may authenticate.
- BR-02: A user requiring a password change cannot enter normal application screens until a valid new password is saved.
- BR-03: Login, password change, and logout use safe error messages and never expose password hashes, session tokens, or stack traces.

- BR-04: Session identity comes from the validated server-side session.
- BR-05: Inactive users cannot authenticate or use an existing session.
- BR-06: A client-supplied requesterId, userId, role, or ownerId cannot replace the authenticated identity or bypass authorization.
- BR-07: Email addresses are unique case-insensitively after trimming.
- BR-08: A password change revokes or rotates the pre-change session before granting normal application access.

Requester ownership and regression

- BR-09: A Ticket requester is the authenticated Requester who created it.
- BR-10: Requester list, detail, Attachment, comment, and resolution queries are scoped to the authenticated Requester in the database query.
- BR-11: Cross-requester resources use a safe not-found response and do not disclose whether another user's resource exists.
- BR-12: Requesters may post Public Comments only on owned Tickets.
- BR-13: Requesters may indicate that a problem appears resolved but cannot formally set Resolved or Closed.

Comments, notes, ownership, priority, and status

- BR-14: Public Comments are visible to the owning Requester, IT Staff, and Administrator.
- BR-15: Internal Notes are visible only to IT Staff and Administrator.
- BR-16: Comments and Notes are append-only in Lab 3.
- BR-17: Content is trimmed, rejects empty or whitespace-only input, has a maximum of 4,000 characters, and is rendered as text rather than unsafe HTML.
- BR-18: Each Comment or Note records its backend author and creation time.
- BR-19: A Ticket has zero or one primary owner. An assigned owner must be an active IT Staff or Administrator account.
- BR-20: New Ticket IT Priority copies Requested Priority.
- BR-21: Only IT Staff or Administrator may change IT Priority; the Requester cannot submit or override it.
- BR-22: Status values are New, Open, In Progress, Waiting for Requester, Resolved, Closed, Reopened, and Cancelled.
- BR-23: Only status transitions in the approved transition matrix are valid.
- BR-24: Transitions to Resolved, Closed, or Cancelled require explicit UI confirmation and an API confirm value.
- BR-25: Actions Taken are not required in Lab 3 and do not block resolution.

Administrator safety

- BR-26: Each User has exactly one permitted role.
- BR-27: Administrators cannot delete users.
- BR-28: Administrators cannot deactivate their own account.
- BR-29: The system must retain at least one active Administrator.
- BR-30: Setting an initial password stores only its hash and sets mustChangePassword=true.
- BR-31: Deactivation prevents future login and protected access but preserves the user's historical Ticket, Comment, and Note authorship.

Data, validation, and failures

- BR-32: Seed operations are safe to run repeatedly.
- BR-33: Required foreign-key relationships and ownership indexes remain valid after migration.
- BR-34: Invalid input returns a field-aware safe response where appropriate.
- BR-35: Unauthenticated, forbidden, invalid, missing, conflict, and unexpected-error cases remain distinguishable by status and error code.
- BR-36: Server failures never return filesystem paths, stack traces, hashes, or

unrelated user data.

7. Authorization Matrix

Operation	Requester	IT Staff	Administrator
Login, logout, current user	Own session	Own session	Own session
Mandatory password change	Own account	Own account	Own account
Requester Ticket create/list/detail	Own identity and Tickets	No	No
Requester Attachments	Own Tickets only	No through requester API	No through requester API
Requester Public Comments	Own Tickets	No through requester API	No through requester API
Problem Appears Resolved	Own Tickets	Read indication	Read indication
IT Staff Queue	No	Yes	No
Staff Ticket Detail	No	Read and operate	Read-only direct detail
Claim/reassign ownership	No	Yes; active IT Staff/Admin targets	No
Update IT Priority	No	Yes	Yes on permitted detail API
Status transitions	No	Yes	No
Public Comments	Own Tickets	Yes	Yes on permitted detail API
Internal Notes	No	Yes	Yes
User Management	No	No	Yes

The Administrator detail and IT Priority permissions are explicit exceptions to the minimum Administrator role so that the owner, priority, and note visibility rules remain testable without granting the Administrator the shared IT Staff Queue.

8. Ticket Status Transition Matrix

From	To	Permitted role	Confirmation
New	Open	IT Staff	No
New	Cancelled	IT Staff	Yes
Open	In Progress	IT Staff	No
Open	Waiting for Requester	IT Staff	No
Open	Resolved	IT Staff	Yes
Open	Cancelled	IT Staff	Yes
In Progress	Waiting for Requester	IT Staff	No
In Progress	Resolved	IT Staff	Yes
In Progress	Cancelled	IT Staff	Yes
Waiting for Requester	In Progress	IT Staff	No
Waiting for Requester	Resolved	IT Staff	Yes
Waiting for Requester	Cancelled	IT Staff	Yes
Resolved	Closed	IT Staff	Yes
Resolved	Reopened	IT Staff	No
Closed	Reopened	IT Staff	No
Reopened	In Progress	IT Staff	No

From	To	Permitted role	Confirmation
Reopened	Waiting for Requester	IT Staff	No
Reopened	Resolved	IT Staff	Yes
Reopened	Cancelled	IT Staff	Yes
Cancelled	Reopened	IT Staff	No

No Requester operation changes the formal Ticket status.

9. Data Model and Migration

User

The canonical Prisma model is User. During this increment it maps to the existing RequesterUser table so existing IDs and foreign keys remain stable. It adds name, email, passwordHash, role, isActive, mustChangePassword, createdAt, and updatedAt.

Session

Session stores id, tokenHash, userId, createdAt, expiresAt, and revokedAt. Only tokenHash is persisted; the raw cookie value is never stored.

Ticket additions

Ticket retains requesterId and all Lab 2 fields. Lab 3 adds nullable ownerId, itPriority, expanded TicketStatus values, and requesterResolutionIndicatedAt. Existing rows keep NEW status and copy requestedPriority into itPriority.

Comments and notes

PublicComment and InternalNote each store id, ticketId, authorId, content, and createdAt. Separate models prevent an Internal Note from being accidentally returned through the Public Comment query.

Migration requirements

- Preserve all existing Category, RelatedSystem, Ticket, and Attachment rows.
- Convert all existing RequesterUser records to role REQUESTER.
- Assign safe local initial-password hashes and require a first-login change.
- Add nullable or defaulted fields before enforcing any new required behavior.
- Add foreign keys and indexes for requester ownership, ticket owner, session

lookup, comments, notes, and queue queries.

- Test the migration and a fresh idempotent seed on the documented disposable

PostgreSQL setup.

Seed requirements

The idempotent seed shall provide at least:

- Four active Requesters and one inactive Requester.
- Three active IT Staff and one inactive IT Staff.
- One active Administrator.
- Tickets distributed across Requesters, statuses, requested/IT priorities,

and assigned/unassigned ownership.

- Non-sensitive Public Comments and Internal Notes.

10. Acceptance Criteria

• AC-01: An active user with valid credentials receives authenticated access, safe identity data, and the permitted role.

- AC-02: Invalid credentials and inactive accounts receive safe failures.
- AC-03: An initial-password user cannot reach normal application screens before saving a valid replacement password.

• AC-04: Logout revokes access and direct protected access is blocked afterward.
 • AC-05: The shell and navigation show only the authenticated role's permitted destinations.

- AC-06: A Requester creates and retrieves Tickets through authenticated

ownership, with no client requesterId override.

- AC-07: Existing Requester Ticket and Attachment functions continue to work without the Development Requester selector.
- AC-08: Cross-requester Ticket and Attachment access is rejected safely.
- AC-09: Public Comments are visible to all required roles and Internal Notes are hidden from Requesters.
- AC-10: A Requester can set Problem Appears Resolved without formally resolving or closing the Ticket.
- AC-11: IT Staff Queue supports the documented search, filters, sorting, and pagination behavior.
- AC-12: Queue loading, empty, no-results, forbidden, and failure states are visible and safe.
- AC-13: IT Staff Ticket Detail shows operational fields, comments, notes, and Attachments without exposing private notes to Requesters.
- AC-14: IT Staff can claim/reassign ownership only to valid active targets.
- AC-15: IT Priority starts from Requested Priority and can be updated only by permitted roles.
- AC-16: The backend accepts only permitted status transitions and requires confirmations where specified.
- AC-17: Comments and notes reject empty content, record backend author/time, and cannot be edited or deleted.
- AC-18: Administrators can list users with the required fields.
- AC-19: Administrator name/email search and optional role filtering work.
- AC-20: Administrators can create a user with one permitted role and initial password.
- AC-21: Duplicate email, invalid role, and invalid input are rejected safely.
- AC-22: Administrators can edit basic account information and activation.
- AC-23: A reset initial password forces a change at the next login.
- AC-24: Self-deactivation and removal of the last active Administrator are prevented.
- AC-25: Non-Administrators cannot use User Management endpoints or screens.
- AC-26: Migration preserves Lab 2 data and seed behavior is idempotent.
- AC-27: All major Lab 3 screens satisfy Zen Green, responsive, keyboard, focus, validation, and overflow requirements.

Every AC must map to at least one planned test in docs/lab-03/tests.md and must have final evidence before it is marked Passed.

11. Product Definition of Done

- The approved specification, API contract, UI specification, and test plan agree with the implementation.
- Authentication, password change, logout, session expiry/revocation, and role navigation are implemented and tested.
- Every protected API operation enforces backend authorization and ownership.
- Lab 2 data survives migration and the authenticated Requester regression passes.
- Queue, Ticket Detail, comments, notes, ownership, priority, and status behavior satisfy the approved matrix.
- Administrator User Management satisfies all required safety rules.
- Required unit/API/UI/style/security/migration/regression/E2E tests pass.
- Responsive screenshots and visual checklists cover all major Lab 3 screens.
- Reviewer and AI-use evidence is complete and traceable.
- Feature PRs are reviewed and merged into lab3-staging, then the release PR is reviewed and merged into main.

- The final PDF contains exactly one concise submission with Answer Part 1 through Answer Part 9, readable screenshots, working links, and honest evidence labels.

12. Assumptions and Decisions to Reconfirm Before Implementation

- The User Prisma model maps the existing RequesterUser SQL table for this increment rather than copying or deleting historical records.
- The session cookie is the only browser credential; no token is exposed to React code.
- Administrator access to direct read-only Ticket Detail, Internal Notes, and IT Priority is intentionally explicit; Administrator does not receive the IT Staff Queue.
- Pagination defaults to page 1 with page sizes 10, 20, or 50. Queue search covers ticket number, summary, description, requester name, and requester email. Queue sorting covers updatedAt, createdAt, ticketNumber, status, requestedPriority, itPriority, and owner.
- Tracking is resolved: Issues #36-#40 map to feature PRs #41-#45, reviewed release #46, correction #47 and release #48; reviewer Punge089. Local follow-up repairs require the same reviewed staging/release workflow. No new GitHub items were created in this audit.

Appendix - Rendered API Specification

Rendered source: docs/lab-03/api-spec.md

Lab 3 REST API Specification

Status: Approved final Lab 3 API contract. Endpoint implementation and automated-test results are traced in docs/lab-03/tests.md and docs/lab-03/final-report.md; evidence limitations are explicitly labelled.

1. Conventions

- Base path: /api.
- JSON is used unless an endpoint explicitly requires multipart/form-data.
- IDs are positive integers.
- Dates are ISO 8601 UTC strings.
- Authentication uses the HttpOnly toktickit_session cookie.
- Unsafe requests require a matching X-CSRF-Token header and CSRF cookie.
- The client never sends X-Requester-Id for ownership.
- The authenticated user is resolved from the server-side session.
- Error responses never include stack traces, filesystem paths, password

hashes, session tokens, or unrelated protected-resource data.

2. Error Envelope

All JSON errors use this shape:

```
{ "error": { "code": "VALIDATION_ERROR", "message": "Request validation failed.", "fields": { "email": "Enter a valid email address." } } }
```

The fields property is included only when field-level feedback is useful.

Status	Error codes	Meaning
400	VALIDATION_ERROR, INVALID_QUERY, INVALID_TRANSITION, INVALID_CSRF	Input or transition is invalid
401	UNAUTHENTICATED, INVALID_CREDENTIALS	No valid session or credentials
403	FORBIDDEN, PASSWORD_CHANGE_REQUIRED, INACTIVE_ACCOUNT	Authenticated but not permitted
404	RESOURCE_NOT_FOUND	Missing or unauthorized protected resource
409	CONFLICT, DUPLICATE_EMAIL, LAST_ADMINISTRATOR	State or uniqueness conflict
413	PAYLOAD_TOO_LARGE	File or configured payload limit exceeded
415	UNSUPPORTED_MEDIA_TYPE	Unsupported upload type
500	INTERNAL_ERROR	Safe unexpected server failure

Cross-owner Ticket, Attachment, Comment, and Note lookups return the same 404 RESOURCE_NOT_FOUND shape as an actually missing resource.

3. User and Session Shapes

Safe User

```
{ "id": 101, "name": "Ariya Somchai", "email": "ariya.somchai@example.com", "role": "REQUESTER", "isActive": true, "mustChangePassword": false, "createdAt": "2026-09-18T10:00:00.000Z", "updatedAt": "2026-09-18T10:00:00.000Z" }
```

The API never returns passwordHash, raw session tokens, CSRF secrets, or initial passwords in a User shape.

Session behavior

- Login sets toktickit_session with HttpOnly, SameSite=Lax, a bounded expiration, and Secure when the environment uses HTTPS.

- Logout clears the cookie and revokes the database session.
- Expired or revoked sessions return 401.
- A session in password-change-only mode can call current-user, change-password, CSRF, and logout endpoints only.

4. CSRF

GET /api/auth/csrf

Returns a fresh CSRF token and sets the matching non-HttpOnly CSRF cookie. The token is not an authentication credential.

Success:

- 200 OK
- Body: { "csrfToken": "..." }

The client sends the returned value in X-CSRF-Token for login, logout, password changes, Ticket mutations, comments, notes, staff operations, and Administrator mutations.

5. Authentication

POST /api/auth/login

Request:

```
{ "email": "ariya.somchai@example.com", "password": "local-development-password" }
```

Success:

- 200 OK
- Sets the session cookie.
- Body: { "user": SafeUser, "mustChangePassword": true|false }

Invalid credentials, nonexistent accounts, and inactive accounts use the same safe 401 INVALID_CREDENTIALS response. Validation of malformed input uses 400.

GET /api/auth/me

Requires a valid session. Returns:

- 200 OK
- Body: { "user": SafeUser, "mustChangePassword": true|false }

A password-change-only session returns the user and the flag but normal application endpoints remain blocked.

POST /api/auth/change-password

Requires a valid session, including a password-change-only session.

Request:

```
{ "currentPassword": "local-development-password", "newPassword": "New-local-password-2026" }
```

The server validates the password policy, hashes the new password, clears mustChangePassword, rotates or revokes the old session, and establishes normal authenticated access.

Success:

- 200 OK
- Body: { "user": SafeUser, "mustChangePassword": false }

Failures:

- 400 VALIDATION_ERROR for password boundaries or missing fields.
- 401 UNAUTHENTICATED for no session.
- 403 INVALID_CREDENTIALS when currentPassword is wrong.

POST /api/auth/logout

Requires the current session when one exists. The operation is idempotent.

Success:

- 204 NO CONTENT
- Revokes the session and clears the cookie.

6. Authenticated Requester API

The existing Lab 2 Ticket and Attachment routes continue under authenticated ownership. Their request and response shapes remain compatible except that the X-Requester-Id header is removed and Lab 3 fields are added.

GET /api/tickets

Permitted: authenticated Requester, for owned Tickets only.

Query parameters:

- search: case-insensitive search in summary and description.
- categoryId: positive integer.
- requestedPriority: LOW, MEDIUM, or HIGH.
- status: one permitted status value.
- sortBy: updatedAt, createdAt, ticketNumber, requestedPriority, or status.
- sortDirection: asc or desc.
- page: one-based page number, default 1.
- pageSize: 10, 20, or 50, default 10.

Success:

- 200 OK
- Body: { "items": [Ticket], "pagination": Pagination }

The requesterId is always taken from the authenticated User. Invalid query parameters return 400 INVALID_QUERY.

POST /api/tickets

Permitted: authenticated Requester.

Content type: multipart/form-data. Fields remain the Lab 2 fields: categoryId, relatedSystemId, requestedPriority, summary, description, and optional attachments.

The server derives requesterId, creates status NEW, and initializes itPriority from requestedPriority.

Success:

- 201 CREATED
- Body: { "ticket": Ticket }

Ownership or authentication failures are returned before persistence. Existing Lab 2 file limits remain: permitted JPEG, PNG, WEBP, and PDF types; 5 MB per file; five active files per Ticket; all-or-nothing validation.

GET /api/tickets/:ticketId

Permitted: authenticated Requester who owns the Ticket.

- 200 OK with { "ticket": Ticket }
- 404 RESOURCE_NOT_FOUND for missing or cross-owner Ticket.

GET /api/tickets/:ticketId/attachments

Permitted: authenticated Requester who owns the Ticket.

- 200 OK with { "attachments": [Attachment] }
- 404 RESOURCE_NOT_FOUND for missing or cross-owner Ticket.

POST /api/tickets/:ticketId/attachments

Permitted: authenticated Requester who owns the Ticket.

Content type: multipart/form-data with one or more attachments.

- 201 CREATED with { "attachments": [Attachment] }
- 400 VALIDATION_ERROR for malformed input.
- 413 PAYLOAD_TOO_LARGE for size/count limits.
- 415 UNSUPPORTED_MEDIA_TYPE for unsupported types.
- 404 RESOURCE_NOT_FOUND for missing or cross-owner Ticket.

GET /api/attachments/:attachmentId/download

Permitted: authenticated Requester who owns the parent Ticket.

- 200 OK with the stored file stream and safe Content-Disposition.
- 404 RESOURCE_NOT_FOUND for missing, removed, or unauthorized Attachment.

The storage directory is never served as a public static directory.

DELETE /api/attachments/:attachmentId

Permitted: authenticated Requester who owns the parent Ticket.

Request:

```
{ "reason": "The screenshot is no longer needed." }
```

The reason must contain 5–250 trimmed characters.

- 200 OK with { "attachment": Attachment }
- 400 VALIDATION_ERROR for an invalid reason.
- 404 RESOURCE_NOT_FOUND for missing, removed, or unauthorized Attachment.

The database metadata remains and the file is no longer downloadable.

7. Requester Comments and Resolution Indication

GET /api/tickets/:ticketId/comments

Permitted: an authenticated owner, IT Staff, or Administrator with the corresponding ticket-detail permission.

- Requesters receive Public Comments for owned Tickets.
- IT Staff and Administrators receive Public Comments for permitted details.
- 200 OK with { "comments": [PublicComment] }.
- 404 RESOURCE_NOT_FOUND for inaccessible Tickets.

POST /api/tickets/:ticketId/comments

Request:

```
{ "content": "The replacement laptop is available." }
```

This endpoint always creates a Public Comment. It rejects empty or whitespace-only content and content longer than 4,000 characters.

- 201 CREATED with { "comment": PublicComment }
- 400 VALIDATION_ERROR for invalid content.
- 403 FORBIDDEN for a role without access.
- 404 RESOURCE_NOT_FOUND for an inaccessible Ticket.

POST /api/tickets/:ticketId/resolution-indication

Permitted: authenticated Requester who owns the Ticket.

Request:

```
{ "appearsResolved": true }
```

Success:

- 200 OK with { "appearsResolved": true, "indicatedAt": "..." }

This operation never changes the formal status.

8. IT Staff Queue and Ticket Operations

GET /api/staff/tickets

Permitted: IT Staff only.

Query parameters:

- search: ticket number, summary, description, requester name, or requester email; case-insensitive.

- status: one TicketStatus value.
- requestedPriority: LOW, MEDIUM, or HIGH.
- itPriority: LOW, MEDIUM, or HIGH.
- ownerId: positive User ID.
- ownership: assigned or unassigned.
- sortBy: updatedAt, createdAt, ticketNumber, status, requestedPriority, itPriority, or owner.

- sortDirection: asc or desc.
- page: one-based page number, default 1.
- pageSize: 10, 20, or 50, default 10.

The default ordering is updatedAt descending followed by id descending.

Success:

- 200 OK
- Body: { "items": [QueueTicket], "pagination": Pagination }

Invalid parameters return 400 INVALID_QUERY. A valid query with no matches returns an empty items array with pagination metadata.

GET /api/staff/tickets/:ticketId

Permitted: IT Staff and Administrator.

- 200 OK with { "ticket": StaffTicketDetail }
- 403 FORBIDDEN only when the role lacks this explicit detail permission.
- 404 RESOURCE_NOT_FOUND for a missing Ticket.

Administrators receive read-only detail access and do not receive the Queue.

PATCH /api/staff/tickets/:ticketId/owner

Permitted: IT Staff.

Request:

```
{ "ownerId": 203 }
```

ownerId may be null to unassign or an active IT Staff/Administrator User ID. The server validates role and activation state.

- 200 OK with { "ticket": StaffTicketDetail }
- 400 VALIDATION_ERROR for malformed IDs.
- 404 RESOURCE_NOT_FOUND for missing Ticket or target User.
- 409 CONFLICT for an inactive or disallowed target.

PATCH /api/staff/tickets/:ticketId/priority

Permitted: IT Staff or Administrator with detail permission.

Request:

```
{ "itPriority": "HIGH" }
```

- 200 OK with { "ticket": StaffTicketDetail }
- 400 VALIDATION_ERROR for an invalid priority.
- 404 RESOURCE_NOT_FOUND for a missing Ticket.

PATCH /api/staff/tickets/:ticketId/status

Permitted: IT Staff.

Request:

```
{ "status": "RESOLVED", "confirm": true }
```

The server evaluates the current status, requested target, role, and confirmation requirement atomically.

- 200 OK with { "ticket": StaffTicketDetail }
- 400 INVALID_TRANSITION or VALIDATION_ERROR.
- 404 RESOURCE_NOT_FOUND for a missing Ticket.
- 409 CONFLICT when the Ticket changed before the update.

GET /api/staff/tickets/:ticketId/notes

Permitted: IT Staff and Administrator.

- 200 OK with { "notes": [InternalNote] }
- 404 RESOURCE_NOT_FOUND for a missing Ticket.

POST /api/staff/tickets/:ticketId/notes

Permitted: IT Staff and Administrator.

Request:

```
{ "content": "Checked device policy and requested a replacement." }
```

- 201 CREATED with { "note": InternalNote }
- 400 VALIDATION_ERROR for invalid content.
- 404 RESOURCE_NOT_FOUND for a missing Ticket.

Internal Notes are never included in Requester responses.

9. Administrator User Management

GET /api/admin/users

Permitted: Administrator.

Query parameters:

- search: case-insensitive name or email search.
- role: optional single REQUESTER, IT_STAFF, or ADMINISTRATOR filter.

Success:

- 200 OK
- Body: { "users": [SafeUser] }

No mandatory pagination or multi-column sorting is required in Lab 3.

POST /api/admin/users

Permitted: Administrator.

Request:

```
{ "name": "New Staff", "email": "new.staff@example.com", "role": "IT_STAFF", "isActive": true, "initialPassword": "Local-password-2026" }
```

The new user receives mustChangePassword=true.

- 201 CREATED with { "user": SafeUser }
- 400 VALIDATION_ERROR for invalid fields.
- 409 DUPLICATE_EMAIL for a case-insensitive duplicate.

PATCH /api/admin/users/:userId

Permitted: Administrator.

Request may contain name, email, role, and isActive. It cannot contain a password or silently assign multiple roles.

- 200 OK with { "user": SafeUser }
- 400 VALIDATION_ERROR for invalid fields.
- 403 FORBIDDEN for self-deactivation.
- 404 RESOURCE_NOT_FOUND for missing User.
- 409 DUPLICATE_EMAIL or LAST_ADMINISTRATOR for conflicts.

POST /api/admin/users/:userId/initial-password

Permitted: Administrator.

Request:

```
{ "initialPassword": "New-local-password-2026" }
```

The hash is replaced and mustChangePassword=true.

- 200 OK with { "user": SafeUser }
- 400 VALIDATION_ERROR for password policy failure.
- 404 RESOURCE_NOT_FOUND for missing User.

10. Response Shapes

Ticket

Ticket responses include the Lab 2 fields plus:

```
{ "owner": { "id": 203, "name": "Somchai Staff", "role": "IT_STAFF" }, "itPriority": "HIGH", "status": "IN_PROGRESS", "appearsResolved": false, "requester": { "id": 101, "name": "Ariya Somchai", "email": "..."}, "category": { "id": 1, "name": "Hardware" }, "relatedSystem": { "id": 1, "name": "Corporate Laptop" }, "attachments": [], "publicCommentCount": 0, "internalNoteCount": 0 }
```

Requester responses omit Internal Notes and do not expose unowned data.

PublicComment

```
{ "id": 9001, "ticketId": 5001, "content": "The replacement laptop is available.", "author": { "id": 203, "name": "Somchai Staff", "role": "IT_STAFF" }, "createdAt": "2026-09-18T10:00:00.000Z" }
```

InternalNote

The InternalNote shape matches PublicComment but is returned only by the staff-note endpoints to IT Staff and Administrators.

Pagination

```
{ "page": 1, "pageSize": 10, "totalItems": 0, "totalPages": 0, "hasPrevious": false, "hasNext": false }
```

11. Authorization Summary

Every protected endpoint uses session middleware followed by an explicit role and ownership check. Frontend hiding is presentation feedback only. Direct requests with altered IDs, roles, query parameters, or bodies must receive the same backend decision as UI requests.

12. Traceability

- Authentication and first-login behavior: FR-01–FR-08, AC-01–AC-05.
- Requester regression and ownership: FR-09–FR-14, AC-06–AC-10.
- Queue and Ticket operations: FR-15–FR-21, AC-11–AC-17.
- User Management: FR-22–FR-28, AC-18–AC-25.
- Migration, failures, and UI evidence: FR-29–FR-33, AC-26–AC-27.

Appendix - Rendered UI Specification

Rendered source: docs/lab-03/ui-spec.md

Lab 3 Zen Green UI Specification

Status: Approved final Lab 3 UI contract. Captured states and verification limits are indexed in docs/lab-03/screenshot-index.md, docs/lab-03/final-report.md, and docs/lab-03/submission-evidence.md.

1. Design Continuity

Lab 3 extends the Lab 2 Zen Green system. Existing tokens, form conventions, cards, badges, buttons, validation placement, responsive rules, and keyboard expectations remain in force. New screens must look like one application.

Design tokens

Token	Value	Use
Primary green	#006B3C	Header, primary actions, strong emphasis
Secondary green	#0B7A46	Links, focus accents, active navigation
Pale green	#EAF6EF	Selected areas, success panels
Page background	#F5F7F6	Application canvas
Surface	#FFFFFF	Cards, panels, forms
Ink	Dark charcoal-green	Primary readable text
Read-only surface	Soft gray-green	System-generated fields
Error	Dark red	Errors and invalid borders
Warning	Amber	Warnings and attention badges
Success	Green	Completion feedback

Use the existing 4px/8px spacing rhythm, restrained card shadows, readable system sans-serif typography, and visible focus outlines.

2. Application Shell

Unauthenticated shell

The Login screen is the only normal unauthenticated destination. It contains:

- TokTickIT identity and IT Service Desk label.
- Email and Password fields.
- Sign in action.
- Field validation, busy state, safe failure, and inactive-account response.
- Keyboard-accessible labels and focus.

Authenticated shell

The shell contains:

- TokTickIT identity.
- Current user's name and role badge.
- Logout action.
- Role-specific navigation.
- Clear active navigation state.
- Responsive mobile navigation.

Navigation rules:

Role	Visible destinations
Requester	My Tickets, Create Ticket
IT Staff	Ticket Queue

Role	Visible destinations
Administrator	User Management

The shell may show a direct Ticket Detail route when reached from an allowed link, but it must not present an unauthorized destination as usable navigation.

3. Mandatory Change Password

The Change Password screen is shown immediately after a successful login when `mustChangePassword=true`. It contains:

- A clear explanation that an initial password must be replaced.
- Current password, New password, and Confirm new password fields.
- Visible password rules.
- Field-level validation and mismatch feedback.
- Busy state that prevents duplicate submission.
- Safe failure feedback that preserves entered values.
- Success feedback followed by normal shell access.

While this screen is required, normal application screens and APIs remain blocked. Logout remains available.

4. Requester Regression

The Lab 2 Requester views continue using the authenticated Requester. The Development Requester selector and Change Requester action are removed.

My Tickets

Keep the Lab 2 list behavior:

- Search, category, requested-priority, and status filters.
- Sorting, page size, pagination, and clear filters.
- Empty and filtered no-results states.
- Owned Ticket rows/cards and Open action.
- Authenticated Requester identity in the shell.

Create Ticket

Keep the Lab 2 fields, validation, attachments, busy state, and generated Ticket success behavior. The Requester field is read-only and comes from the authenticated user. The client never sends a `requesterId` override.

Requester Ticket Detail

Preserve read-only Ticket information and Attachment lifecycle. Add:

- Public Comment list and composer.
- Problem Appears Resolved action with clear current state.
- A visible explanation that only IT Staff formally resolve or close Tickets.
- No IT Priority, owner assignment, Internal Note, or status controls.

Requester comments and resolution indication must be visually separate from staff-only controls and must not reveal Internal Notes.

5. IT Staff Ticket Queue

The Queue is a work-finding screen, not an unreadable mega-grid.

Desktop information hierarchy

The default desktop table or grid includes:

- Ticket Number.
- Created or last-updated date.
- Summary.
- Category.
- Requester.
- Requested Priority badge.
- IT Priority badge.
- Current Status badge.
- Ticket Owner or Unassigned.
- Open Detail action.

The final visible set must be justified in the final evidence. Secondary data may be placed in the detail screen.

Controls

- Search input with an accessible label and clear action.
- Status filter.
- Requested Priority filter.
- IT Priority filter.
- Owner/assigned filter.
- Sort field and direction.
- Page-size selector and pagination.
- Apply and Clear actions where the chosen interaction model needs them.

Queue states

- Loading: visible progress message and non-jarring disabled controls.
- Empty: no seeded or existing work, with a next-step explanation.
- No results: query has no matches, with Clear filters.
- Forbidden: safe role message with no ticket data.
- Failure: safe API error and Retry action.
- Success: readable rows/cards with ownership, status, and priority badges.

Responsive representation

At tablet width, retain the important columns while allowing controlled wrapping. At mobile width, convert each row to a card or stacked record with Ticket Number, Summary, status, priorities, owner, and Open Detail visible without horizontal scrolling.

6. IT Staff Ticket Detail

Group the screen into:

1. Ticket identity and dates. 2. Requester and classification. 3. Ownership and operational controls. 4. Description and existing Attachments. 5. Public Comments. 6. Internal Notes.

Operational controls:

- Claim as current IT Staff user.
- Assign or reassign to an active permitted owner.
- IT Priority selector.
- Status selector containing only permitted target transitions.
- Confirmation for Resolved, Closed, and Cancelled.
- Save feedback that identifies the changed field.

Public Comments and Internal Notes must use distinct headings, containers, labels, colors, and submit actions. The Internal Note composer must never share the Public Comment submit control or label.

The detail view shows the Requester resolution indication but does not let the Requester formally change staff status. Administrators may receive read-only detail, Internal Note, and IT Priority presentation according to the authorization contract, without receiving the Queue navigation.

7. Administrator User Management

Use one intentionally simple screen.

User list

Display:

- Name.
- Email.
- Role badge.
- Active/Inactive status badge.
- Edit action.

Controls:

- Search by name or email.
- Optional one-role filter.
- Create User action.
- Retry and safe API failure feedback.

Create and edit

Create fields:

- Name.
- Email.
- One role.
- Active state.
- Initial password.

Edit fields:

- Name.
- Email.
- One role.
- Active state.

Initial password reset is a clearly separate action and explains the required next-login password change. The UI gives explicit feedback for duplicate email, invalid role, self-deactivation, and last-active-Administrator protection.

The screen does not include deletion, bulk actions, import/export, multiple roles, departments, account-history screens, or advanced recovery controls.

8. Shared Feedback and Modes

Every screen identifies its relevant modes:

Mode	Required behavior
Initial	Clear editable or read-only structure
Loading	Visible progress and no duplicate actions
Saving	Busy action, disabled duplicate submit
Validation	Field-level message near the invalid control
Success	Specific result and next action
Empty	Explain why no records exist
No results	Explain query mismatch and offer clearing
Forbidden	Role-safe message without protected data
Not found	Safe resource message
Conflict	Explain duplicate/state conflict and preserve input
Failure	Safe API message, retry where useful

Do not use color alone to communicate status, priority, role, error, or success.

9. Responsive Rules

- Desktop: 992px and above; centered content and multi-column layouts where they improve scanability.
- Tablet: 768–991px; controlled two-column forms and wrapped queue fields.
- Mobile: below 768px; stacked forms, touch-sized controls, cards or stacked queue records, and no horizontal page scrolling.
- Long summaries, emails, filenames, and comments must wrap or truncate with accessible full text.
- No labels, controls, validation messages, badges, or buttons may clip, overlap, or disappear at the required widths.

10. Accessibility

- Every control has an associated label or accessible name.
- Required state and validation are announced to assistive technology.
- Important error and success feedback uses alert/status semantics.

- Focus remains visible and moves predictably after navigation or submission.
- Busy controls are disabled without trapping focus.
- Keyboard users can complete login, password change, Requester regression,

Queue filters, detail operations, comments, notes, and User Management.

- Contrast is readable for text, borders, focus rings, status badges, and disabled controls.
- Public Comment and Internal Note controls have unambiguous names.

11. Reusable Components

Reuse or extend the Lab 2 conventions for:

- Application shell and role navigation.
- Page header and role badge.
- Form field and validation message.
- Read-only field.
- Loading, error, empty, and no-results panels.
- Status, priority, role, and account-state badges.
- Search/filter toolbar and pagination.
- Attachment list.
- Comment and Internal Note panels.
- Confirmation dialog.
- User list and edit form.

12. Screenshot and Visual Evidence Matrix

Required screenshot root:

artifacts/lab-03/screenshots/

Required groups from the handout:

- authentication/
- staff-queue/
- staff-ticket-detail/
- user-management/

Requester regression screenshots may be placed under authentication/requester-regression/ when they show the authenticated shell, My Tickets, Create Ticket, or Requester Ticket Detail.

Capture at:

Viewport	Size	Minimum checks
Desktop	1280 × 900	hierarchy, alignment, readable data
Tablet	900 × 900	wrapping, no clipping, usable controls
Mobile	390 × 844	stacked/card layout, no horizontal overflow

Each evidence record must include screen, viewport, date, branch/SHA, scenario, and visible state. A screenshot is not Passed until it is visibly captured and indexed.

13. Visual Checklist

For each major screen and viewport, verify:

- Zen Green tokens and component conventions match Lab 2.
- Authenticated identity and role are clear.
- Role navigation does not expose unauthorized destinations.
- Status, requested priority, IT Priority, role, and account-state badges are consistent.

are understandable.

- Editable and read-only fields are distinct.
- Required markers and validation messages are correctly placed.
- Loading, saving, empty, no-results, forbidden, success, and failure states

- Public Comments and Internal Notes cannot be confused.
- Focus, keyboard order, contrast, wrapping, clipping, overlap, and horizontal overflow are acceptable.

Appendix - Rendered Test DD and Traceability

Rendered source: docs/lab-03/tests.md

Lab 3 Test Plan, Traceability, and Results

Test DD was created before implementation. The original numbered matrix is retained below with current local execution status. Supplemental traceability identifies newly added assertions and proof boundaries. Current results: 190 server tests, 36 client tests and 16 E2E scenarios passed on 2026-10-01 in the dirty local tree; not final-main.

1. Test Principles

- Tests are written from the approved specification and acceptance criteria, not reconstructed after implementation.
- Server tests verify authentication, authorization, ownership, validation, safe errors, migration behavior, and data visibility directly at the API.
- Client tests verify role navigation, form behavior, feedback, accessibility semantics, and Zen Green presentation.
- E2E tests verify integrated browser behavior against the real application and disposable PostgreSQL data.
- Responsive evidence uses the required desktop, tablet, and mobile sizes.
- A planned test is not evidence. Results are labeled Planned, Passed, Failed, Deferred, or Unavailable only after observation.

2. Planned Automated Test Files

Server/API

Required server files:

- server/tests/lab-03/auth.api.test.ts
- server/tests/lab-03/authorization.api.test.ts
- server/tests/lab-03/staff-queue.api.test.ts
- server/tests/lab-03/staff-ticket-detail.api.test.ts
- server/tests/lab-03/comments-notes.api.test.ts
- server/tests/lab-03/users-admin.api.test.ts

Additional focused migration coverage:

- server/tests/lab-03/migration-regression.test.ts

Client/UI

Required client files:

- client/tests/lab-03/Login.test.tsx
- client/tests/lab-03/ChangePassword.test.tsx
- client/tests/lab-03/StaffTicketQueue.test.tsx
- client/tests/lab-03/StaffTicketDetail.test.tsx
- client/tests/lab-03/UserManagement.test.tsx

Additional shared visual assertions:

- client/tests/lab-03/ui-style.test.tsx

End to end

- e2e/lab-03/authentication.spec.ts
- e2e/lab-03/staff-ticket-flow.spec.ts
- e2e/lab-03/user-administration.spec.ts

3. Test Matrix

ID	Type	Requirement / AC	What it tests	Automated file	Current local result
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ID	Type	Requirement / AC	What it tests	Automated file	Current local result
UNIT-01	Unit	BR-03, BR-07, AC-01–AC-03	Email normalization and password boundaries/composition	server/tests/lab-03/auth.unit.test.ts	Passed locally; supplemental scope applies. Not main.
UNIT-02	Unit	BR-03	Salted script digest, no plaintext, correct/wrong/malformed verification	server/tests/lab-03/auth.unit.test.ts	Passed locally; supplemental scope applies. Not main.
API-01	API	FR-01, AC-01	Valid-CSRF login returns safe user data	server/tests/lab-03/auth.api.test.ts	Passed locally; supplemental scope applies. Not main.
API-02	API	FR-02, AC-02	Missing and inactive accounts share safe 401 behavior	server/tests/lab-03/auth.api.test.ts	Passed locally; supplemental scope applies. Not main.
API-03	API	FR-05, AC-03	Mandatory password gate and old-session revocation after password change	server/tests/lab-03/auth.api.test.ts	Passed locally; supplemental scope applies. Not main.
API-04	Security	FR-09, AC-06, AC-08	Ticket creation uses session identity and ignores requesterId input	server/tests/lab-03/authorization.api.test.ts	Passed locally; supplemental scope applies. Not main.
API-05	Security	FR-10, AC-07–AC-08	Cross-requester ticket access returns safe not-found response	server/tests/lab-03/authorization.api.test.ts	Passed locally; supplemental scope applies. Not main.
API-06	API	FR-12, FR-14, AC-09–AC-10	Public comments and Requester resolution indication	server/tests/lab-03/authorization.api.test.ts	Passed locally; supplemental scope applies. Not main.
API-07	API	FR-15–FR-16, AC-11	Staff-only filtered paginated Queue response	server/tests/lab-03/staff-queue.api.test.ts	Passed locally; supplemental scope applies. Not main.
API-08	Validation	FR-16, AC-11	Invalid queue query rejected before ticket query	server/tests/lab-03/staff-queue.api.test.ts	Passed locally; supplemental scope applies. Not main.
API-09	Security	FR-15, AC-12, AC-25	Requester and Administrator denied Queue access	server/tests/lab-03/staff-queue.api.test.ts	Passed locally; supplemental scope applies. Not main.
API-10	API	FR-20, AC-16	Backend status transition/confirmation requirements	server/tests/lab-03/staff-ticket-detail.api.test.ts	Passed locally; supplemental scope applies. Not main.
API-11	API	FR-20, AC-16	Status update checks current state before changing it	server/tests/lab-03/staff-ticket-detail.api.test.ts	Passed locally; supplemental scope applies. Not main.
API-12	Security	FR-18–FR-19, AC-14–AC-15	Requester owner rejected; permitted IT Priority update accepted	server/tests/lab-03/staff-ticket-detail.api.test.ts	Passed locally; supplemental scope applies. Not main.
API-13	API	FR-21, AC-09, AC-17	Staff Notes endpoint records session author	server/tests/lab-03/comments-notes.api.test.ts	Passed locally; supplemental scope applies. Not main.

ID	Type	Requirement / AC	What it tests	Automated file	Current local result
API-14	Security	FR-21, AC-09, AC-17	Requester cannot read Internal Notes	server/tests/lab-03/comments-notes.api.test.ts	Passed locally; supplemental scope applies. Not main.
API-15	API	FR-22–FR-23, AC-18–AC-19	Safe Administrator list, case-insensitive search, one role filter	server/tests/lab-03/users-admin.api.test.ts	Passed locally; supplemental scope applies. Not main.
API-16	Security	FR-22–FR-23, AC-25	Non-Administrator and invalid role-filter rejection	server/tests/lab-03/users-admin.api.test.ts	Passed locally; supplemental scope applies. Not main.
API-17	API	FR-24, AC-20	User creation hashes initial password and sets forced-change state	server/tests/lab-03/users-admin.api.test.ts	Passed locally; supplemental scope applies. Not main.
API-18	Validation	FR-24, AC-20–AC-22	Duplicate email and invalid create input rejected safely	server/tests/lab-03/users-admin.api.test.ts	Passed locally; supplemental scope applies. Not main.
API-19	API	FR-25–FR-26, AC-21–AC-23	Permitted edit fields and initial-password reset	server/tests/lab-03/users-admin.api.test.ts	Passed locally; supplemental scope applies. Not main.
API-20	Security	FR-28, AC-24–AC-25	Self-deactivation and last-active-Administrator protection	server/tests/lab-03/users-admin.api.test.ts	Passed locally; supplemental scope applies. Not main.
MIG-01	Migration	FR-33, AC-26	Static schema/SQL assertions plus Prisma migration deploy/status	server/prisma/prepare-lab3-e2e.ts; database-run2.txt	Passed locally; supplemental scope applies. Not main.
MIG-02	Regression	FR-09–FR-14, AC-06–AC-07	Existing RequesterUser, Ticket, Attachment IDs, ownership, and continuity through migration	server/prisma/prepare-lab3-e2e.ts; database-run2.txt	Passed locally; supplemental scope applies. Not main.
MIG-03	Seed	FR-33, AC-26	Seed fixture counts, idempotence, roles/ownership/authorship, and attachment continuity	server/prisma/prepare-lab3-e2e.ts; database-run2.txt	Passed locally; supplemental scope applies. Not main.
UI-01	UI	FR-01–FR-04, AC-01–AC-02	Authenticated Requester shell, safe login error, pending/busy feedback	client/tests/lab-03/Login.test.tsx	Passed locally; supplemental scope applies. Not main.
UI-02	UI	FR-05, AC-03	Mandatory password gate, valid continuation, mismatch disabled	client/tests/lab-03/ChangePassword.test.tsx	Passed locally; supplemental scope applies. Not main.
UI-03	UI	FR-09–FR-14, AC-06–AC-10	Requester Public Comment submission and resolution indication	client/tests/lab-03/RequesterWorkflow.test.tsx	Passed locally; supplemental scope applies. Not main.

ID	Type	Requirement / AC	What it tests	Automated file	Current local result
UI-04	UI	FR-15–FR-16, AC-11–AC-12	Queue filters, empty/no-results feedback	client/tests/lab-03/StaffTicketQueue.test.tsx; e2e/lab-03/staff-ticket-flow.spec.ts	Passed locally; supplemental scope applies. Not main.
UI-05	UI	FR-17–FR-21, AC-13–AC-17	Public/Private panel separation, claim, status action, Admin read-only detail	client/tests/lab-03/StaffTicketDetail.test.tsx	Passed locally; supplemental scope applies. Not main.
UI-06	UI	FR-22–FR-28, AC-18–AC-25	User list/filter, create, edit, reset confirmation	client/tests/lab-03/UserManagement.test.tsx	Passed locally; supplemental scope applies. Not main.
STYLE-01	UI style	FR-31–FR-32, AC-27	Palette tokens, focus rule, mobile media rule, sign-in labels/button class	client/tests/lab-03/ui-style.test.tsx	Passed locally; supplemental scope applies. Not main.
STYLE-02	Responsive	FR-32, AC-27	Viewport behavior and overflow	e2e/lab-03/fixtures.ts; actual responsive captures	Passed locally; supplemental scope applies. Not main.
E2E-01	E2E	AC-01–AC-05	Login, initial password, role shell, logout, direct blocking	e2e/lab-03/authentication.spec.ts	Passed locally; supplemental scope applies. Not main.
E2E-02	E2E	AC-06–AC-10	Requester create/list/detail/comment/resolution and regression	e2e/lab-03/authentication.spec.ts	Passed locally; supplemental scope applies. Not main.
E2E-03	E2E	AC-11–AC-17	Queue, detail, claim/reassign, priority/status, comments/notes, Attachment	e2e/lab-03/staff-ticket-flow.spec.ts	Passed locally; supplemental scope applies. Not main.
E2E-04	E2E	AC-18–AC-25	Admin management, reset, and direct-route restrictions	e2e/lab-03/user-administration.spec.ts; server/tests/lab-03/users-admin.api.test.ts	Passed locally; supplemental scope applies. Not main.
E2E-05	Responsive	AC-27	Desktop/tablet/mobile E2E captures and overflow assertions	e2e/lab-03/	Passed locally; supplemental scope applies. Not main.
A11Y-01	Accessibility	FR-31–FR-32, AC-27	Keyboard/focus audit	e2e/lab-03/z-accessibility-evidence.spec.ts	Passed locally; supplemental scope applies. Not main.

4. Acceptance-Criterion Traceability

AC	Planned/observed coverage	Current status
AC-01	API-01, UI-01, E2E-01	Passed within current assertions/captures; final-main rerun required. See supplemental traceability.

AC	Planned/observed coverage	Current status
AC-02	API-02, UI-01, E2E-01	Passed within current assertions/captures; final-main rerun required. See supplemental traceability.
AC-03	API-03, UI-02, E2E-01	Passed within current assertions/captures; final-main rerun required. See supplemental traceability.
AC-04	API-03, UI-01, E2E-01	Passed within current assertions/captures; final-main rerun required. See supplemental traceability.
AC-05	UI-01, STYLE-01, E2E-01	Passed within current assertions/captures; final-main rerun required. See supplemental traceability.
AC-06	API-04, UI-03, E2E-02	Passed within current assertions/captures; final-main rerun required. See supplemental traceability.
AC-07	API-05, MIG-02, E2E-02	Passed within current assertions/captures; final-main rerun required. See supplemental traceability.
AC-08	API-04–API-05, E2E-02	Passed within current assertions/captures; final-main rerun required. See supplemental traceability.
AC-09	API-06, API-13–API-14, UI-03, UI-05	Passed within current assertions/captures; final-main rerun required. See supplemental traceability.
AC-10	API-06, UI-03, E2E-02	Passed within current assertions/captures; final-main rerun required. See supplemental traceability.
AC-11	API-07–API-08, UI-04, E2E-03	Passed within current assertions/captures; final-main rerun required. See supplemental traceability.
AC-12	API-09, UI-04, STYLE-02, E2E-03	Passed within current assertions/captures; final-main rerun required. See supplemental traceability.
AC-13	API-10–API-12, UI-05, E2E-03	Passed within current assertions/captures; final-main rerun required. See supplemental traceability.

AC	Planned/observed coverage	Current status
AC-14	API-12, UI-05, E2E-03	Passed within current assertions/captures; final-main rerun required. See supplemental traceability.
AC-15	API-12, UI-05, E2E-03	Passed within current assertions/captures; final-main rerun required. See supplemental traceability.
AC-16	API-10–API-11, UI-05, E2E-03	Passed within current assertions/captures; final-main rerun required. See supplemental traceability.
AC-17	API-13–API-14, UI-05, E2E-03	Passed within current assertions/captures; final-main rerun required. See supplemental traceability.
AC-18	API-15, UI-06, E2E-04	Passed within current assertions/captures; final-main rerun required. See supplemental traceability.
AC-19	API-15, UI-06, E2E-04	Passed within current assertions/captures; final-main rerun required. See supplemental traceability.
AC-20	API-17–API-18, UI-06, E2E-04	Passed within current assertions/captures; final-main rerun required. See supplemental traceability.
AC-21	API-19, UI-06, E2E-04	Passed within current assertions/captures; final-main rerun required. See supplemental traceability.
AC-22	API-18–API-19, UI-06, E2E-04	Passed within current assertions/captures; final-main rerun required. See supplemental traceability.
AC-23	API-19, UI-06, E2E-04	Passed within current assertions/captures; final-main rerun required. See supplemental traceability.
AC-24	API-20, UI-06, E2E-04	Passed within current assertions/captures; final-main rerun required. See supplemental traceability.
AC-25	API-09, API-16, UI-05, E2E-04	Passed within current assertions/captures; final-main rerun required. See supplemental traceability.

AC	Planned/observed coverage	Current status
AC-26	MIG-01–MIG-03	Passed within current assertions/captures; final-main rerun required. See supplemental traceability.
AC-27	STYLE-01–STYLE-02, A11Y-01, E2E-05	Passed within current assertions/captures; final-main rerun required. See supplemental traceability.

5. Commands to Record

The following are required verification commands. Current dirty-tree commands and available output are recorded in the final appendix; complete historical final-main output is not retained.

```
cd server npm test -- --run tests/lab-03 npm run build npx prisma validate
```

```
cd ..\client npm test -- --run tests/lab-03 npm run build
```

```
cd .. npx playwright test e2e/lab-03
```

Historical summary in the 2026-09-21 release ledger: migration/status/seed and nine final-main Playwright tests were reported as passing after a persistent-database attempt failed eight tests because prior runs changed the seeded password-gate state. Raw historical output is not retained, and that summary does not prove legacy-row preservation. Separately, the 2026-09-29 dirty-tree audit applied the migration to a real pre-Lab-3 disposable database, verified preservation of Requester/Ticket/Attachment rows, ran seed twice with stable assertions, and passed 15 integrated E2E tests. These newer results are not final-main.

6. Evidence Rules

- Record date, branch, commit SHA, command, exit status, and relevant output.
- Store E2E screenshots under artifacts/lab-03/screenshots/ using stable

screen and viewport names.

- Keep failed, deferred, unavailable, and partially evidenced checks visible.
- Keep setup failures and partial visual/a11y evidence visible.
- A final result is accepted only when the exact test or visual capture exists

on the documented branch.

Appendix - Rendered AI Use and Reflection

Rendered source: docs/lab-03/ai-use.md

Lab 3 AI Use and Reflection

Status: Final working record for the Lab 3 specification, implementation support, review preparation, and evidence audit.

1. Tool and boundary

The Codex coding agent was used as a development assistant for repository inspection, requirement separation, specification drafting, test planning, implementation support, verification, and evidence organization. GitHub Issues, branches, Pull Requests, peer-review decisions, merges, board updates, and the final release remained controlled workflow actions and were checked against the real GitHub record.

The Lab 3 handouts and workflow guide were treated as requirements and reference material, not new user messages. The original Lab 2 worktree and report were preserved; authenticated regression tests were adapted only in the Lab 3 worktree.

2. Representative prompts used

1. Read every Lab 3 file and the repository workflow before editing; separate document instructions from the direct request. 2. Inspect the repository structure, branch state, Lab 2 baseline, Lab 3 folder, package files, tests, and scripts without making GitHub changes. 3. Summarize Lab 3 objectives, deliverables, coding work, tests, evidence, workflow, naming rules, completed work, and risks. 4. Propose an implementation and evidence plan and stop for approval before consequential changes. 5. Use an isolated Lab 3 specification branch so the existing Lab 2 checkout is not modified. 6. Draft the authentication, authorization, migration, API, UI, and Test DD contracts before implementation, with unexecuted evidence marked Planned. 7. When GitHub tracking begins, create only the approved Issues first, use one feature branch per unit, link PRs, obtain the friend review, prepare copy-ready replies, and merge only after approval.

3. Reflection

The most useful AI contribution was maintaining traceability between the specification, API/UI contracts, test matrix, implementation files, and the five-unit GitHub workflow. Separating document instructions from direct user instructions prevented the planning templates from being mistaken for real Issues, reviews, approvals, or test results.

The specification decisions that mattered most were session-derived identity, the mandatory initial-password gate, explicit role boundaries, Requester ownership checks, separate public Comments and Internal Notes, and the Administrator safety rules. These were checked in server tests and through the review questions rather than accepted as prose alone.

The current audit retained complete output and real evidence. New tests exposed the combined-owner-filter defect and missing Staff resolution indicator before fixes. Migration snapshots and seed-twice checks used actual disposable PostgreSQL, while separate controlled empty/failure images are labelled as UI-state evidence. The repaired local runs passed 190 server, 36 client and 16 E2E tests. These are not committed-main results; human review and final-main verification remain required. The exact earlier model name was not recorded.

Human review remained decisive: Punge089 reviewed every feature PR and the release PR, requested or discussed evidence where appropriate, approved, and merged. The agent prepared and checked documentation; it did not substitute for the reviewer or invent an approval.

4. Evidence rules

- Keep exact commands, branch/commit context, and observed results.
- Keep actual Issue, PR, review, reply, approval, merge, and board links.
- Do not claim an AI prompt, test, screenshot, or review happened unless it is

visible in the repository or external workflow record.

- Keep unavailable, partial, and prior-integrated evidence visibly distinct.

Appendix - Rendered Reviewer and Workflow Evidence

Rendered source: docs/lab-03/reviewer.md

Lab 3 Reviewer and Workflow Evidence

Observed workflow through PR #48: the original five units and reviewed follow-up/release completed. The local 2026-10-01 repairs have not created new Issues, PRs or approvals.

1. Workflow rules applied

- Five approved Issues were created and added to the existing Kanban board.
- One feature branch was used for each approved unit.
- Each feature PR targeted `lab3-staging` and was linked to its Issue through

GitHub Development metadata.

- The designated reviewer was Punge089 on every required PR.
- The PR author replied to review comments; requested fixes stayed on the same

branch and PR.

- Punge089 approved and merged the feature PRs. The release PR was reviewed and merged by Punge089 into `main`.

- Issues were closed only after the matching reviewed PR was merged.
- Issue #38 briefly showed `No status` after closure because of a project-board

automation transition. It was corrected to `Done` and verified on 2026-09-20.

2. Approved Issue and PR map

Unit	Issue	Feature branch	Feature PR	Target	Merge commit	Board / Issue state
Lab 3 contract and Test DD	#36 (https://github.com/book6349/toktickit/issues/36)	feature/lab3-specification	#41 (https://github.com/book6349/toktickit/pull/41)	lab3-staging	95cbdfc	Done / closed
Authentication, migration, authorization, and Requester regression	#37 (https://github.com/book6349/toktickit/issues/37)	feature/lab3-auth-requester	#42 (https://github.com/book6349/toktickit/pull/42)	lab3-staging	dbc2511	Done / closed
IT Staff Queue, Ticket Detail, and ticket operations	#38 (https://github.com/book6349/toktickit/issues/38)	feature/lab3-staff-workflow	#43 (https://github.com/book6349/toktickit/pull/43)	lab3-staging	3b55635	Done / closed
Administrator User Management	#39 (https://github.com/book6349/toktickit/issues/39)	feature/lab3-admin-users	#44 (https://github.com/book6349/toktickit/pull/44)	lab3-staging	160343f	Done / closed
Integrated E2E, visual evidence, and Lab 3 release integration	#40 (https://github.com/book6349/toktickit/issues/40)	feature/lab3-integration-evidence	#45 (https://github.com/book6349/toktickit/pull/45)	lab3-staging	71139cc	Original unit merged; follow-up #47 and release #48 merged; Done / closed

Initial release #46 merged as 493758b. Reviewed correction #47 merged to staging as 7d8ee7a; reviewed release #48 merged to main as 0353a40. Existing Issue #40 is recorded Done/closed.

3. Review, reply, approval, and merge evidence

PR	Review comment	Author reply	Approval	Merge event
#41	review (https://github.com/book6349/toktickit/pull/41#pullrequestreview-5255126909)	reply (https://github.com/book6349/toktickit/pull/41#issuecomment-5740540405)	approval (https://github.com/book6349/toktickit/pull/41#pullrequestreview-5255132969)	merge (https://github.com/book6349/toktickit/pull/41#event-31438804235)
#42	review (https://github.com/book6349/toktickit/pull/42#pullrequestreview-5255270367)	reply (https://github.com/book6349/toktickit/pull/42#issuecomment-5740877811)	approval (https://github.com/book6349/toktickit/pull/42#pullrequestreview-5255271960)	merge (https://github.com/book6349/toktickit/pull/42#event-31440367588)
#43	review (https://github.com/book6349/toktickit/pull/43#pullrequestreview-5255366757)	reply (https://github.com/book6349/toktickit/pull/43#issuecomment-5741064874)	approval (https://github.com/book6349/toktickit/pull/43#pullrequestreview-5255369288)	merge (https://github.com/book6349/toktickit/pull/43#event-31441319578)
#44	review (https://github.com/book6349/toktickit/pull/44#pullrequestreview-5255671320)	reply (https://github.com/book6349/toktickit/pull/44#issuecomment-5741727724)	approval (https://github.com/book6349/toktickit/pull/44#pullrequestreview-5255676942)	merge (https://github.com/book6349/toktickit/pull/44#event-31443781271)
#45	review (https://github.com/book6349/toktickit/pull/45#pullrequestreview-5255979494)	reply (https://github.com/book6349/toktickit/pull/45#issuecomment-5742558454)	approval (https://github.com/book6349/toktickit/pull/45#pullrequestreview-5255984540)	merge (https://github.com/book6349/toktickit/pull/45#event-31447113555)
#46 release	review (https://github.com/book6349/toktickit/pull/46#pullrequestreview-5256029152)	reply (https://github.com/book6349/toktickit/pull/46#issuecomment-5742711093)	approval (https://github.com/book6349/toktickit/pull/46#pullrequestreview-5256029917)	merge (https://github.com/book6349/toktickit/pull/46#event-31452435418)
#47 post-audit follow-up (https://github.com/book6349/toktickit/pull/47)	review (https://github.com/book6349/toktickit/pull/47#pullrequestreview-5361902956)	reply (https://github.com/book6349/toktickit/pull/47#issuecomment-5904847057)	approval (https://github.com/book6349/toktickit/pull/47#pullrequestreview-5361915030)	merge commit (https://github.com/book6349/toktickit/commit/7d8ee7acb09c76e06ca1bec5b1f5410452b8f2d7)
#48 final release (https://github.com/book6349/toktickit/pull/48)	review (https://github.com/book6349/toktickit/pull/48#pullrequestreview-5368368814)	reply (https://github.com/book6349/toktickit/pull/48#issuecomment-5914347504)	approval (https://github.com/book6349/toktickit/pull/48#pullrequestreview-5368393338)	merge commit (https://github.com/book6349/toktickit/commit/0353a4052c4a6d7ead9ddfb9b7c682d3e7958d02)

4. Release review result

Punge089 approved and merged the documented staging/release PRs through #48. A read-only remote check on 2026-10-01 confirmed main 0353a40. The additional local fixes are uncommitted and are not part of those completed PR events.

Clean-main results from 2026-09-30 and later local results are distinct. The current appendix contains full 2026-10-01 local output, including the 16-test E2E run after the Staff indicator fix. Only a subsequent clean-main run after reviewed integration may be labelled final-main.

Appendix - Rendered Screenshot Index

Rendered source: docs/lab-03/screenshot-index.md

Lab 3 Screenshot and Visual Evidence Index

Current screenshot provenance: inspected 2026-10-01 local captures, real authentication gates, controlled empty/failure responses explicitly labelled, and post-fix Staff-resolution evidence. The complete image inventory is retained in evidence-20261001-step4; final report figures are selected readable images, not all 81 inventory files.

Historical capture branch recorded by the prior report: main

Historical capture commit recorded by the prior report: 493758be65850504de7b970e50f7a61b846584f0

Historical final release commit: 493758be65850504de7b970e50f7a61b846584f0

Current integrated capture command: npx --no-install playwright test e2e/lab-03; 16 scenarios passed locally. Full output is appended.

Additional native browser capture scripts and exact outputs are retained in evidence-20261001-step4 and step5. Controlled response captures are labelled; they are not real database failures.

Required viewport sizes:

- Desktop: 1280 x 900
- Tablet: 900 x 900
- Mobile: 390 x 844

Required captures

Figure	File	Viewport	Scenario/caption	Status / boundary
1	output\docs\evidence-20261001-step4\run3\screenshots\authentication\requester\my-tickets-1280x900.png	1280x1462	Figure 1. Requester My Tickets shell after authentication at desktop width Captured 2026-10-01, dirty local tree.	2026-10-01 local; not main
2	output\docs\evidence-20261001-step4\run3\screenshots\authentication\requester\my-tickets-900x900.png	900x1400	Figure 2. Requester My Tickets shell at tablet width Captured 2026-10-01, dirty local tree.	2026-10-01 local; not main
3	output\docs\evidence-20261001-step4\run3\screenshots\authentication\requester\my-tickets-section-01-390px.png	390x760	Figure 3. Requester My Tickets shell at mobile width	2026-10-01 local; not main
4	output\docs\evidence-20261001-step5\staff-resolution-fixed\desktop-queue-viewport.png	1280x900	Figure 4. Queue native viewport at 1280x900, captured 2026-10-01 locally. Full-page originals and readable mobile continuations retained separately	2026-10-01 local; not main
5	output\docs\evidence-20261001-step5\staff-resolution-fixed\tablet-queue-viewport.png	900x900	Figure 5. Queue native viewport at 900x900, captured 2026-10-01 locally. Full-page originals and readable mobile continuations retained separately	2026-10-01 local; not main
6	output\docs\evidence-20261001-step4\run3\screenshots\staff-queue\populated-section-01-390px.png	390x760	Figure 6. IT Staff Queue at mobile width (390 x 844 viewport)	2026-10-01 local; not main

Figure	File	Viewport	Scenario/caption	Status / boundary
7	output\docs\evidence-20261001-step5\staff-resolution-fixed\desktop-detail-viewport.png	1280x900	Figure 7. Staff Detail native viewport at 1280x900, captured 2026-10-01 locally. Full-page originals and readable mobile continuations retained separately	2026-10-01 local; not main
8	output\docs\evidence-20261001-step5\staff-resolution-fixed\tablet-detail-viewport.png	900x900	Figure 8. Staff Detail native viewport at 900x900, captured 2026-10-01 locally. Full-page originals and readable mobile continuations retained separately	2026-10-01 local; not main
9	output\docs\evidence-20261001-step4\run3\screenshots\staff-ticket-detail\public-private-status-attachment-section-01-390px.png	390x760	Figure 9. IT Staff Ticket Detail at mobile width (390 x 844 viewport)	2026-10-01 local; not main
10	output\docs\evidence-20261001-step5\staff-resolution-fixed\desktop-admin-viewport.png	1280x900	Figure 10. Administrator User Management native viewport at 1280x900, captured 2026-10-01 locally. Full-page originals and readable mobile continuations retained separately	2026-10-01 local; not main
11	output\docs\evidence-20261001-step5\staff-resolution-fixed\tablet-admin-viewport.png	900x900	Figure 11. Administrator User Management native viewport at 900x900, captured 2026-10-01 locally. Full-page originals and readable mobile continuations retained separately	2026-10-01 local; not main
12	output\docs\evidence-20261001-step4\run4-admin\screenshots\user-management\list-section-01-390px.png	390x760	Figure 12. Administrator User Management at mobile width (390 x 844 viewport)	2026-10-01 local; not main

Additional state captures

File	State shown	Evidence type	Status
output\docs\evidence-20261001-step4\run3\screenshots\authentication\invalid-login-1280x900.png	Additional state capture — Safe invalid-login failure Captured 2026-10-01, dirty local tree.	Actual browser	Local; not main
output\docs\evidence-20261001-step4\run3\screenshots\authentication\inactive-login-1280x900.png	Additional state capture — Safe inactive-account failure Captured 2026-10-01, dirty local tree.	Actual browser	Local; not main
output\docs\evidence-20261001-step4\run3\screenshots\authentication\mandatory-change-password-1280x900.png	Additional state capture - Real mandatory password gate; 2026-10-01 local capture, not keyboard-traversal proof.	Actual browser	Local; not main
output\docs\evidence-20261001-step4\run3\screenshots\authentication\logout-api-blocked-1280x900.png	Additional state capture — Login gate after logout Captured 2026-10-01, dirty local tree.	Actual browser	Local; not main
output\docs\evidence-20261001-step4\run3\screenshots\staff-queue\no-results-1280x900.png	Additional state capture — Filtered queue with no matches Captured 2026-10-01, dirty local tree.	Actual browser	Local; not main

File	State shown	Evidence type	Status
output\docs\evidence-20261001-step4\run4-admin\screenshots\staff-queue\controlled-empty-1280x900.png	Additional state capture — Empty queue state Captured 2026-10-01, dirty local tree. CONTROLLED MOCKED API STATE; UI presentation only.	Controlled API	Local; not main
output\docs\evidence-20261001-step4\run4-admin\screenshots\staff-queue\controlled-failure-1280x900.png	Additional state capture — Queue API failure with retry Captured 2026-10-01, dirty local tree. CONTROLLED MOCKED API STATE; UI presentation only.	Controlled API	Local; not main
output\docs\evidence-20261001-step4\run3\screenshots\staff-ticket-detail\claimed-migrated-attachment-1280x900.png	Additional state capture — Claim, IT Priority, public comment, and Internal Note operations Captured 2026-10-01, dirty local tree.	Actual browser	Local; not main
output\docs\evidence-20261001-step4\run3\screenshots\staff-ticket-detail\invalid-owner-safe-error-1280x900.png	Additional state capture — Safe owner-validation failure Captured 2026-10-01, dirty local tree.	Actual browser	Local; not main
output\docs\evidence-20261001-step4\run4-admin\screenshots\user-management\create-form-1280x900.png	Additional state capture — Administrator create-user form Captured 2026-10-01, dirty local tree.	Actual browser	Local; not main
output\docs\evidence-20261001-step4\run4-admin\screenshots\user-management\duplicate-email-1280x900.png	Additional state capture — Duplicate-email validation Captured 2026-10-01, dirty local tree.	Actual browser	Local; not main
output\docs\evidence-20261001-step4\run4-admin\screenshots\user-management\no-results-1280x900.png	Additional state capture — Administrator search with no matches Captured 2026-10-01, dirty local tree.	Actual browser	Local; not main
output\docs\evidence-20261001-step4\run4-admin\screenshots\user-management\edit-form-1280x900.png	Additional state capture — Administrator edit form Captured 2026-10-01, dirty local tree.	Actual browser	Local; not main
output\docs\evidence-20261001-step4\run5-supplement\screenshots\user-management\reset-success-1280x900.png	Additional state capture - Persistent reset confirmation, real API; 2026-10-01 local capture.	Actual browser	Local; not main
output\docs\evidence-20261001-step4\run5-supplement\screenshots\user-management\reset-next-login-password-gate-1280x900.png	Additional state capture — Reset user reaches the mandatory password-change screen at next login Captured 2026-10-01, dirty local tree.	Actual browser	Local; not main

Visual checklist

Checklist item	Evidence	Status
Desktop/tablet/mobile	Actual major-screen captures and native mobile sections	Inspected locally; not final-main
Authentication gates	Real API-backed Login/Change Password at three widths	Captured and overflow checked
Role navigation/badges/read-only fields	Requester/Staff/Admin views, readonly Admin Detail	Inspected; operation authorization also tested

Checklist item	Evidence	Status
Validation and feedback	Real invalid login/owner/password/duplicate; controlled failure states labelled	Observed in identified captures
Public/private and resolution display	Separate content panels; Figures 43-45 after Staff fix	Locally verified; formal status unchanged
Keyboard/focus	Three real E2E journeys plus busy flow	Passed locally; not complete accessibility certification
Clipping/overflow/readability	Width assertions and native sections; QA inspection	Passed for captured states; no numeric contrast audit

Appendix - Repository README

Rendered source: README.md

TokTickIT - IT Service Desk Application

Full-stack IT service desk application built for CPE 334 Introduction to Software Engineering.

Tech Stack

- **Frontend***: React + TypeScript + Vite + Bootstrap 5
- **Backend***: Node.js + Express + TypeScript
- **Database & OR***: PostgreSQL + Prisma ORM
- **Testin***: Vitest, Supertest, and Playwright

Getting Started

1. Prerequisites

- Node.js (v18+)
- Docker Desktop (recommended for the PostgreSQL 18 test database)

Start a disposable local database from PowerShell:

```
docker run --name toktickit-postgres --restart unless-stopped -e POSTGRES_USER=toktickit -e POSTGRES_PASSWORD=toktickit -e POSTGRES_DB=toktickit -p 5432:5432 -v toktickit-pgdata18:/var/lib/postgresql -d postgres:18
```

If the container already exists, use `docker start toktickit-postgres`.

2. Environment Setup

Client Configuration Copy `client/.env.example` to `client/.env`:

```
VITE_API_URL="http://localhost:3000"
```

Server Configuration Copy `server/.env.example` to `server/.env`:

```
DATABASE_URL="postgresql://toktickit:toktickit@localhost:5432/toktickit?schema=public"
PORT=3000
```

3. Installation & Local Development

Frontend (Client)

```
cd client
npm install
npm run dev
```

Backend (Server)

```
cd server
npm install
npx prisma db push --accept-data-loss
npx prisma migrate resolve --applied 20260824000000_lab2_foundation
npm run prisma:seed
npm run dev
```

The three Prisma commands above are the current disposable-database baseline. The committed Lab 2 migration history assumes a pre-existing Lab 1 Category table on a brand-new database, so `prisma migrate deploy` is not currently a fresh-database setup path. Do not use `--accept-data-loss` against a shared database.

4. Running Tests

Run these as separate PowerShell commands:

```
cd server
npm test -- --run
```

```
cd ../client
npm test -- --run
```

```
cd ..
npm install
npx playwright test e2e/lab-02/requester-ticket-flow.spec.ts
```

The Playwright command starts the Vite client and Express API using the committed `playwright.config.ts`; Docker PostgreSQL must be running first.

Build checks:

```
cd server
npm run build
```

```
cd ../client
npm run build
```

Appendix - Repository gitignore

Rendered source: .gitignore

dependencies

node_modules/

env & secrets

.env *.env !.env.example

build output

dist/ build/

prisma

server/prisma/*.db

private uploaded files

server/storage/

Playwright runtime output

test-results/ artifacts/lab-03/playwright-report/

logs & OS

*.log .DS_Store

Appendix - Committed Repository Tree at Main 0353a40

Committed main 0353a4052c4a6d7ead9ddfb9b7c682d3e7958d02: 218 paths; all committed paths included; untracked/ignored local files excluded

.gitattributes

.gitignore

README.md

artifacts/lab-02/screenshots-real/create-ticket/desktop-1280x900.png

artifacts/lab-02/screenshots-real/create-ticket/mobile-390x844.png

artifacts/lab-02/screenshots-real/create-ticket/tablet-900x900.png

artifacts/lab-02/screenshots-real/my-tickets/desktop-1280x900.png

artifacts/lab-02/screenshots-real/my-tickets/mobile-390x844.png

artifacts/lab-02/screenshots-real/my-tickets/tablet-900x900.png

artifacts/lab-02/screenshots-real/run-notes.json

artifacts/lab-02/screenshots-real/ticket-detail/desktop-1280x900.png

artifacts/lab-02/screenshots-real/ticket-detail/mobile-390x844.png

artifacts/lab-02/screenshots-real/ticket-detail/tablet-900x900.png

artifacts/lab-02/screenshots/committed-playwright-run.json

artifacts/lab-02/screenshots/create-ticket/desktop-1280x900.png

artifacts/lab-02/screenshots/create-ticket/mobile-390x844.png

artifacts/lab-02/screenshots/create-ticket/tablet-900x900.png

artifacts/lab-02/screenshots/my-tickets/desktop-1280x900.png

artifacts/lab-02/screenshots/my-tickets/mobile-390x844.png

artifacts/lab-02/screenshots/my-tickets/tablet-900x900.png

artifacts/lab-02/screenshots/run-notes.json

artifacts/lab-02/screenshots/ticket-detail/desktop-1280x900.png

artifacts/lab-02/screenshots/ticket-detail/mobile-390x844.png

artifacts/lab-02/screenshots/ticket-detail/tablet-900x900.png

artifacts/lab-03/screenshots/accessibility/admin-create-keyboard-focus-1280x900.png

artifacts/lab-03/screenshots/accessibility/login-keyboard-focus-1280x900.png

artifacts/lab-03/screenshots/accessibility/requester-navigation-keyboard-focus-1280x900.png

artifacts/lab-03/screenshots/accessibility/staff-detail-keyboard-focus-1280x900.png

artifacts/lab-03/screenshots/authentication/requester-regression/desktop-1280x900-my-tickets.png

artifacts/lab-03/screenshots/authentication/requester-regression/first-login-change-password-1280x900.png

Committed main 0353a4052c4a6d7ead9ddfb9b7c682d3e7958d02: 218 paths; all committed paths included; untracked/ignored local files excluded

artifacts/lab-03/screenshots/authentication/requester-regression/inactive-account-1280x900.png

artifacts/lab-03/screenshots/authentication/requester-regression/invalid-login-1280x900.png

artifacts/lab-03/screenshots/authentication/requester-regression/logout-gate-1280x900.png

artifacts/lab-03/screenshots/authentication/requester-regression/mobile-390x844-my-tickets.png

artifacts/lab-03/screenshots/authentication/requester-regression/tablet-900x900-my-tickets.png

artifacts/lab-03/screenshots/authentication/requester-resolution-indicated-1280x900.png

artifacts/lab-03/screenshots/authentication/responsive-auth-gates/change-password-desktop-1280x900.png

artifacts/lab-03/screenshots/authentication/responsive-auth-gates/change-password-mobile-390x844.png

artifacts/lab-03/screenshots/authentication/responsive-auth-gates/change-password-tablet-900x900.png

artifacts/lab-03/screenshots/authentication/responsive-auth-gates/login-desktop-1280x900.png

artifacts/lab-03/screenshots/authentication/responsive-auth-gates/login-mobile-390x844.png

artifacts/lab-03/screenshots/authentication/responsive-auth-gates/login-tablet-900x900.png

artifacts/lab-03/screenshots/authentication/sign-in-success-1280x900.png

artifacts/lab-03/screenshots/authentication/signing-in-busy-1280x900.png

artifacts/lab-03/screenshots/mobile-report-sections/admin-users-mobile-section-01-of-05.png

artifacts/lab-03/screenshots/mobile-report-sections/admin-users-mobile-section-02-of-05.png

artifacts/lab-03/screenshots/mobile-report-sections/admin-users-mobile-section-03-of-05.png

artifacts/lab-03/screenshots/mobile-report-sections/admin-users-mobile-section-04-of-05.png

artifacts/lab-03/screenshots/mobile-report-sections/admin-users-mobile-section-05-of-05.png

artifacts/lab-03/screenshots/mobile-report-sections/staff-queue-mobile-section-01-of-03.png

artifacts/lab-03/screenshots/mobile-report-sections/staff-queue-mobile-section-01-of-05.png

artifacts/lab-03/screenshots/mobile-report-sections/staff-queue-mobile-section-02-of-03.png

artifacts/lab-03/screenshots/mobile-report-sections/staff-queue-mobile-section-02-of-05.png

artifacts/lab-03/screenshots/mobile-report-sections/staff-queue-mobile-section-03-of-03.png

artifacts/lab-03/screenshots/mobile-report-sections/staff-queue-mobile-section-03-of-05.png

artifacts/lab-03/screenshots/mobile-report-sections/staff-queue-mobile-section-04-of-05.png

artifacts/lab-03/screenshots/mobile-report-sections/staff-queue-mobile-section-05-of-05.png

artifacts/lab-03/screenshots/mobile-report-sections/staff-ticket-detail-mobile-section-01-of-04.png

artifacts/lab-03/screenshots/mobile-report-sections/staff-ticket-detail-mobile-section-02-of-04.png

artifacts/lab-03/screenshots/mobile-report-sections/staff-ticket-detail-mobile-section-03-of-04.png

artifacts/lab-03/screenshots/mobile-report-sections/staff-ticket-detail-mobile-section-04-of-04.png

Committed main 0353a4052c4a6d7ead9ddfb9b7c682d3e7958d02: 218 paths; all committed paths included; untracked/ignored local files excluded

artifacts/lab-03/screenshots/staff-queue/desktop-1280x900-queue.png

artifacts/lab-03/screenshots/staff-queue/empty-1280x900.png

artifacts/lab-03/screenshots/staff-queue/failure-1280x900.png

artifacts/lab-03/screenshots/staff-queue/mobile-390x844-queue.png

artifacts/lab-03/screenshots/staff-queue/no-results-1280x900.png

artifacts/lab-03/screenshots/staff-queue/pagination-page-2-1280x900.png

artifacts/lab-03/screenshots/staff-queue/tablet-900x900-queue.png

artifacts/lab-03/screenshots/staff-ticket-detail/desktop-1280x900-detail.png

artifacts/lab-03/screenshots/staff-ticket-detail/migration-preservation-operations-1280x900.png

artifacts/lab-03/screenshots/staff-ticket-detail/mobile-390x844-detail.png

artifacts/lab-03/screenshots/staff-ticket-detail/operations-1280x900.png

artifacts/lab-03/screenshots/staff-ticket-detail/tablet-900x900-detail.png

artifacts/lab-03/screenshots/staff-ticket-detail/validation-1280x900.png

artifacts/lab-03/screenshots/user-management/create-form-1280x900.png

artifacts/lab-03/screenshots/user-management/desktop-1280x900-list.png

artifacts/lab-03/screenshots/user-management/duplicate-validation-1280x900.png

artifacts/lab-03/screenshots/user-management/edit-form-1280x900.png

artifacts/lab-03/screenshots/user-management/mobile-390x844-list.png

artifacts/lab-03/screenshots/user-management/no-results-1280x900.png

artifacts/lab-03/screenshots/user-management/reset-first-login-1280x900.png

artifacts/lab-03/screenshots/user-management/reset-list-1280x900.png

artifacts/lab-03/screenshots/user-management/reset-success-1280x900.png

artifacts/lab-03/screenshots/user-management/tablet-900x900-list.png

client/.env.example

client/index.html

client/package-lock.json

client/package.json

client/src/App.tsx

client/src/admin.tsx

client/src/api.ts

client/src/main.tsx

Committed main 0353a4052c4a6d7ead9ddfb9b7c682d3e7958d02: 218 paths; all committed paths included; untracked/ignored local files excluded

client/src/staff.tsx

client/src/styles.css

client/src/vite-env.d.ts

client/tests/lab-01/App.test.tsx

client/tests/lab-02/AttachmentSection.test.tsx

client/tests/lab-02/CreateTicket.test.tsx

client/tests/lab-02/MyTickets.test.tsx

client/tests/lab-02/RequesterContext.test.tsx

client/tests/lab-02/RequesterTicketDetail.test.tsx

client/tests/lab-02/ui-style.test.tsx

client/tests/lab-03/ChangePassword.test.tsx

client/tests/lab-03/Login.test.tsx

client/tests/lab-03/RequesterWorkflow.test.tsx

client/tests/lab-03/StaffTicketDetail.test.tsx

client/tests/lab-03/StaffTicketQueue.test.tsx

client/tests/lab-03/UserManagement.test.tsx

client/tests/lab-03/ui-style.test.tsx

client/tests/setup.ts

client/tsconfig.json

client/vite.config.ts

docs/lab-01/ai_use.md

docs/lab-01/reviewer.md

docs/lab-01/tests.md

docs/lab-02/ai-use.md

docs/lab-02/api-spec.md

docs/lab-02/e2e-visual-plan.md

docs/lab-02/release-evidence.md

docs/lab-02/reviewer.md

docs/lab-02/specification.md

docs/lab-02/tests.md

docs/lab-02/ui-smoke-evidence.md

Committed main 0353a4052c4a6d7ead9ddfb9b7c682d3e7958d02: 218 paths; all committed paths included; untracked/ignored local files excluded

docs/lab-02/ui-spec.md

docs/lab-03/ai-use.md

docs/lab-03/api-spec.md

docs/lab-03/final-report.md

docs/lab-03/github/issue-01-engineering-contract.md

docs/lab-03/github/issue-02-auth-requester.md

docs/lab-03/github/issue-03-staff-workflow.md

docs/lab-03/github/issue-04-admin-users.md

docs/lab-03/github/issue-05-integration-release.md

docs/lab-03/github/pr-01-approval.txt

docs/lab-03/github/pr-01-author-reply.txt

docs/lab-03/github/pr-01-body.md

docs/lab-03/github/pr-01-review-comment.txt

docs/lab-03/github/pr-02-approval.txt

docs/lab-03/github/pr-02-author-reply.txt

docs/lab-03/github/pr-02-body.md

docs/lab-03/github/pr-02-review-comment.txt

docs/lab-03/github/pr-03-approval.txt

docs/lab-03/github/pr-03-author-reply.txt

docs/lab-03/github/pr-03-body.md

docs/lab-03/github/pr-03-review-comment.txt

docs/lab-03/github/pr-04-approval.txt

docs/lab-03/github/pr-04-author-reply.txt

docs/lab-03/github/pr-04-body.md

docs/lab-03/github/pr-04-review-comment.txt

docs/lab-03/github/pr-05-approval.txt

docs/lab-03/github/pr-05-author-reply.txt

docs/lab-03/github/pr-05-body.md

docs/lab-03/github/tracking-plan.md

docs/lab-03/release-evidence.md

docs/lab-03/reviewer.md

docs/lab-03/screenshot-index.md

docs/lab-03/specification.md

Committed main 0353a4052c4a6d7ead9ddfb9b7c682d3e7958d02: 218 paths; all committed paths included; untracked/ignored local files excluded

docs/lab-03/submission-evidence.md
docs/lab-03/tests.md
docs/lab-03/ui-spec.md
e2e/lab-02/requester-ticket-flow.spec.ts
e2e/lab-03/authentication.spec.ts
e2e/lab-03/fixtures.ts
e2e/lab-03/staff-ticket-flow.spec.ts
e2e/lab-03/user-administration.spec.ts
e2e/lab-03/z-accessibility-evidence.spec.ts
output/docs/Lab3_Final_Report.pdf
output/docs/Lab3_Final_Report_Editable.docx
output/docs/evidence-20260929/auth-responsive-capture.txt
output/docs/evidence-20260929/capture-auth-responsive.mjs
output/docs/evidence-20260929/client-build.txt
output/docs/evidence-20260929/client-lab3-tests.txt
output/docs/evidence-20260929/docker-verification.txt
output/docs/evidence-20260929/e2e-lab3-final.txt
output/docs/evidence-20260929/e2e-lab3-fixture-failure.txt
output/docs/evidence-20260929/prisma-validate.txt
output/docs/evidence-20260929/run-context.txt
output/docs/evidence-20260929/server-build.txt
output/docs/evidence-20260929/server-lab3-tests.txt
output/pdf/Lab3_Final_Report.pdf
package-lock.json
package.json
playwright.config.ts
server/.env.example
server/package-lock.json
server/package.json
server/prisma/migrations/20260824000000_lab2_foundation/migration.sql
server/prisma/migrations/20260919000000_lab3_auth_requester/migration.sql
server/prisma/migrations/migration_lock.toml
server/prisma/schema.prisma
server/prisma/seed.ts
server/src/admin-users.ts
server/src/app.ts
server/src/auth.ts
server/src/index.ts
server/src/lab2.ts
server/src/prisma.ts
server/src/requester-regression.ts
server/src/staff-workflow.ts
server/tests/lab-01/categories.test.ts
server/tests/lab-01/health.test.ts
server/tests/lab-02/attachments.api.test.ts
server/tests/lab-02/create-ticket.api.test.ts
server/tests/lab-02/my-tickets.api.test.ts
server/tests/lab-02/reference-data.api.test.ts
server/tests/lab-02/requester-context.api.test.ts
server/tests/lab-02/ticket-detail.api.test.ts
server/tests/lab-02/ticket-number.unit.test.ts
server/tests/lab-02/validation.unit.test.ts

Committed main 0353a4052c4a6d7ead9ddfb9b7c682d3e7958d02: 218 paths; all committed paths included; untracked/ignored local files excluded

server/tests/lab-03/auth.api.test.ts
server/tests/lab-03/auth.unit.test.ts
server/tests/lab-03/authorization.api.test.ts
server/tests/lab-03/comments-notes.api.test.ts
server/tests/lab-03/migration-regression.test.ts
server/tests/lab-03/staff-queue.api.test.ts
server/tests/lab-03/staff-ticket-detail.api.test.ts
server/tests/lab-03/users-admin.api.test.ts
server/tsconfig.json
server/vitest.config.ts

Appendix - Readable Mobile Screenshot Sections

Tall mobile evidence uses native browser sections with 100px overlap. Unchanged full-page originals are retained alongside the readable sections. No application images were generated or edited.

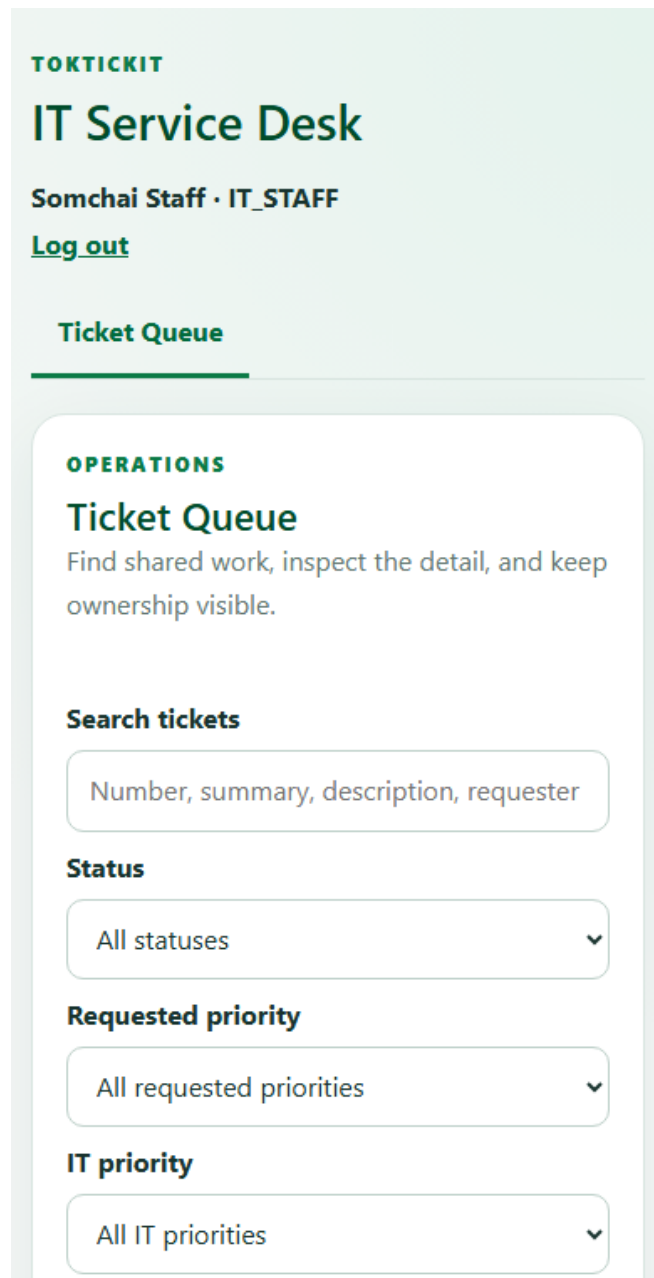


Figure 20. Queue native mobile section 1/6; 2026-10-01 local browser capture. Final section follows as Figure 46.

IT priority

All IT priorities

Owner

All ownership

Sort by

Last updated

Sort direction

Descending

Tickets per page

10

Apply filters

Clear filters

TT-DOCKER-PRELAB3-001

Pre-Lab 3 preservation fixture

Hardware · Ariya Somchai · Updated 10/1/2026, 1:19:54 AM

Requested HIGH

IT LOW

OPEN

Owner: Kanya Staff

Open Detail

Figure 21. Queue native mobile section 2/6; 2026-10-01 local browser capture. Final section follows as Figure 46.

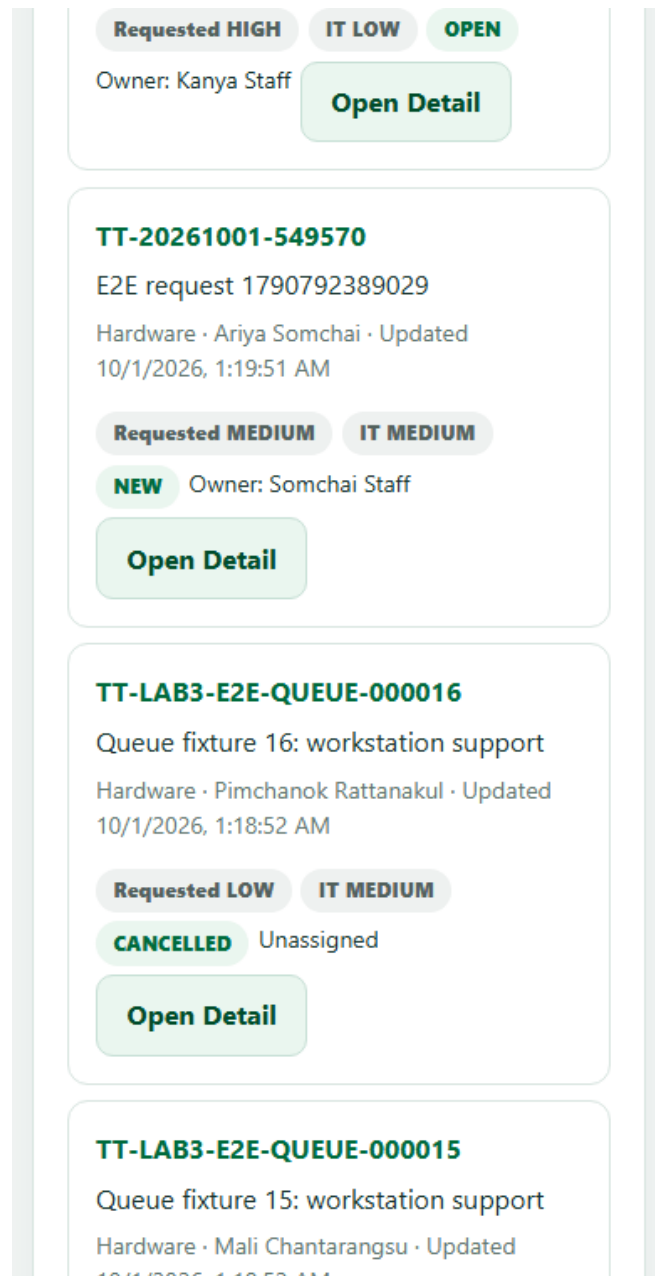


Figure 22. Queue native mobile section 3/6; 2026-10-01 local browser capture. Final section follows as Figure 46.

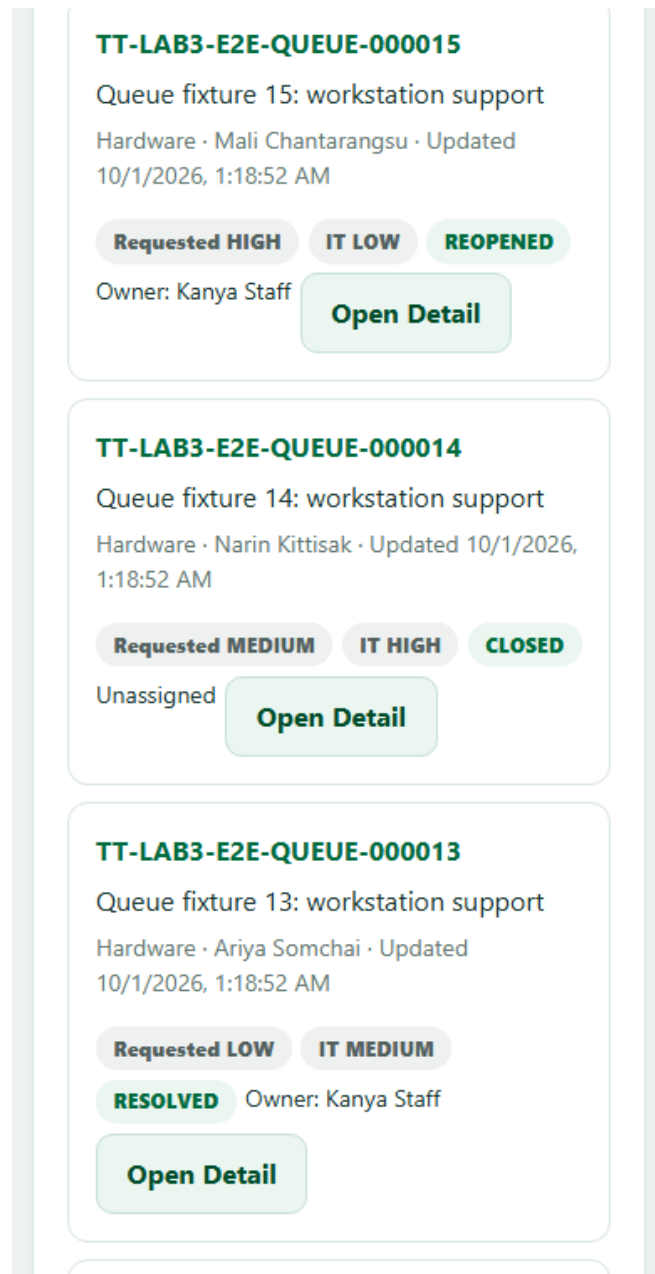


Figure 23. Queue native mobile section 4/6; 2026-10-01 local browser capture. Final section follows as Figure 46.

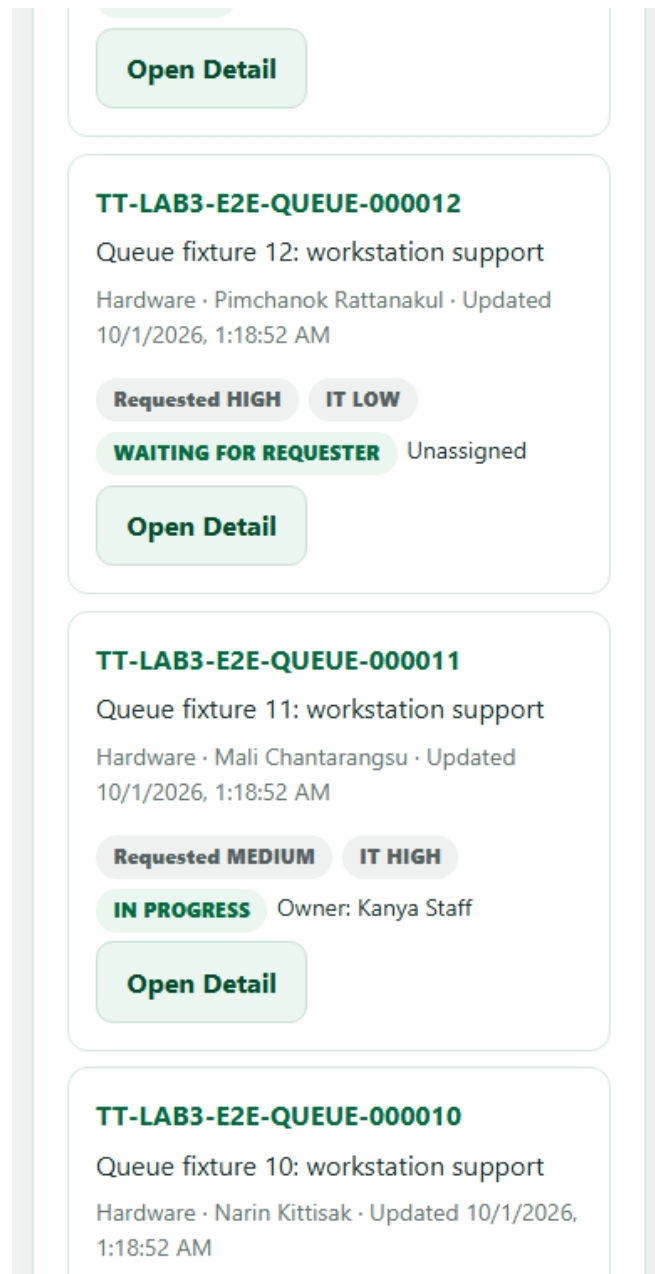


Figure 24. Queue native mobile section 5/6; 2026-10-01 local browser capture. Final section follows as Figure 46.

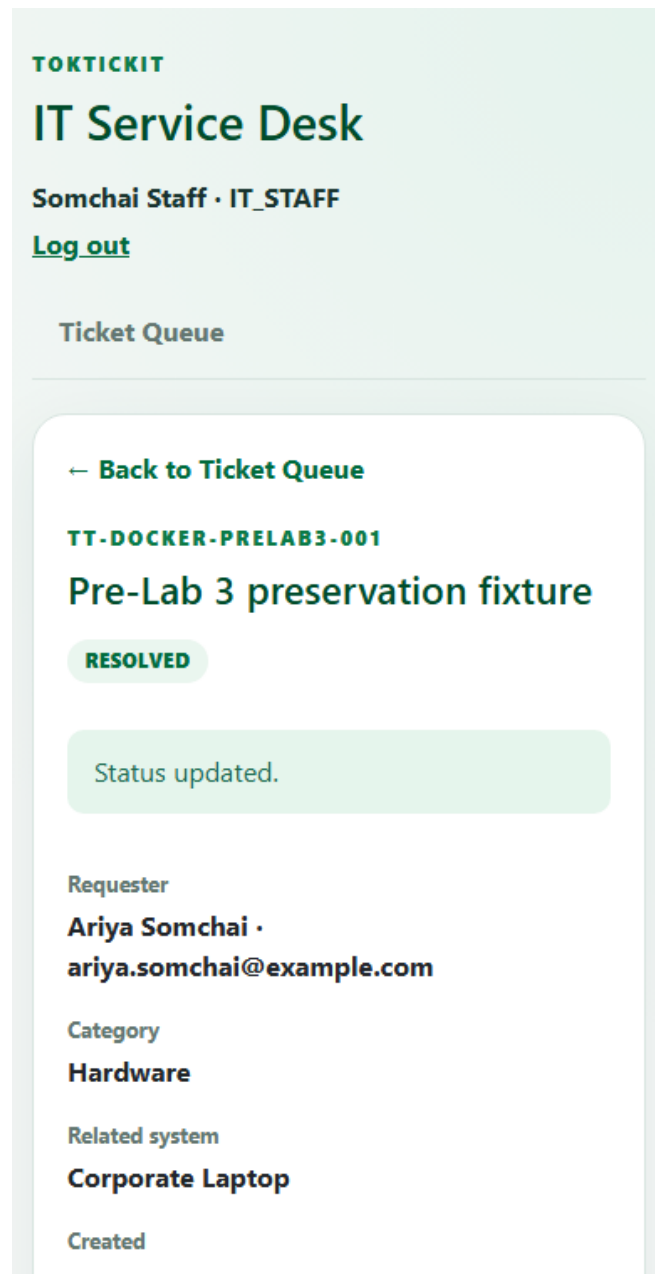


Figure 25. Staff Detail native mobile section 1/5; 2026-10-01 local browser capture. Final section follows as Figure 47.

Related system

Corporate Laptop

Created

10/1/2026, 1:18:48 AM

Updated

10/1/2026, 1:31:48 AM

Requested priority

HIGH

IT priority

LOW

Owner

Somchai Staff ·
somchai.staff@example.com

Description

Synthetic migration fixture for local verification.

Ownership

Only active IT Staff or Administrators may own a Ticket.

Claim as me

Unassign

Figure 26. Staff Detail native mobile section 2/5; 2026-10-01 local browser capture. Final section follows as Figure 47.

Only active IT Staff or Administrators may own a Ticket.

Claim as me

Unassign

Assign owner ID

7

Assign owner

IT Priority and status

IT Priority

LOW

Save IT Priority

Next status

Choose a permitted transition

Update status

Attachments

Existing Ticket files.

Figure 27. Staff Detail native mobile section 3/5; 2026-10-01 local browser capture. Final section follows as Figure 47.

Attachments

Existing Ticket files.

baseline.txt
Uploaded 10/1/2026, 1:18:48 AM

Public Comments

Visible to Requesters, IT Staff, and Administrators.

1

Somchai Staff
10/1/2026, 1:31:47 AM
Evidence public update: investigation in progress.

Add a public comment

Add public comment

Internal Notes

Figure 28. Staff Detail native mobile section 4/5; 2026-10-01 local browser capture. Final section follows as Figure 47.

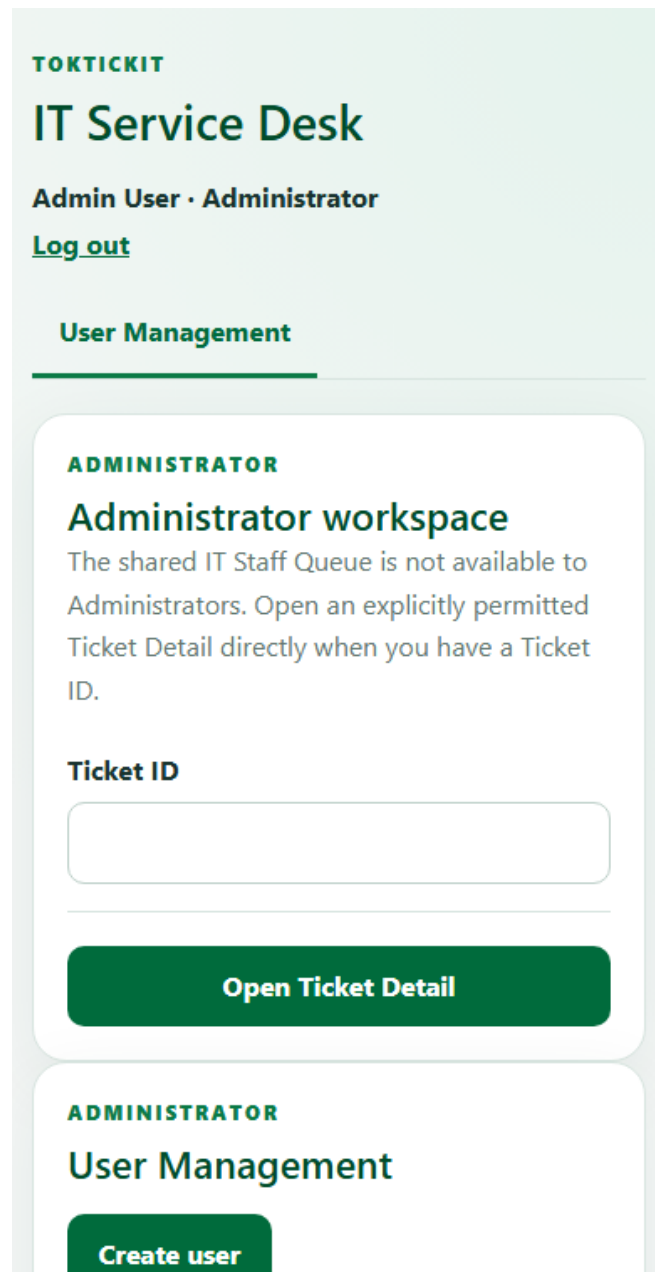


Figure 29. Administrator native mobile section 1/6; 2026-10-01 local browser capture. Final section follows as Figure 48.

User Management

[Create user](#)

Manage the minimum account fields required by Lab 3. Password reset is a separate action and requires a change at the user's next login.

[Apply filters](#)[Clear filters](#)

Admin User

admin@example.com

Updated 10/1/2026, 1:19:54 AM

[Administrator](#)[Active](#)[Edit](#)

Ariya Somchai

ariya.somchai@example.com

Updated 10/1/2026, 1:19:47 AM

Figure 30. Administrator native mobile section 2/6; 2026-10-01 local browser capture. Final section follows as Figure 48.

Ariya Somchai
ariya.somchai@example.com
Updated 10/1/2026, 1:19:47 AM

Requester

Active

Edit

E2E Administrator Test Updated
e2e-1790792394973@example.com
Updated 10/1/2026, 1:19:56 AM

Requester

Active

Edit

Inactive Requester
inactive@example.com
Updated 10/1/2026, 1:18:52 AM

Requester

Inactive

Edit

Inactive Staff
inactive.staff@example.com

Figure 31. Administrator native mobile section 3/6; 2026-10-01 local browser capture. Final section follows as Figure 48.

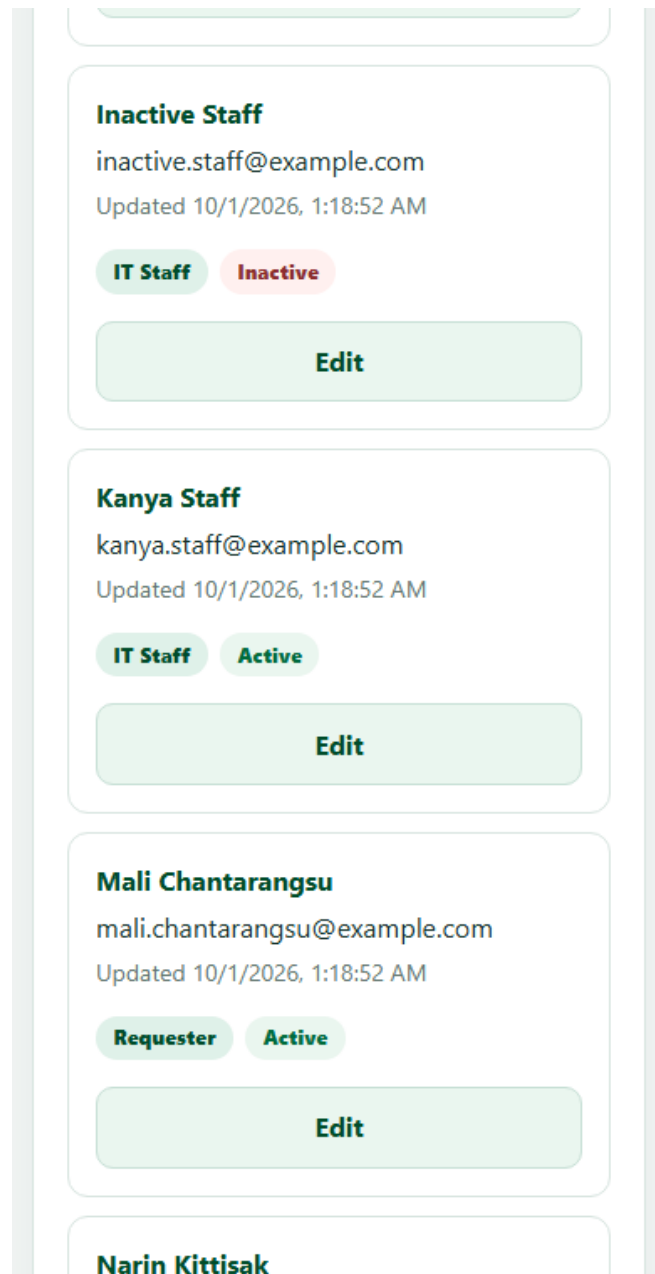


Figure 32. Administrator native mobile section 4/6; 2026-10-01 local browser capture. Final section follows as Figure 48.

Edit

Narin Kittisak

narin.kittisak@example.com

Updated 10/1/2026, 1:18:52 AM

Requester

Active

Edit

Nopparat Staff

nopparat.staff@example.com

Updated 10/1/2026, 1:18:52 AM

IT Staff

Active

Edit

Pimchanok Rattanakul

pimchanok.rattanakul@example.com

Updated 10/1/2026, 1:18:52 AM

Requester

Active

Edit

Figure 33. Administrator native mobile section 5/6; 2026-10-01 local browser capture. Final section follows as Figure 48.

Appendix - Responsive Authentication Gate Screenshots

These Login and mandatory Change Password captures use real local API sessions on the disposable database, not mocked authentication responses. All three required viewports were captured with width/overflow assertions; they remain local evidence, not final-main.

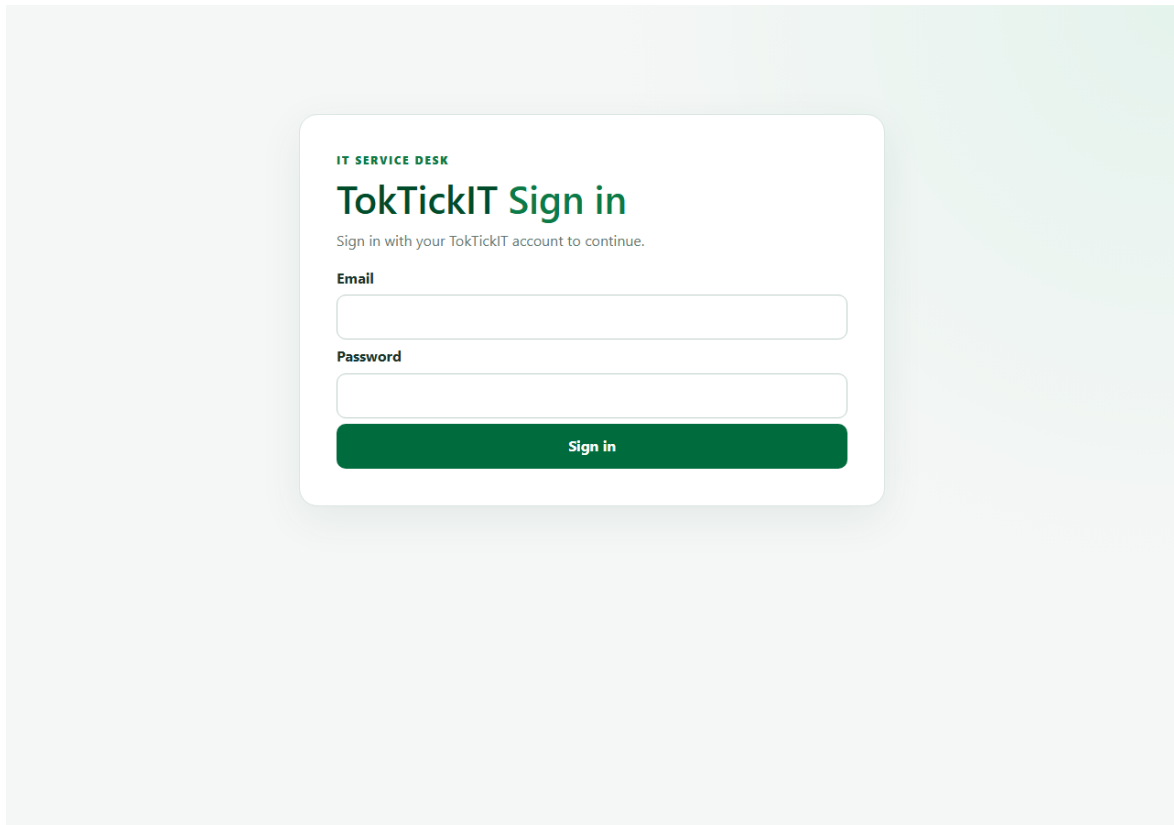


Figure 34. Real login UI at 1280x900; actual API session, 2026-10-01 dirty local capture, not final-main.

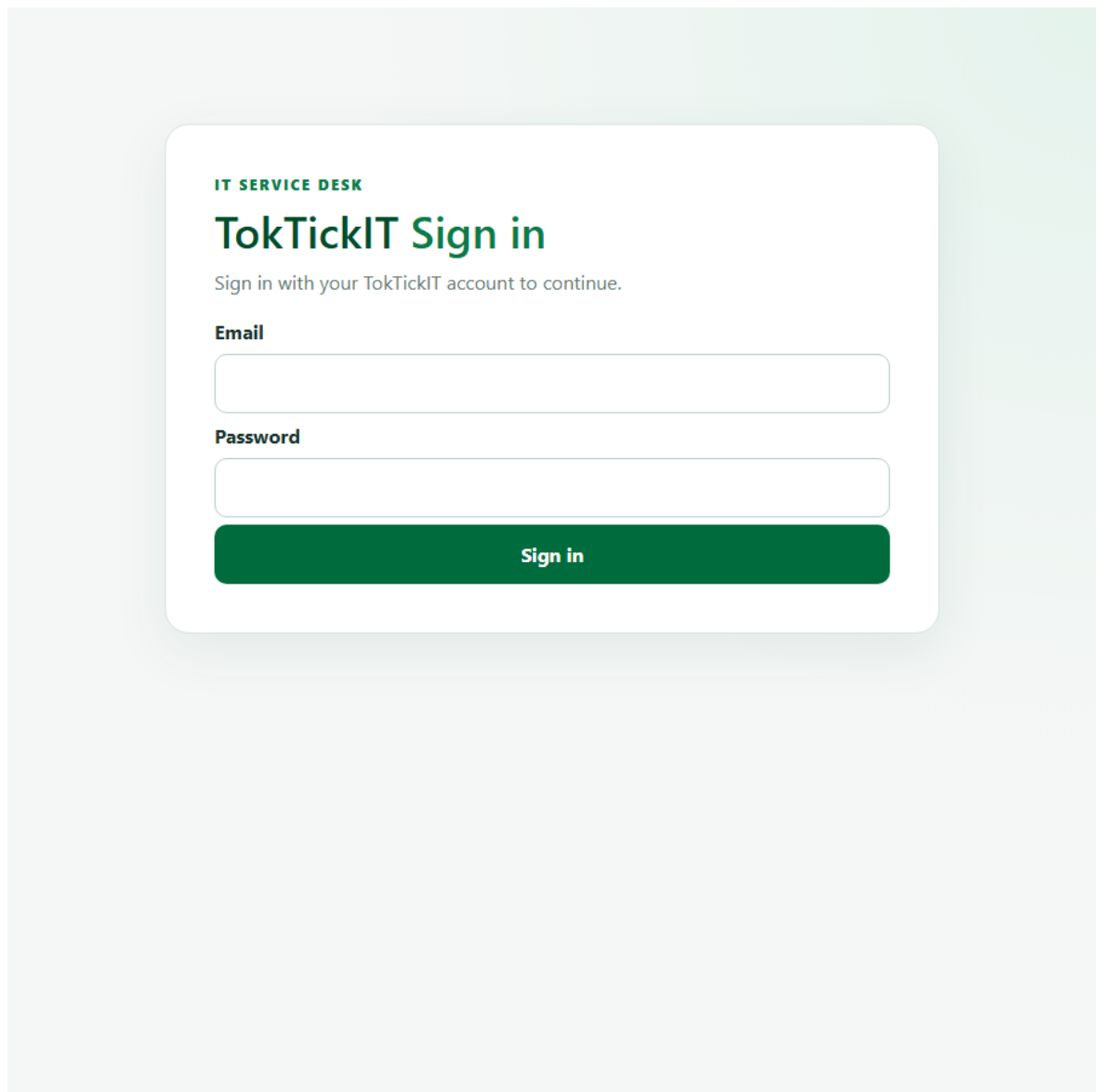
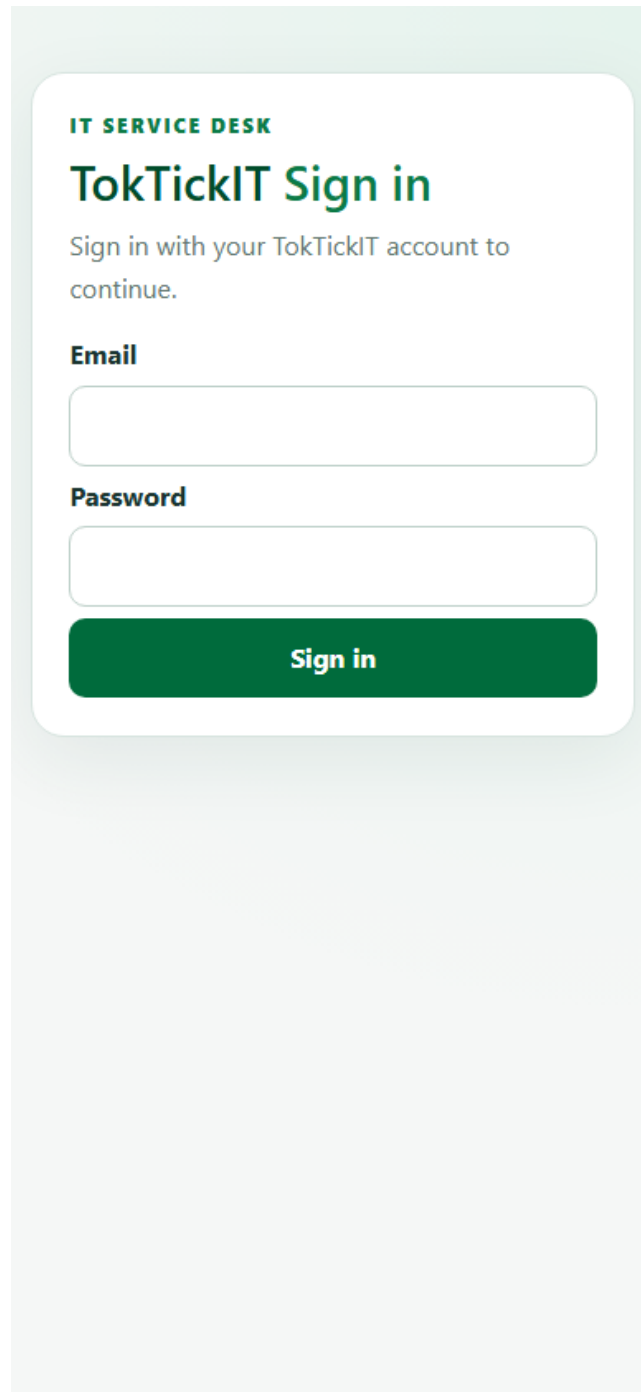


Figure 35. Real login UI at 900x900; actual API session, 2026-10-01 dirty local capture, not final-main.

The image shows a mobile application login screen. At the top, it says "IT SERVICE DESK" in a small, dark green font. Below that is the title "TokTickIT Sign in" in a large, bold, dark green font. Under the title is a line of text: "Sign in with your TokTickIT account to continue." Below this text are two input fields. The first is labeled "Email" and the second is labeled "Password". Both labels are in a bold, dark green font. The input fields are white with a thin grey border. At the bottom of the form is a dark green button with the text "Sign in" in white. The entire form is set against a light green background.

IT SERVICE DESK

TokTickIT Sign in

Sign in with your TokTickIT account to continue.

Email

Password

Sign in

Figure 36. Real login UI at 390x844; actual API session, 2026-10-01 dirty local capture, not final-main.

ACCOUNT SECURITY

Change your password

Your initial password must be replaced before you can open the service desk.

Signed in as narin.kittisak@example.com. Use 12-128 characters with at least one letter and one number.

Current password

New password

12-128 characters; include a letter and a number.

Confirm new password

[Log out](#)

Save password

Figure 37. Real mandatory change password UI at 1280x900; actual API session, 2026-10-01 dirty local capture, not final-main.

ACCOUNT SECURITY

Change your password

Your initial password must be replaced before you can open the service desk.

Signed in as narin.kittisak@example.com. Use 12–128 characters with at least one letter and one number.

Current password

New password

12–128 characters; include a letter and a number.

Confirm new password

Save password

[Log out](#)

Figure 38. Real mandatory change password UI at 900x900; actual API session, 2026-10-01 dirty local capture, not final-main.

ACCOUNT SECURITY

Change your password

Your initial password must be replaced before you can open the service desk.

Signed in as narin.kittisak@example.com. Use 12–128 characters with at least one letter and one number.

Current password

New password

12–128 characters; include a letter and a number.

Confirm new password

Save password

[Log out](#)

Figure 39. Real mandatory change password UI at 390x844; actual API session, 2026-10-01 dirty local capture, not final-main.

Appendix - Executed Interaction Evidence

TOKTICKIT

IT Service Desk

Somchai Staff - IT_STAFF [Log out](#)

Ticket Queue

OPERATIONS

Ticket Queue

Find shared work, inspect the detail, and keep ownership visible.

Search tickets

Number, summary, description, requester

Status

All statuses

Requested priority

All requested priority

IT priority

All IT priorities

Owner

All ownership

Sort by

Last updated

Sort direction

Descending

Tickets per page

10

Apply filters

Clear filters

TT-LAB3-E2E-QUEUE-000007

Queue fixture 7: workstation support

Hardware - Mail Chantarangsu - Updated 10/1/2026, 1:43:57 AM

Requested LOW

IT MEDIUM

RESOLVED

Owner: Kanya Staff

Open Detail

TT-LAB3-E2E-QUEUE-000006

Queue fixture 6: workstation support

Hardware - Nain Kittak - Updated 10/1/2026, 1:43:57 AM

Requested HIGH

IT LOW

CLOSED

Unassigned

Open Detail

TT-LAB3-E2E-QUEUE-000005

Queue fixture 5: workstation support

Hardware - Aniya Somchai - Updated 10/1/2026, 1:43:57 AM

Requested MEDIUM

IT HIGH

RESOLVED

Owner: Kanya Staff

Open Detail

TT-LAB3-E2E-QUEUE-000004

Queue fixture 4: workstation support

Hardware - Pinchanok Kottarakul - Updated 10/1/2026, 1:43:57 AM

Requested LOW

IT MEDIUM

WAITING FOR REQUESTER

Unassigned

Open Detail

TT-LAB3-E2E-QUEUE-000003

Queue fixture 3: workstation support

Hardware - Mail Chantarangsu - Updated 10/1/2026, 1:43:57 AM

Requested HIGH

IT LOW

IN PROGRESS

Owner: Kanya Staff

Open Detail

TT-LAB3-E2E-QUEUE-000002

Queue fixture 2: workstation support

Hardware - Nain Kittak - Updated 10/1/2026, 1:43:57 AM

Requested MEDIUM

IT HIGH

OPEN

Unassigned

Open Detail

TT-LAB3-E2E-QUEUE-000001

Queue fixture 1: workstation support

Hardware - Aniya Somchai - Updated 10/1/2026, 1:43:57 AM

Requested LOW

IT MEDIUM

NEW

Owner: Kanya Staff

Open Detail

TT-20260919-000003

VPN access request

Network - Mail Chantarangsu - Updated 10/1/2026, 1:43:57 AM

Requested LOW

IT LOW

NEW

Unassigned

Open Detail

TT-20260919-000002

Email access request

Account and Access - Nain Kittak - Updated 10/1/2026, 1:43:57 AM

Requested MEDIUM

IT MEDIUM

IN PROGRESS

Owner: Kanya Staff

Open Detail

TT-20260919-000001

Laptop cannot connect to Wi-Fi

Hardware - Aniya Somchai - Updated 10/1/2026, 1:43:57 AM

Requested HIGH

IT HIGH

OPEN

Owner: Somchai Staff

Open Detail

Previous

Page 2 of 3

Next

Figure 40. Populated second Queue page in the passing 16-test E2E run on 2026-10-01; local disposable database, not final-main.

TOKTICKIT

IT Service Desk

Somchai Staff · IT_STAFF [Log out](#)

Ticket Queue

[← Back to Ticket Queue](#)

TT-DOCKER-PRELAB3-001

OPEN

Pre-Lab 3 preservation fixture

Status updated.

Requester

Ariya Somchai ·
ariya.somchai@example.com

Category

Hardware

Related system

Corporate Laptop

Created

10/1/2026, 1:43:53 AM

Updated

10/1/2026, 1:45:15 AM

Requested priority

HIGH

IT priority

LOW

Owner

Kanya Staff ·
kanya.staff@example.com

Description

Synthetic migration fixture for local verification.

Ownership

Only active IT Staff or Administrators may own a Ticket.

Claim as me

Unassign

Assign owner ID

8

Assign owner

IT Priority and status

IT Priority

LOW

Save IT Priority

Next status

Choose a permitted transition

Update status

Attachments

Existing Ticket files.

baseline.txt

Uploaded 10/1/2026, 1:43:53 AM

Public Comments

Visible to Requesters, IT Staff, and Administrators.

No public comments.

Add a public comment

Add public comment

Internal Notes

Private to IT Staff and Administrators: never shown in the Requester workspace.

No Internal Notes.

Add an Internal Note

Add Internal Note

Figure 41. Migrated Attachment after reassignment, LOW IT Priority and NEW-to-OPEN update; 2026-10-01 passing local E2E.

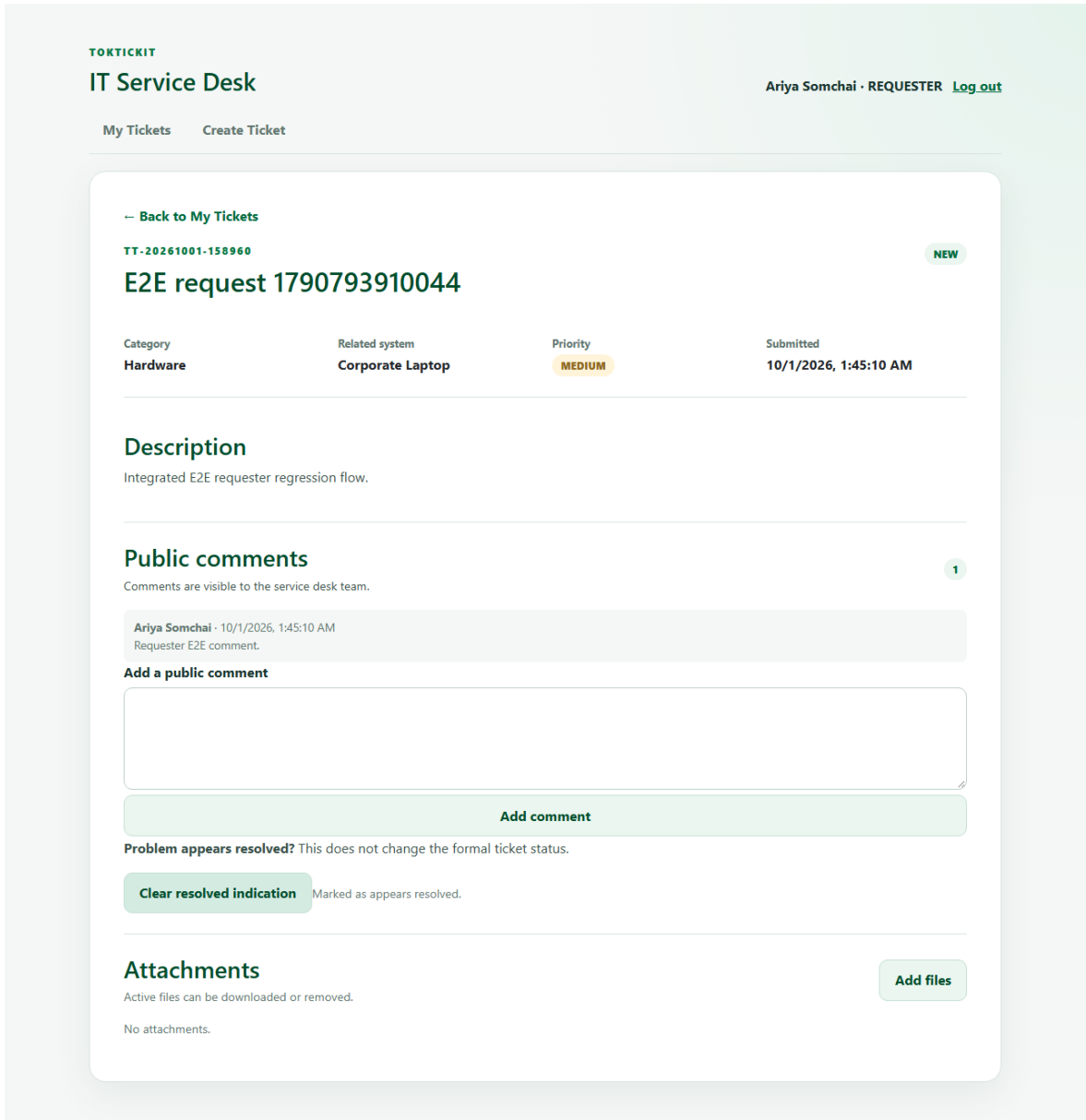


Figure 42. Requester indication is separate from formal status; 2026-10-01 passing local E2E. Staff rendering follows in Figures 43-45.

Current local supplementary evidence

All following images are actual application/browser captures, not generated images. Source is the dirty local tree, not final-main. Historical failed captures and tests remain retained in the evidence folders.

Requester resolution indication

Requester indicates that the problem appears resolved.

Reported 10/1/2026, 1:45:10 AM. Formal ticket status is still NEW. Only IT Staff can formally resolve or close the Ticket.

Figure 43. Native browser region from fixed Staff Detail at 1280x900: API indication shown, formal status still NEW; passing 16-test local run 2026-10-01. Full-page captures retained separately.

Requester resolution indication

Requester indicates that the problem appears resolved.

Reported 10/1/2026, 1:45:10 AM. Formal ticket status is still NEW. Only IT Staff can formally resolve or close the Ticket.

Figure 44. Native browser region from fixed Staff Detail at 900x900: API indication shown, formal status still NEW; passing 16-test local run 2026-10-01. Full-page captures retained separately.

Requester resolution indication

Requester indicates that the problem appears resolved.

Reported 10/1/2026, 1:45:10 AM. Formal ticket status is still NEW. Only IT Staff can formally resolve or close the Ticket.

Figure 45. Native browser region from fixed Staff Detail at 390x844: API indication shown, formal status still NEW; passing 16-test local run 2026-10-01. Full-page captures retained separately.

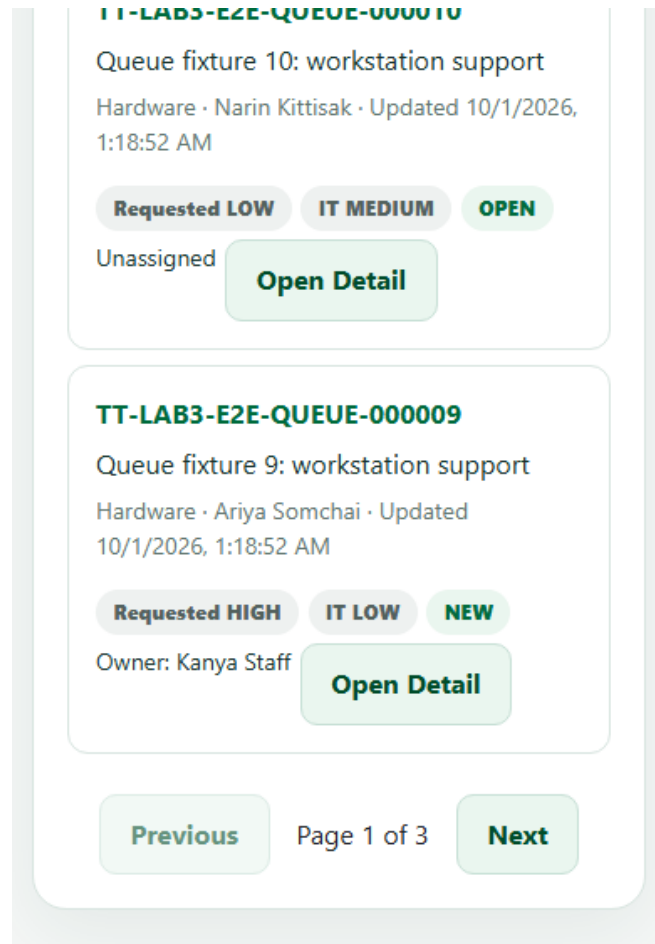


Figure 46. Queue final native mobile section 6/6, pagination controls; 2026-10-01 local browser capture, 390x844 viewport; native sections overlap by 100px.

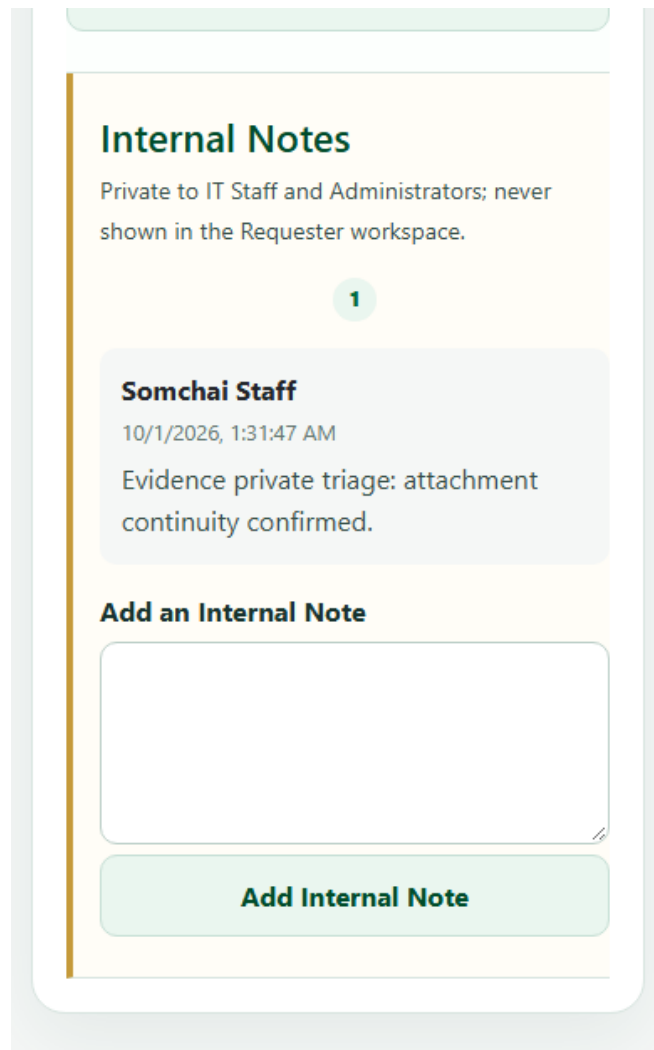


Figure 47. Staff Detail final native mobile section 5/5, Internal Notes; 2026-10-01 local browser capture, 390x844 viewport; native sections overlap by 100px.

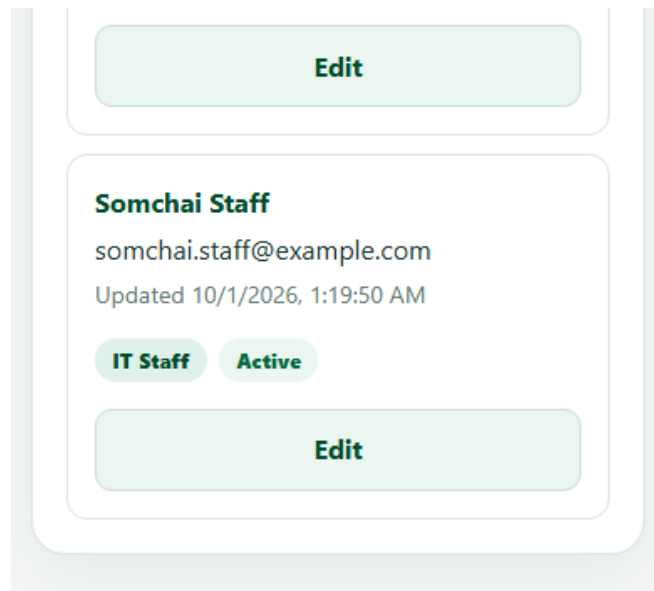


Figure 48. Administrator final native mobile section 6/6; 2026-10-01 local browser capture, 390x844 viewport; native sections overlap by 100px.

TOKTICKIT

IT Service Desk

Admin User · Administrator [Log out](#)

User Management

← Back to User Management

EDIT ACCOUNT

User #11

Edit user

Name

Admin User

Email

admin@example.com

Role

Administrator

☒ Active account

You cannot deactivate your own Administrator account.

Passwords use 12–128 characters with at least one letter and one number. New and reset passwords require a change at next login.

Save changes

Cancel

Reset initial password

This is separate from account editing and forces a password change at next login.

New initial password

Reset password

Figure 49. Own Administrator activation control disabled with explanation; real API rejection is included in the live-check output. Local capture.

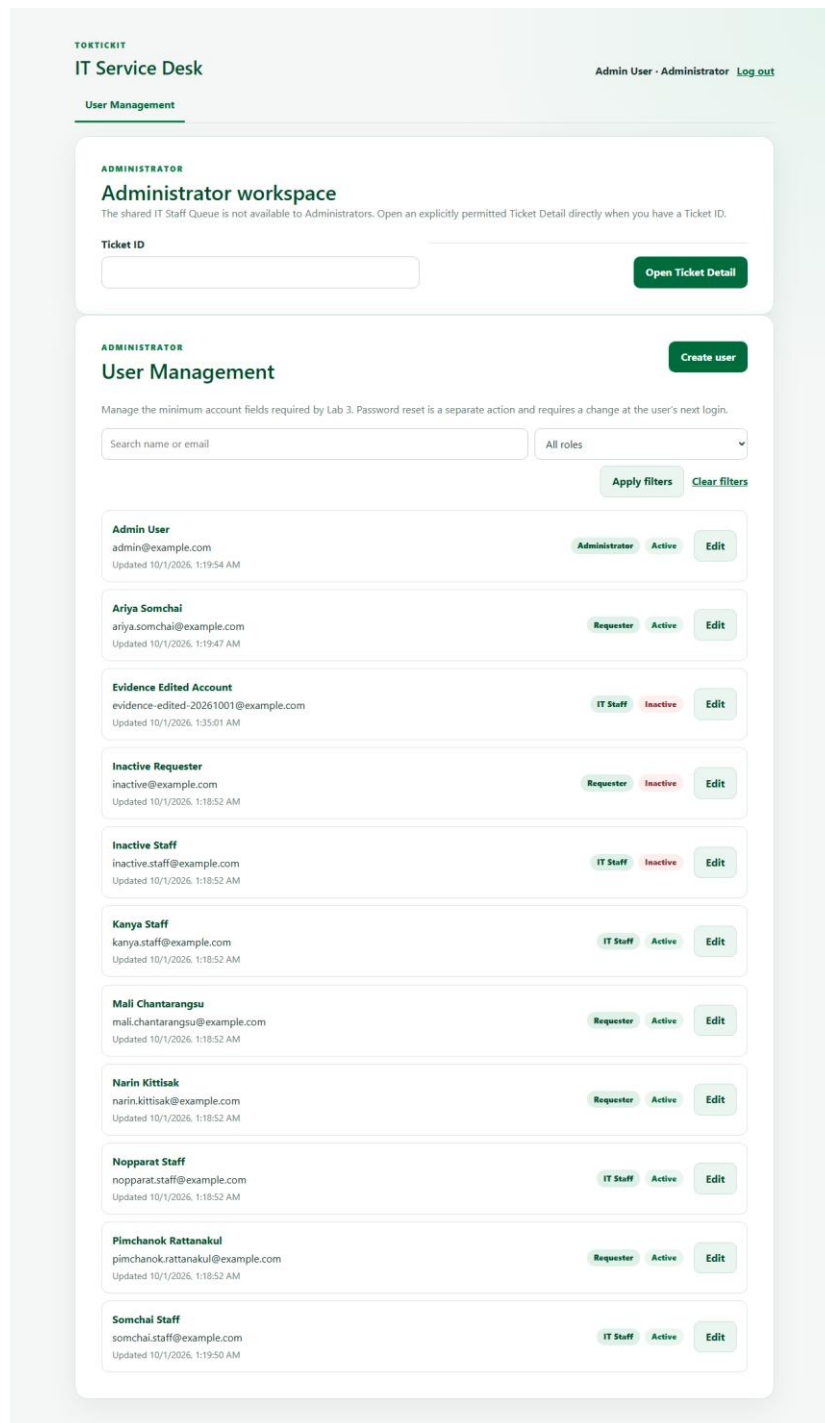


Figure 50. Real synthetic account updated across Name, Email, Role and activation; later restored to active Requester for reset-gate evidence. Local capture.

User Management

[← Back to Administrator workspace](#)

TT-DOCKER-PRELAB3-001

RESOLVED

Pre-Lab 3 preservation fixture

Requester Ariya Somchai · ariya.somchai@example.com	Category Hardware	Related system Corporate Laptop	Created 10/1/2026, 1:18:48 AM
Updated 10/1/2026, 1:31:48 AM	Requested priority HIGH	IT priority LOW	Owner Somchai Staff · somchai.staff@example.com

Description

Synthetic migration fixture for local verification.

Ownership

Only active IT Staff or Administrators may own a Ticket.

Administrator access is read-only. Current owner: Somchai Staff · somchai.staff@example.com.

IT Priority and status

IT Priority

LOW

Next status

Choose a permitted transition

Attachments

Existing Ticket files.

baseline.txt

Uploaded 10/1/2026, 1:18:48 AM

Public Comments

Visible to Requesters, IT Staff, and Administrators.

Somchai Staff

10/1/2026, 1:31:47 AM

Evidence public update: investigation in progress.

Internal Notes

Private to IT Staff and Administrators; never shown in the Requester workspace.

Somchai Staff

10/1/2026, 1:31:47 AM

Evidence private triage: attachment continuity confirmed.

Figure 51. Administrator direct Detail is read-only; no Queue/claim controls, IT Priority disabled. Local capture.

Current supplemental Test DD

ID	Type	Requirement / AC	Actual assertion and path	Current result
AUTH-04	API authorization	FR-04, AC-04	Expired/revoked/inactive sessions rejected; logout revokes token; server/tests/lab-03/auth.api.test.ts	Passed locally, included in 190-test run
CONTENT-01	API validation/security	AC-09, AC-17	Comment/note boundaries, session authorship, append-only endpoints; server/tests/lab-03/comments-notes.api.test.ts; safe text rendering client/tests/lab-03/RequesterWorkflow.test.tsx	Passed locally
QUEUE-04	API queries	AC-11	Seven sort fields in both directions, owner/filter combinations and invalid queries; server/tests/lab-03/staff-queue.api.test.ts	Passed locally; combined owner bug repaired
STATUS-04	API operations	AC-14–AC-16	All 64 status pairs, terminal confirmation, active owner and role rules; server/tests/lab-03/staff-ticket-detail.api.test.ts	Passed locally with mocked delegates; browser exercises representative transitions
ADMIN-04	API authorization	AC-24–AC-25	Self/last-active-admin guards, demotion and every non-admin management operation; server/tests/lab-03/users-admin.api.test.ts	Passed locally; real self-removal/denial output retained separately
MIG-04	Real PostgreSQL	AC-26	Actual deploy, before/after baseline fields/IDs/relationships, seed twice counts/authorship/ownership and attachment bytes; server/prisma/prepare-lab3-e2e.ts	Executed; database-run2.txt exit 0; synthetic disposable data only
REG-01	Regression	AC-06–AC-10	Authenticated Requester legacy tests plus client/tests/lab-03/RequesterWorkflow.test.tsx and e2e/lab-03/authentication.spec.ts	Passed locally; UI-03 is not attributed to Login tests
UI-07	UI and E2E	AC-10, AC-13	Indication shown for Staff/Admin when present, absent when null, no status mutation; client/tests/lab-03/StaffTicketDetail.test.tsx; real Staff browser case e2e/lab-03/staff-ticket-flow.spec.ts	Added before fix: 2 failed; after fix full client 36 passed and E2E 16 passed
STYLE-03	UI style/visual	AC-27	Actual client/tests/lab-03/ui-style.test.tsx; role/viewport/browser checks plus real Login/Change Password captures	Passed within asserted/captured states; not universal accessibility certification

Complete current local verification output

2026-10-01, Asia/Bangkok. Branch feature/lab3-submission-corrections; base 1fb8df040eb12ac443bb4246a314babfde1564e9, dirty. Complete stdout/stderr below is local verification only. Reviewed integration and clean-main output are still required. The 190 server/36 client tests and 16 E2E checks passed with exit 0; the separate red test and earlier failures are retained, not erased.

Server unit API authorization regression tests

Raw output source: output\docs\evidence-20261001-step5\server-tests.txt. Exit 0. Directory: server or client as identified by npm package output; E2E/preparer/live script run from repository root.

```
> toktickit-server@1.0.0 test
> vitest run --reporter=verbose
```

```
RUN v2.1.9 C:/Users/Jitvoottikrai/OneDrive/เดสก์ท็อป/SoftwareEn/LAB1/Lab1_Starter_Scaffold/toktickit-lab3-specification/server
```

```
✓ tests/lab-03/migration-regression.test.ts > Lab 3 migration and seed contract > maps the canonical User model to the Lab 2 table and preserves ownership fields
✓ tests/lab-03/migration-regression.test.ts > Lab 3 migration and seed contract > keeps role fixtures and local-only initial credentials in the idempotent seed
✓ tests/lab-02/validation.unit.test.ts > Lab 2 ticket validation > trims values and reports field-level boundaries
✓ tests/lab-02/validation.unit.test.ts > Lab 2 ticket validation > accepts valid fields and rejects invalid references and priority
✓ tests/lab-02/validation.unit.test.ts > Lab 2 ticket validation > enforces attachment type, size, and active-count limits
✓ tests/lab-02/ticket-number.unit.test.ts > Lab 2 ticket number date component > formats the official YYYYMMDD date segment
✓ tests/lab-03/auth.unit.test.ts > Lab 3 authentication primitives > normalizes email consistently
✓ tests/lab-03/auth.unit.test.ts > Lab 3 authentication primitives > enforces the documented password boundaries and composition
✓ tests/lab-03/auth.unit.test.ts > Lab 3 authentication primitives > stores a salted scrypt digest rather than the supplied password
✓ tests/lab-03/auth.unit.test.ts > Lab 3 authentication primitives > verifies the correct password and rejects wrong or malformed digests
✓ tests/lab-01/categories.test.ts > GET /api/categories > returns the four seeded categories in id order
✓ tests/lab-01/health.test.ts > GET /api/health > returns 200 with status ok and the service name
✓ tests/lab-02/reference-data.api.test.ts > Lab 2 reference data failures > returns active related systems in the documented wrapper
✓ tests/lab-02/reference-data.api.test.ts > Lab 2 reference data failures > returns a safe 500 instead of fabricated categories when the database fails
✓ tests/lab-02/requester-context.api.test.ts > Lab 2 requester context and reference data > returns active requesters only
✓ tests/lab-02/attachments.api.test.ts > Lab 2 attachment lifecycle > lists attachments only after confirming ticket ownership
✓ tests/lab-02/create-ticket.api.test.ts > Lab 2 ticket creation > returns field errors without persisting an invalid ticket
✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > requires the approved confirmation and transition matrix on the backend
✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > updates a permitted status only when the current state still matches
✓ tests/lab-03/authorization.api.test.ts > Lab 3 Requester authorization > uses session identity for Ticket creation and ignores requesterId input
✓ tests/lab-03/authorization.api.test.ts > Lab 3 Requester authorization > returns the same safe not-found response for cross-requester Ticket access
✓ tests/lab-02/my-tickets.api.test.ts > Lab 2 My Tickets list > applies filters and pagination inside the authenticated requester scope
✓ tests/lab-02/my-tickets.api.test.ts > Lab 2 My Tickets list > rejects invalid list controls before querying tickets
✓ tests/lab-02/ticket-detail.api.test.ts > Lab 2 owned Ticket Detail > returns the detail only within the authenticated requester scope
✓ tests/lab-02/ticket-detail.api.test.ts > Lab 2 owned Ticket Detail > returns the same safe not-found response for another requester's ticket
✓ tests/lab-03/comments-notes.api.test.ts > Lab 3 Public Comments and Internal Notes > keeps Internal Notes on separate staff endpoints and records the session author
```

- ✓ tests/lab-03/comments-notes.api.test.ts > Lab 3 Public Comments and Internal Notes > does not allow Requesters to read staff notes
- ✓ tests/lab-02/requester-context.api.test.ts > Lab 2 requester context and reference data > rejects missing sessions and inactive accounts without accepting the old requester header
- ✓ tests/lab-02/attachments.api.test.ts > Lab 2 attachment lifecycle > soft-removes an owned attachment and retains its audit metadata
- ✓ tests/lab-02/attachments.api.test.ts > Lab 2 attachment lifecycle > requires a useful removal reason before touching the database
- ✓ tests/lab-02/create-ticket.api.test.ts > Lab 2 ticket creation > creates a new ticket with the official number and NEW status
- ✓ tests/lab-03/authorization.api.test.ts > Lab 3 Requester authorization > returns Public Comments only and records a requester resolution indication
- ✓ tests/lab-03/comments-notes.api.test.ts > Lab 3 Public Comments and Internal Notes > rejects blank, overlength, and non-string comment content before database writes
- ✓ tests/lab-03/e2e-fixtures.unit.test.ts > Lab 3 disposable E2E preparation safeguards > rejects an absent, shared, remote, or invalid database URL (undefined)
- ✓ tests/lab-03/e2e-fixtures.unit.test.ts > Lab 3 disposable E2E preparation safeguards > rejects an absent, shared, remote, or invalid database URL (postgresql://local:local@localhost:5432/toktickit)
- ✓ tests/lab-03/e2e-fixtures.unit.test.ts > Lab 3 disposable E2E preparation safeguards > rejects an absent, shared, remote, or invalid database URL (postgresql://local:local@example.com:5432/toktickit_lab3_e2e_unit)
- ✓ tests/lab-03/e2e-fixtures.unit.test.ts > Lab 3 disposable E2E preparation safeguards > rejects an absent, shared, remote, or invalid database URL (postgresql://local:local@localhost:5432/toktickit_lab3_e2e_bad%22name)
- ✓ tests/lab-03/e2e-fixtures.unit.test.ts > Lab 3 disposable E2E preparation safeguards > rejects an absent, shared, remote, or invalid database URL (postgresql://local:local@localhost:5432/toktickit_lab3_e2e_XX)
- ✓ tests/lab-03/e2e-fixtures.unit.test.ts > Lab 3 disposable E2E preparation safeguards > allows an explicitly named local disposable database
- ✓ tests/lab-03/e2e-fixtures.unit.test.ts > Lab 3 disposable E2E preparation safeguards > refuses a shared database before any database operation
- ✓ tests/lab-03/e2e-fixtures.unit.test.ts > Lab 3 disposable E2E preparation safeguards > refuses previously exercised fixtures without resetting accounts or deleting data
- ✓ tests/lab-03/e2e-fixtures.unit.test.ts > Lab 3 disposable E2E preparation safeguards > requires the preexisting migration-baseline attachment before adding pagination data
- ✓ tests/lab-03/e2e-fixtures.unit.test.ts > Lab 3 disposable E2E preparation safeguards > rejects an already changed first-login account before creating fixtures
- ✓ tests/lab-03/e2e-fixtures.unit.test.ts > Lab 3 disposable E2E preparation safeguards > uses actual email-based IDs, preserves accounts, and populates stable realistic Queue fixtures 398ms
- ✓ tests/lab-03/comments-notes.api.test.ts > Lab 3 Public Comments and Internal Notes > rejects blank, overlength, and non-string note content before database writes
- ✓ tests/lab-03/comments-notes.api.test.ts > Lab 3 Public Comments and Internal Notes > accepts the 4,000-character comment boundary and uses the backend author and timestamp
- ✓ tests/lab-03/comments-notes.api.test.ts > Lab 3 Public Comments and Internal Notes > accepts the 4,000-character note boundary and uses the backend author and timestamp
- ✓ tests/lab-03/comments-notes.api.test.ts > Lab 3 Public Comments and Internal Notes > keeps comments append-only: collection and entry edit/delete requests cannot mutate stored content
- ✓ tests/lab-03/comments-notes.api.test.ts > Lab 3 Public Comments and Internal Notes > keeps notes append-only: collection and entry edit/delete requests cannot mutate stored content
- ✓ tests/lab-03/staff-queue.api.test.ts > Lab 3 IT Staff Queue > enforces staff-only access and returns filtered paginated queue data
- ✓ tests/lab-03/staff-queue.api.test.ts > Lab 3 IT Staff Queue > rejects invalid queries before touching ticket data
- ✓ tests/lab-03/staff-queue.api.test.ts > Lab 3 IT Staff Queue > does not expose the Queue to REQUESTER
- ✓ tests/lab-03/staff-queue.api.test.ts > Lab 3 IT Staff Queue > does not expose the Queue to ADMINISTRATOR
- ✓ tests/lab-03/staff-queue.api.test.ts > Lab 3 IT Staff Queue > applies the documented 'updatedAt' 'asc' ordering with a deterministic tie-breaker
- ✓ tests/lab-03/staff-queue.api.test.ts > Lab 3 IT Staff Queue > applies the documented 'updatedAt' 'desc' ordering with a deterministic tie-breaker
- ✓ tests/lab-03/staff-queue.api.test.ts > Lab 3 IT Staff Queue > applies the documented 'createdAt' 'asc' ordering with a deterministic tie-breaker
- ✓ tests/lab-03/staff-queue.api.test.ts > Lab 3 IT Staff Queue > applies the documented 'createdAt' 'desc' ordering with a deterministic tie-breaker
- ✓ tests/lab-03/staff-queue.api.test.ts > Lab 3 IT Staff Queue > applies the documented 'ticketNumber' 'asc' ordering with a deterministic tie-breaker
- ✓ tests/lab-03/staff-queue.api.test.ts > Lab 3 IT Staff Queue > applies the documented 'ticketNumber' 'desc' ordering with a deterministic tie-breaker
- ✓ tests/lab-03/staff-queue.api.test.ts > Lab 3 IT Staff Queue > applies the documented 'status' 'asc' ordering with a deterministic tie-breaker
- ✓ tests/lab-03/staff-queue.api.test.ts > Lab 3 IT Staff Queue > applies the documented 'status' 'desc' ordering with a deterministic tie-breaker

✓ tests/lab-03/staff-queue.api.test.ts > Lab 3 IT Staff Queue > applies the documented 'requestedPriority' 'asc' ordering with a deterministic tie-breaker
 ✓ tests/lab-03/staff-queue.api.test.ts > Lab 3 IT Staff Queue > applies the documented 'requestedPriority' 'desc' ordering with a deterministic tie-breaker
 ✓ tests/lab-03/staff-queue.api.test.ts > Lab 3 IT Staff Queue > applies the documented 'itPriority' 'asc' ordering with a deterministic tie-breaker
 ✓ tests/lab-03/staff-queue.api.test.ts > Lab 3 IT Staff Queue > applies the documented 'itPriority' 'desc' ordering with a deterministic tie-breaker
 ✓ tests/lab-03/staff-queue.api.test.ts > Lab 3 IT Staff Queue > applies the documented 'owner' 'asc' ordering with a deterministic tie-breaker
 ✓ tests/lab-03/staff-queue.api.test.ts > Lab 3 IT Staff Queue > applies the documented 'owner' 'desc' ordering with a deterministic tie-breaker
 ✓ tests/lab-03/staff-queue.api.test.ts > Lab 3 IT Staff Queue > keeps count and item retrieval in the same 'status=waiting_for_requester' filter scope
 ✓ tests/lab-03/staff-queue.api.test.ts > Lab 3 IT Staff Queue > keeps count and item retrieval in the same 'requestedPriority=high' filter scope
 ✓ tests/lab-03/staff-queue.api.test.ts > Lab 3 IT Staff Queue > keeps count and item retrieval in the same 'itPriority=low' filter scope
 ✓ tests/lab-03/staff-queue.api.test.ts > Lab 3 IT Staff Queue > keeps count and item retrieval in the same 'ownerId=7' filter scope
 ✓ tests/lab-03/staff-queue.api.test.ts > Lab 3 IT Staff Queue > keeps count and item retrieval in the same 'ownership=assigned' filter scope
 ✓ tests/lab-03/staff-queue.api.test.ts > Lab 3 IT Staff Queue > keeps count and item retrieval in the same 'ownership=unassigned' filter scope
 ✓ tests/lab-03/staff-queue.api.test.ts > Lab 3 IT Staff Queue > keeps count and item retrieval in the same 'ownerId=7&ownership=assigned' filter scope
 ✓ tests/lab-03/staff-queue.api.test.ts > Lab 3 IT Staff Queue > keeps count and item retrieval in the same 'ownerId=7&ownership=unassigned' filter scope
 ✓ tests/lab-03/staff-queue.api.test.ts > Lab 3 IT Staff Queue > keeps count and item retrieval in the same 'status=OPEN&requestedPriority=HIGH&it...' filter scope
 ✓ tests/lab-03/staff-queue.api.test.ts > Lab 3 IT Staff Queue > searches every documented field case-insensitively
 ✓ tests/lab-03/staff-queue.api.test.ts > Lab 3 IT Staff Queue > rejects page=0 before querying Queue data
 ✓ tests/lab-03/staff-queue.api.test.ts > Lab 3 IT Staff Queue > rejects page=-1 before querying Queue data
 ✓ tests/lab-03/staff-queue.api.test.ts > Lab 3 IT Staff Queue > rejects pageSize=15 before querying Queue data
 ✓ tests/lab-03/staff-queue.api.test.ts > Lab 3 IT Staff Queue > rejects sortBy=unknown before querying Queue data
 ✓ tests/lab-03/staff-queue.api.test.ts > Lab 3 IT Staff Queue > rejects sortDirection=sideways before querying Queue data
 ✓ tests/lab-03/staff-queue.api.test.ts > Lab 3 IT Staff Queue > rejects status=UNKNOWN before querying Queue data
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > rejects requester owners and allows a permitted priority update
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > enforces 'NEW' -> 'NEW' (allowed=false) from the approved matrix
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > enforces 'NEW' -> 'OPEN' (allowed=true) from the approved matrix
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > enforces 'NEW' -> 'IN_PROGRESS' (allowed=false) from the approved matrix
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > enforces 'NEW' -> 'WAITING_FOR_REQUESTER' (allowed=false) from the approved matrix
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > enforces 'NEW' -> 'RESOLVED' (allowed=false) from the approved matrix
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > enforces 'NEW' -> 'CLOSED' (allowed=false) from the approved matrix
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > enforces 'NEW' -> 'REOPENED' (allowed=false) from the approved matrix
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > enforces 'NEW' -> 'CANCELLED' (allowed=true) from the approved matrix
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > enforces 'OPEN' -> 'NEW' (allowed=false) from the approved matrix
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > enforces 'OPEN' -> 'OPEN' (allowed=false) from the approved matrix
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > enforces 'OPEN' -> 'IN_PROGRESS' (allowed=true) from the approved matrix
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > enforces 'OPEN' -> 'WAITING_FOR_REQUESTER' (allowed=true) from the approved matrix
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > enforces 'OPEN' -> 'RESOLVED' (allowed=true) from the approved matrix
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > enforces 'OPEN' -> 'CLOSED' (allowed=false) from the approved matrix
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > enforces 'OPEN' -> 'REOPENED' (allowed=false) from the approved matrix

✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > enforces 'CLOSED' -> 'REOPENED' (allowed=true) from the approved matrix
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > enforces 'CLOSED' -> 'CANCELLED' (allowed=false) from the approved matrix
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > enforces 'REOPENED' -> 'NEW' (allowed=false) from the approved matrix
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > enforces 'REOPENED' -> 'OPEN' (allowed=false) from the approved matrix
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > enforces 'REOPENED' -> 'IN_PROGRESS' (allowed=true) from the approved matrix
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > enforces 'REOPENED' -> 'WAITING_FOR_REQUESTER' (allowed=true) from the approved matrix
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > enforces 'REOPENED' -> 'RESOLVED' (allowed=true) from the approved matrix
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > enforces 'REOPENED' -> 'CLOSED' (allowed=false) from the approved matrix
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > enforces 'REOPENED' -> 'REOPENED' (allowed=false) from the approved matrix
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > enforces 'REOPENED' -> 'CANCELLED' (allowed=true) from the approved matrix
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > enforces 'CANCELLED' -> 'NEW' (allowed=false) from the approved matrix
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > enforces 'CANCELLED' -> 'OPEN' (allowed=false) from the approved matrix
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > enforces 'CANCELLED' -> 'IN_PROGRESS' (allowed=false) from the approved matrix
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > enforces 'CANCELLED' -> 'WAITING_FOR_REQUESTER' (allowed=false) from the approved matrix
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > enforces 'CANCELLED' -> 'RESOLVED' (allowed=false) from the approved matrix
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > enforces 'CANCELLED' -> 'CLOSED' (allowed=false) from the approved matrix
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > enforces 'CANCELLED' -> 'REOPENED' (allowed=true) from the approved matrix
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > enforces 'CANCELLED' -> 'CANCELLED' (allowed=false) from the approved matrix
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > requires explicit confirmation for 'NEW' -> 'CANCELLED'
 ✓ tests/lab-03/auth.api.test.ts > Lab 3 authentication contract > uses the same safe 401 for missing and inactive accounts
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > requires explicit confirmation for 'OPEN' -> 'RESOLVED'
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > requires explicit confirmation for 'OPEN' -> 'CANCELLED'
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > requires explicit confirmation for 'IN_PROGRESS' -> 'RESOLVED'
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > requires explicit confirmation for 'IN_PROGRESS' -> 'CANCELLED'
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > requires explicit confirmation for 'WAITING_FOR_REQUESTER' -> 'RESOLVED'
 ✓ tests/lab-03/users-admin.api.test.ts > Lab 3 Administrator User Management API > lists safe users with case-insensitive search and one role filter 417ms
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > requires explicit confirmation for 'WAITING_FOR_REQUESTER' -> 'CANCELLED'
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > requires explicit confirmation for 'RESOLVED' -> 'CLOSED'
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > requires explicit confirmation for 'REOPENED' -> 'RESOLVED'
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > requires explicit confirmation for 'REOPENED' -> 'CANCELLED'
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > returns a safe conflict when the Ticket status changes before persistence
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > owner assignment accepts only active permitted accounts: 'IT_STAFF' active=true
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > owner assignment accepts only active permitted accounts: 'ADMINISTRATOR' active=true
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > owner assignment accepts only active permitted accounts: 'REQUESTER' active=true
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > owner assignment accepts only active permitted accounts: 'IT_STAFF' active=false

```

✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > owner assignment accepts only
active permitted accounts: 'ADMINISTRATOR' active=false
✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > permits unassignment without
looking up an owner account
✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > denies REQUESTER ownership and
status mutations before reading Ticket data
✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > denies ADMINISTRATOR ownership
and status mutations before reading Ticket data
✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > applies the IT Priority
authorization rule to IT_STAFF
✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > applies the IT Priority
authorization rule to ADMINISTRATOR
✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > applies the IT Priority
authorization rule to REQUESTER
✓ tests/lab-03/auth.api.test.ts > Lab 3 authentication contract > gates normal routes until password change and
revokes the old session
✓ tests/lab-03/auth.api.test.ts > Lab 3 authentication contract > rejects a expired session before querying
protected Ticket data
✓ tests/lab-03/auth.api.test.ts > Lab 3 authentication contract > rejects a revoked session before querying
protected Ticket data
✓ tests/lab-03/users-admin.api.test.ts > Lab 3 Administrator User Management API > rejects non-Administrators and
invalid role filters 349ms
✓ tests/lab-03/auth.api.test.ts > Lab 3 authentication contract > rejects a inactive session before querying
protected Ticket data
✓ tests/lab-03/auth.api.test.ts > Lab 3 authentication contract > logout clears the cookie, revokes the session,
and blocks reuse of the old token
✓ tests/lab-03/users-admin.api.test.ts > Lab 3 Administrator User Management API > creates an active user with a
hashed initial password and forced change
✓ tests/lab-03/users-admin.api.test.ts > Lab 3 Administrator User Management API > rejects duplicate email and
invalid create input safely
✓ tests/lab-03/users-admin.api.test.ts > Lab 3 Administrator User Management API > edits permitted account fields
and resets the initial password
✓ tests/lab-03/users-admin.api.test.ts > Lab 3 Administrator User Management API > blocks self-deactivation and
removal of the last active Administrator
✓ tests/lab-03/users-admin.api.test.ts > Lab 3 Administrator User Management API > prevents demotion of the last
active Administrator
✓ tests/lab-03/users-admin.api.test.ts > Lab 3 Administrator User Management API > denies REQUESTER every User
Management operation before user queries or writes
✓ tests/lab-03/users-admin.api.test.ts > Lab 3 Administrator User Management API > denies IT_STAFF every User
Management operation before user queries or writes

Test Files 19 passed (19)
Tests 190 passed (190)
Start at 01:43:13
Duration 3.73s (transform 1.48s, setup 0ms, collect 13.08s, tests 5.87s, environment 4ms, prepare 5.10s)

```

Client UI style and regression tests

Raw output source: output\docs\evidence-20261001-step5\client-tests.txt. Exit 0. Directory: server or client as identified by npm package output; E2E/preparer/live script run from repository root.

```

> toktickit-client@1.0.0 test
> vitest run --reporter=verbose

```

```

RUN v2.1.9 C:/Users/Jitvoottikrai/OneDrive/เอกสารที่เก็บ/SoftwareEn/LAB1/Lab1_Starter_Scaffold/toktickit-lab3-
specification/client

```

```

✓ tests/lab-02/RequesterContext.test.tsx > Lab 2 requester context > restores the authenticated Requester and
ignores stale development-selector state
✓ tests/lab-03/ui-style.test.tsx > Lab 3 Zen Green styling and responsive rules > defines the documented palette,
visible focus treatment, and mobile layout
✓ tests/lab-01/App.test.tsx > App > renders the TokTickIT sign-in heading after checking the session
✓ tests/lab-01/App.test.tsx > App > retains the system-health API and seeded categories after the shell migration
✓ tests/lab-01/App.test.tsx > App > retains a safe system-health error when the API is unavailable
✓ tests/lab-03/ui-style.test.tsx > Lab 3 Zen Green styling and responsive rules > renders accessible sign-in
controls with the primary action hierarchy
✓ tests/lab-03/StaffTicketDetail.test.tsx > Lab 3 Staff Ticket Detail > shows a Requester resolution indication
without changing formal status for 'IT_STAFF'

```

```

✓ tests/lab-03/Login.test.tsx > Lab 3 Login > shows the login form and enters the authenticated Requester shell
without a selector
✓ tests/lab-03/StaffTicketDetail.test.tsx > Lab 3 Staff Ticket Detail > shows a Requester resolution indication
without changing formal status for 'ADMINISTRATOR'
✓ tests/lab-02/CreateTicket.test.tsx > Lab 2 Create Ticket > shows field-level validation without calling the API
✓ tests/lab-03/ChangePassword.test.tsx > Lab 3 mandatory Change Password > blocks the normal shell until a valid
replacement password is saved 303ms
✓ tests/lab-03/UserManagement.test.tsx > Lab 3 Administrator User Management > lists users, applies search and
role filters, and keeps the direct detail entry point 322ms
✓ tests/lab-02/ui-style.test.tsx > Lab 2 Zen Green UI conventions > exposes the required theme tokens and visible
button hierarchy
✓ tests/lab-03/StaffTicketDetail.test.tsx > Lab 3 Staff Ticket Detail > does not show an indication when none
exists for 'IT_STAFF'
✓ tests/lab-03/Login.test.tsx > Lab 3 Login > shows a safe login failure
✓ tests/lab-03/ChangePassword.test.tsx > Lab 3 mandatory Change Password > shows confirmation mismatch and keeps
the save action disabled
✓ tests/lab-02/MyTickets.test.tsx > Lab 2 My Tickets > sends the selected filters and sort controls to the
requester-scoped API 357ms
✓ tests/lab-03/RequesterWorkflow.test.tsx > Lab 3 Requester ticket regression > adds a public comment to the
owned ticket and never renders Internal Notes 372ms
✓ tests/lab-02/AttachmentSection.test.tsx > Lab 2 AttachmentSection > allows upload and removal for active files
while hiding actions for removed files 399ms
✓ tests/lab-03/StaffTicketDetail.test.tsx > Lab 3 Staff Ticket Detail > does not show an indication when none
exists for 'ADMINISTRATOR'
✓ tests/lab-03/StaffTicketQueue.test.tsx > Lab 3 IT Staff Ticket Queue > loads the authenticated Staff Queue and
sends the documented filters 392ms
✓ tests/lab-02/CreateTicket.test.tsx > Lab 2 Create Ticket > rejects an invalid attachment in the browser
✓ tests/lab-02/ui-style.test.tsx > Lab 2 Zen Green UI conventions > keeps active navigation and focusable labeled
controls visible
✓ tests/lab-03/Login.test.tsx > Lab 3 Login > shows and disables the visible sign-in busy state while
authentication is pending
✓ tests/lab-03/UserManagement.test.tsx > Lab 3 Administrator User Management > creates a user with clear initial-
password behavior
✓ tests/lab-03/StaffTicketQueue.test.tsx > Lab 3 IT Staff Ticket Queue > distinguishes an empty Queue from a
filtered no-result state
✓ tests/lab-03/RequesterWorkflow.test.tsx > Lab 3 Requester ticket regression > shows the Requester resolution
indication after the API confirms it
✓ tests/lab-02/RequesterTicketDetail.test.tsx > Lab 2 owned Ticket Detail > renders an owned detail as read-only
and navigates back to My Tickets 456ms
✓ tests/lab-02/RequesterTicketDetail.test.tsx > Lab 2 owned Ticket Detail > shows a safe error when the detail is
not owned or no longer exists
✓ tests/lab-02/MyTickets.test.tsx > Lab 2 My Tickets > supports pagination and distinguishes a filtered no-
results state
✓ tests/lab-03/RequesterWorkflow.test.tsx > Lab 3 Requester ticket regression > renders comment markup as text
without creating executable HTML
✓ tests/lab-02/CreateTicket.test.tsx > Lab 2 Create Ticket > disables duplicate submission and shows the backend
ticket number on success
✓ tests/lab-03/UserManagement.test.tsx > Lab 3 Administrator User Management > edits an account and exposes reset
initial password as a separate action
✓ tests/lab-03/StaffTicketDetail.test.tsx > Lab 3 Staff Ticket Detail > keeps public comments and Internal Notes
separate and supports staff operations
✓ tests/lab-02/CreateTicket.test.tsx > Lab 2 Create Ticket > keeps entered values after an API failure
✓ tests/lab-03/StaffTicketDetail.test.tsx > Lab 3 Staff Ticket Detail > keeps Administrators out of the Queue and
exposes only direct read-only detail

Test Files 14 passed (14)
Tests 36 passed (36)
Start at 01:43:18
Duration 3.61s (transform 816ms, setup 3.42s, collect 6.46s, tests 6.34s, environment 19.89s, prepare 3.47s)

```

Server build

Raw output source: output\docs\evidence-20261001-step5\server-build.txt. Exit 0. Directory: server or client as identified by npm package output; E2E/preparer/live script run from repository root.

```

> tokticket-server@1.0.0 build
> tsc

```

Client build

Raw output source: output\docs\evidence-20261001-step5\client-build.txt. Exit 0. Directory: server or client as identified by npm package output; E2E/preparer/live script run from repository root.

```
> toktickit-client@1.0.0 build
> tsc && vite build

vite v6.4.3 building for production...
transforming...
✓ 31 modules transformed.
rendering chunks...
computing gzip size...
dist/index.html          0.41 kB | gzip: 0.28 kB
dist/assets/index-LW7_0rpz.css 241.29 kB | gzip: 33.52 kB
dist/assets/index-DZ4gmmR4.js 201.88 kB | gzip: 58.81 kB
✓ built in 1.00s
```

Prisma validation

Raw output source: output\docs\evidence-20261001-step5\prisma-validation.txt. Exit 0. Directory: server or client as identified by npm package output; E2E/preparer/live script run from repository root.

```
Prisma schema loaded from prisma\schema.prisma
The schema at prisma\schema.prisma is valid 🍀
```

Real migration preservation and seed twice

Raw output source: output\docs\evidence-20261001-step5\database-run2.txt. Exit 0. Directory: server or client as identified by npm package output; E2E/preparer/live script run from repository root.

```
> toktickit-lab3-integration@1.0.0 test:e2e:prepare
> npm --prefix server run test:e2e:prepare
```

```
> toktickit-server@1.0.0 test:e2e:prepare
> tsx prisma/prepare-lab3-e2e.ts
```

```
New disposable database: toktickit_lab3_e2e_20261001_run2
server: npx --no-install prisma db execute --schema prisma/schema.prisma --stdin
Script executed successfully.
Synthetic pre-Lab-3 snapshot: {"requesters":[{"id":1,"name":"Ariya Somchai","email":"ariya.somchai@example.com","isActive":true,"createdAt":"2026-09-30T18:43:53.478Z"}],"tickets":[{"id":1,"ticketNumber":"TT-DOCKER-PRELAB3-001","requesterId":1,"categoryId":1,"relatedSystemId":1,"requestedPriority":"HIGH","status":"NEW","summary":"Pre-Lab 3 preservation fixture","description":"Synthetic migration fixture for local verification."},"attachments":[{"id":1,"ticketId":1,"originalFilename":"baseline.txt","storageKey":"toktickit_lab3_e2e_20261001_run2-baseline.txt","mimeType":"text/plain","sizeBytes":17,"removedAt":null}],"categories":[{"id":1,"name":"Hardware","isActive":true,"createdAt":"2026-09-30T18:43:53.478Z"}],"systems":[{"id":1,"name":"Corporate Laptop","isActive":true,"createdAt":"2026-09-30T18:43:53.478Z"}]}
server: npx --no-install prisma migrate resolve --applied 20260824000000_lab2_foundation
Prisma schema loaded from prisma\schema.prisma
Datasource "db": PostgreSQL database "toktickit_lab3_e2e_20261001_run2", schema "public" at "localhost:5432"
```

Migration 20260824000000_lab2_foundation marked as applied.

```
server: npx --no-install prisma migrate deploy
Prisma schema loaded from prisma\schema.prisma
Datasource "db": PostgreSQL database "toktickit_lab3_e2e_20261001_run2", schema "public" at "localhost:5432"
```

2 migrations found in prisma/migrations

Applying migration `20260919000000\_lab3\_auth\_requester`

The following migration(s) have been applied:

```
migrations/
├─ 20260919000000_lab3_auth_requester/
│   └─ migration.sql
```

All migrations have been successfully applied.

```
server: npx --no-install prisma migrate status
Prisma schema loaded from prisma/schema.prisma
Datasource "db": PostgreSQL database "toktickit_lab3_e2e_20261001_run2", schema "public" at "localhost:5432"
```

2 migrations found in prisma/migrations

Database schema is up to date!

```
Post-migration snapshot: {"requesters":[{"id":1,"name":"Ariya Somchai","email":"ariya.somchai@example.com","isActive":true,"createdAt":"2026-09-30T18:43:53.478Z"}],"tickets":[{"id":1,"ticketNumber":"TT-DOCKER-PRELAB3-001","requesterId":1,"categoryId":1,"relatedSystemId":1,"requestedPriority":"HIGH","status":"NEW","summary":"Pre-Lab 3 preservation fixture","description":"Synthetic migration fixture for local verification."}],"attachments":[{"id":1,"ticketId":1,"originalFilename":"baseline.txt","storageKey":"toktickit_lab3_e2e_20261001_run2-baseline.txt","mimeType":"text/plain","sizeBytes":17,"removedAt":null}],"categories":[{"id":1,"name":"Hardware","isActive":true,"createdAt":"2026-09-30T18:43:53.478Z"}],"systems":[{"id":1,"name":"Corporate Laptop","isActive":true,"createdAt":"2026-09-30T18:43:53.478Z"}]}
server: npm run prisma:seed (run 1)
```

```
> toktickit-server@1.0.0 prisma:seed
> tsx prisma/seed.ts
```

Seeding completed successfully: 10 users and 3 tickets.

Seed 1 assertions:

```
{"users":10,"tickets":4,"attachments":1,"comments":3,"notes":3,"activeRequesters":4,"inactiveRequesters":1,"activeStaff":3,"inactiveStaff":1,"activeAdmins":1,"seededRequesterOwnership":3,"seededStaffOwnership":2,"commentAuthors":3,"noteAuthors":3,"baselineAttachment":1}
server: npm run prisma:seed (run 2)
```

```
> toktickit-server@1.0.0 prisma:seed
> tsx prisma/seed.ts
```

Seeding completed successfully: 10 users and 3 tickets.

Seed 2 assertions:

```
{"users":10,"tickets":4,"attachments":1,"comments":3,"notes":3,"activeRequesters":4,"inactiveRequesters":1,"activeStaff":3,"inactiveStaff":1,"activeAdmins":1,"seededRequesterOwnership":3,"seededStaffOwnership":2,"commentAuthors":3,"noteAuthors":3,"baselineAttachment":1}
Baseline attachment SHA256: cc7098632c30d31ed81ff5936a75a027f3136024cb96efd35e8fe3f9b4d3cfaf
Prepared E2E fixture IDs/count:
{"databaseName":"toktickit_lab3_e2e_20261001_run2","baselineTicketId":1,"reassignmentOwnerId":8,"queueTotal":20}
```

Integrated E2E responsive and keyboard checks

Raw output source: output/docs/evidence-20261001-step5/e2e-run2.txt. Exit 0. Directory: server or client as identified by npm package output; E2E/preparer/live script run from repository root.

Running 16 tests using 1 worker

```
(node:25232) Warning: The 'NO_COLOR' env is ignored due to the 'FORCE_COLOR' env being set.
(Use `node --trace-warnings ...` to show where the warning was created)
ok 1 [chromium] > e2e\lab-03\authentication.spec.ts:15:3 > Lab 3 authentication and Requester regression > valid login, first-login change, role shell, logout, and direct blocking (2.3s)
ok 2 [chromium] > e2e\lab-03\authentication.spec.ts:33:3 > Lab 3 authentication and Requester regression > invalid and inactive accounts receive safe sign-in failure (570ms)
ok 3 [chromium] > e2e\lab-03\authentication.spec.ts:50:3 > Lab 3 authentication and Requester regression > Requester can create, open, comment, and indicate resolution (931ms)
ok 4 [chromium] > e2e\lab-03\authentication.spec.ts:73:3 > Lab 3 authentication and Requester regression > Requester shell is captured at required responsive sizes (704ms)
ok 5 [chromium] > e2e\lab-03\staff-ticket-flow.spec.ts:5:3 > Lab 3 IT Staff Queue and Ticket Detail > IT Staff can search Queue, open Detail, and see separate comments and notes (1.0s)
ok 6 [chromium] > e2e\lab-03\staff-ticket-flow.spec.ts:23:3 > Lab 3 IT Staff Queue and Ticket Detail > IT Staff Queue and Detail are captured at required responsive sizes (1.4s)
ok 7 [chromium] > e2e\lab-03\staff-ticket-flow.spec.ts:30:3 > Lab 3 IT Staff Queue and Ticket Detail > IT Staff can advance to a populated second Queue page (654ms)
ok 8 [chromium] > e2e\lab-03\staff-ticket-flow.spec.ts:46:3 > Lab 3 IT Staff Queue and Ticket Detail > preserves migrated Attachment and verifies reassignment, priority, and status operations (992ms)
ok 9 [chromium] > e2e\lab-03\staff-ticket-flow.spec.ts:75:3 > Lab 3 IT Staff Queue and Ticket Detail > Staff sees the Requester resolution indication without a formal status change (913ms)
ok 10 [chromium] > e2e\lab-03\user-administration.spec.ts:5:3 > Lab 3 Administrator User Management > Administrator can list, search, create, edit, reset, and require the new initial password at next login (2.5s)
ok 11 [chromium] > e2e\lab-03\user-administration.spec.ts:60:3 > Lab 3 Administrator User Management > Administrator navigation excludes Queue and direct Ticket Detail remains explicit (543ms)
ok 12 [chromium] > e2e\lab-03\user-administration.spec.ts:70:3 > Lab 3 Administrator User Management > Administrator User Management is captured at required responsive sizes (822ms)
```

```
ok 13 [chromium] › e2e\lab-03\z-accessibility-evidence.spec.ts:24:3 › Lab 3 accessibility and transient-state
evidence › captures the visible sign-in busy state and successful destination (1.9s)
ok 14 [chromium] › e2e\lab-03\z-accessibility-evidence.spec.ts:40:3 › Lab 3 accessibility and transient-state
evidence › completes a keyboard-only sign-in and reaches requester navigation (918ms)
ok 15 [chromium] › e2e\lab-03\z-accessibility-evidence.spec.ts:55:3 › Lab 3 accessibility and transient-state
evidence › traverses the IT Staff Queue and opens Ticket Detail with the keyboard (1.0s)
ok 16 [chromium] › e2e\lab-03\z-accessibility-evidence.spec.ts:67:3 › Lab 3 accessibility and transient-state
evidence › traverses Administrator User Management and opens Create User with the keyboard (779ms)
```

16 passed (20.4s)

Live API protection and attachment byte assertions

Raw output source: output\docs\evidence-20261001-step3\live-api-checks.txt. Exit 0. Directory: server or client as identified by npm package output; E2E/preparer/live script run from repository root.

Executed live API checks: 2026-09-30T18:22:22.628Z database: toktickit_lab3_e2e_20261001_run1

Own Administrator removal rejected: {"isActive":false} 403 {"error":{"code":"FORBIDDEN","message":"You cannot deactivate or demote your own Administrator account."},"fields":{"isActive":"Keep your own Administrator account active."}}

Own Administrator removal rejected: {"role":"IT_STAFF"} 403 {"error":{"code":"FORBIDDEN","message":"You cannot deactivate or demote your own Administrator account."},"fields":{"isActive":"Keep your own Administrator account active."}}

Last-active-Administrator invariant retained in real database. These requests execute the self-protection guard, not the separate LAST_ADMINISTRATOR branch.

Non-Administrator User Management denial: ariya.somchai@example.com 403

{"error":{"code":"FORBIDDEN","message":"You do not have permission to perform this action."}}

Migrated Attachment download: 200 17 bytes; SHA256:

cc7098632c30d31ed81ff5936a75a027f3136024cb96efd35e8fe3f9b4d3cfaf

Non-Administrator User Management denial: somchai.staff@example.com 403

{"error":{"code":"FORBIDDEN","message":"You do not have permission to perform this action."}}

All live assertions passed. No pre-existing application database was used.